2 Separated sources remove local contact counterterms exactly

Let $Z[J]$ be a finite-cutoff generating functional normalized by $Z[0] = 1$. Choose smooth compactly supported $f_1, \dots, f_n$ with pairwise positively separated supports. For $A \subseteq [n]$ write

$$J_A = \sum_{i \in A} z_i f_i, \qquad \mathcal{C}_n(z) = \sum_{A \subseteq [n]} (-1)^{n-|A|} \log Z[J_A].$$

At each finite cutoff the logarithms have compatible branches near zero. No cutoff-independent logarithm domain is assumed unless stated.

Proposition 2.1 (Exact cancellation)**.** *For $n \geq 2$, replace $\log Z[J]$ by $\log Z[J] - L[J]$, where*

$$L[J] = \int \ell(x, J(x), \partial J(x), \dots, \partial^m J(x))\, \mathrm{d}x, \qquad \ell(x, 0, \dots, 0) = 0.$$

Whenever this local functional is defined, $\mathcal{C}_n$ is unchanged. Moreover

$$\partial_{z_1} \cdots \partial_{z_n} \mathcal{C}_n(0) = \kappa_n(\mathcal{O}(f_1), \dots, \mathcal{O}(f_n)).$$

Proof. At any point x, at most one source and its derivatives is nonzero. Therefore $L[J_A] = \sum_{i \in A} L[z_i f_i]$. For each fixed i, the coefficient of this term in the alternating subset sum is $(1-1)^{n-1} = 0$. Differentiating once in every source variable annihilates every proper-subset term; the remaining derivative of $\log Z$ is the connected correlation by the definition of cumulants. □

The proof also applies to multiple source species and to finite-range lattice-local counterterms when the source supports are farther apart than their range. It does not apply to a bilocal counterterm.

Corollary 2.2 (A precise analytic estimate that would suffice locally)**.** *Suppose $\mathcal{C}_{n,j}$ is holomorphic on one polydisc $|z_i| < r_i$, uniformly over cutoff index j. If*

$$\sup |\mathcal{C}_{n,j}| \leq M_n,$$

then

$$|\kappa_{n,j}(f_1, \dots, f_n)| \leq M_n \prod_i r_i^{-1}.$$

If also $\sup |\mathcal{C}_{n,j+1} - \mathcal{C}_{n,j}| \leq \varepsilon_j$ and $\sum_j \varepsilon_j < \infty$, the connected correlations converge, with tail bounded by $\prod_i r_i^{-1} \sum_{j \geq J} \varepsilon_j$.

Proof. Apply the multidimensional Cauchy integral formula on smaller concentric polydiscs and let their radii increase to r_i. Apply the same bound to successive differences and sum. □

This result isolates an analytic task. It does not establish a zero-free source domain or the stated bounds for Yang–Mills. The algebraic cancellation of divergent local terms is exact; the required uniform estimates on the remaining connected functional are the hard part.

2.1 Compatibility with reflection positivity

Let θ reflect Euclidean time. Suppose Z has positive source Gram matrices

$$K_{ij} = Z[\theta J_i + J_j]$$

for real sources J_i supported a positive distance inside $t > 0$. Suppose L is local, real on real sources, reflection invariant, and $L[0] = 0$. For $Z_R[J] = e^{-L[J]}Z[J]$, disjointness across the reflection plane gives

$$L[\theta J_i + J_j] = L[J_i] + L[J_j], \qquad K^R = D^* K D, \qquad D_{ii} = e^{-L[J_i]}.$$

Consequently K^R is positive. Complex sources require the corresponding conjugation rule on the reflected side. Differentiated polynomial vectors are obtained as limits of finite differences of these source vectors, so their Gram positivity is retained whenever the derivatives and limits exist.

This explains why local contact subtraction can preserve the physical reflection-positive inner product even if Z_R is not the moment-generating functional of a positive classical random distribution. It supplies no convergence estimate by itself.

3 A complete conditional ultraviolet-iteration lemma

Here the renormalization direction is from fine to coarse scales. The inverse squared coupling decreases. Let $\mathcal{B}$ be a real Banach space of irrelevant interactions and let

$$x_{k+1} = x_k - b + E_k(x_k, K_k), \qquad K_{k+1} = T_k(x_k, K_k), \qquad b > 0, \tag{2}$$

where $x = g^{-2}$. Constants below must be uniform in the number of ultraviolet steps, volume, and the frozen choices defining the maps.

Theorem 3.1 (First-passage closure and source tangent)**.** *Fix $0 \le q < 1$, $\delta, p > 0$, $C \ge 1$. On admitted neighborhoods with x sufficiently large and $\|K\| \le Dx^{-p}$, assume*

$$\begin{aligned} |E_k| &\le Cx^{-\delta} + C\|K\|, & \|T_k\| &\le q\|K\| + Cx^{-p}, \\ |\partial_x E_k| &\le C(x^{-1-\delta} + x^{-1-p}), & \|D_K E_k\| &\le C, \\ \|\partial_x T_k\| &\le Cx^{-1-p}, & \|D_K T_k\| &\le q. \end{aligned} \tag{3}$$

Take $D = 2C/(1-q)$, $x_0 > X$, $K_0 = 0$, and stop at the first index N with $x_N < X$. For X sufficiently large:

1. *$\|K_k\| \le Dx_k^{-p}$ and $b/2 \le x_k - x_{k+1} \le 3b/2$ for $k < N$;*
2. *$X - 3b/2 \le x_N < X$ and $N \to \infty$ as $x_0 \to \infty$;*
3. *for every $s > 1$,*
$$\sum_{k<N} x_k^{-s} \le X^{-s} + \frac{2X^{1-s}}{b(s-1)};$$
4. *at every fixed finite prefix, with $y_k = \partial x_k/\partial x_0$ and $z_k = \partial K_k/\partial x_0$,*
$$e^{-AX^{-\epsilon}} \le y_k \le e^{AX^{-\epsilon}}, \qquad \|z_k\| \le Rx_k^{-1-p} y_k, \quad \epsilon = \min(\delta, p),\ R = \frac{2C}{1-q}.$$

The stopping index is not differentiated.

Proof. Choose $X \geq 3b$ so that $CX^{-\delta} + CDX^{-p} \leq b/2$. If the invariant bound holds at step k, the bound on E_k gives the decrement bounds. Also

$$\|K_{k+1}\| \leq (qD + C)x_k^{-p} \leq Dx_k^{-p} \leq Dx_{k+1}^{-p}.$$

This closes the induction and establishes finite first passage. The endpoint lies in the stated interval. In particular

$$\frac{2(x_0 - X)}{3b} < N \leq 1 + \frac{2(x_0 - X)}{b}.$$

Reading backward from the last admitted step gives $x_{N-1-j} \geq X + bj/2$. Comparison with an integral proves the sum bound.

The derivative recursions are

$$y_{k+1} = (1 + \partial_x E_k)y_k + (D_K E_k)z_k, \qquad z_{k+1} = (\partial_x T_k)y_k + (D_K T_k)z_k.$$

Start with $y_0 = 1$, $z_0 = 0$. Suppose $y_k > 0$ and $\|z_k\| / y_k \leq Rx_k^{-1-p}$. Define

$$\eta_X = C(X^{-1-\delta} + X^{-1-p}) + CRX^{-1-p}.$$

Choose X so $\eta_X \leq \min(1/2, (1-q)/4)$. Then $y_{k+1}/y_k \geq 1 - \eta_X > 0$, and

$$\frac{\|z_{k+1}\|}{y_{k+1}} \leq \frac{C + qR}{1 - \eta_X} x_k^{-1-p} \leq Rx_{k+1}^{-1-p}.$$

Thus the induction closes. The finer estimate $|y_{k+1}/y_k - 1| \leq C_1 x_k^{-1-\epsilon}$ gives

$$|\log y_{k+1} - \log y_k| \leq 2C_1 x_k^{-1-\epsilon}.$$

Sum and use the preceding bound with $s = 1 + \epsilon$. □

Proportional complex analyticity domains can imply the displayed x-derivative estimates by Cauchy bounds. The contraction and the correct model-specific domain are additional inputs. They do not follow from a bound on a scalar partition function.

3.1 Correct scale dependence

Summing (2) backward for j steps yields

$$x_{N-j} = x_N + bj + O\left(\sum_{i=1}^{j} (X + ci)^{-\epsilon}\right).$$

The error is $O(j^{1-\epsilon})$ for $\epsilon < 1$, $O(\log(1+j))$ for $\epsilon = 1$, and $O(1)$ for $\epsilon > 1$. Thus $x_{N-j}/j \to b$ uniformly over the endpoint window. With blocking factor $L > 1$ and $r = L^{-j}$ in terminal-scale units,

$$g(r)^2 \log(1/r) \longrightarrow \frac{\log L}{b}.$$

In particular $\sum g_k^2$ is generally divergent over arbitrarily many ultraviolet steps, whereas $\sum g_k^{2+\eta}$ is summable for every $\eta > 0$. A proof must extract the marginal term before applying a summability argument.

If the leading decrement is b_k with bounded accumulated discrepancy $|\sum_{j<k}(b_j - b)| \leq A_0$, the change of coordinate $u_k = x_k + \sum_{j<k}(b_j - b)$ reduces it to the same statement at a slightly larger threshold. This requires the discrepancy to be independent of the differentiated bare parameter when using the tangent result.

3.2 What the tangent actually proves

Suppose exact source-independent kernels push the unnormalized density to

$$\rho_k(V; x_0) = \exp[c_k(x_0) - x_k(x_0)\mathcal{A}_k(V) - K_k(V; x_0)].$$

A constant bare insertion $e^{t\mathcal{A}_0}$ is exactly the shift $x_0 \mapsto x_0 - t$. After differentiation and centering, its transported first insertion is

$$y_k(\mathcal{A}_k - \langle \mathcal{A}_k \rangle) + z_k - \langle z_k \rangle.$$

The leading coefficient cannot disappear under the hypotheses of the theorem. This is a first-insertion identity, not a two-point theorem: transporting a product is not multiplying transported conditional expectations. Second derivatives retain conditional covariance and further source terms. Nonconstant sources also require a separate local extension. Source-dependent cutoff tests or changes of variables cannot be differentiated without their extra terms.

4.1 The constraint correction is real

Let E be the exact-curvature subspace of a finite lattice and let P_E be its orthogonal projection. Coarse constraints restrict the fluctuation subspace to $E_C \subseteq E$. Its canonical Gaussian covariance is P_{E_C}, not P_E. If Q_C describes the constraints on E, then

$$P_{E_C} = P_E - P_E Q_C^*(Q_C P_E Q_C^*)^+ Q_C P_E,$$

where $+$ denotes the Moore–Penrose inverse. Singular-value decomposition proves that the subtracted term is an orthogonal projection. It need not be small in operator norm.

The following estimate controls its effect on separated quadratic observables after an exact translation average. It does not require a pointwise smoothness bound on the correction kernel.

Theorem 4.1 (Averaged constraint correction)**.** *Let a periodic four-dimensional lattice of spacing a have N_s sites and a fixed finite curvature-component space at each site. Let P be a translation-invariant orthogonal projection, and let $P_s \le P$ be orthogonal projections with*

$$D_s = P - P_s = T_s D_0 T_s^*, \qquad \operatorname{rank} D_0 \le C N_s (a/H)^4.$$

Average s over all fine-lattice translations, or one coarse cell if the projections have coarse periodicity. Let $f, h \ge 0$, $M_f = \|f\|_\infty$, $M_h = \|h\|_\infty$, and suppose f occupies at most $C_f (d/a)^4$ sites. Set

$$V_0 = \frac{1}{2} \sum_{x,y} f_x h_y \, \|P(x,y)\|_{\mathrm{HS}}^2 , \quad V_s = \frac{1}{2} \sum_{x,y} f_x h_y \, \|P_s(x,y)\|_{\mathrm{HS}}^2 .$$

Then, with $R_D = \sum_{x,y} f_x h_y \mathbb{E}_s \, \|D_s(x,y)\|_{\mathrm{HS}}^2$,

$$R_D \le C C_f M_f M_h (d/H)^4, \tag{4}$$

$$|\mathbb{E}_s V_s - V_0| \le \sqrt{2 V_0 R_D} + R_D/2. \tag{5}$$

If $V_0 \ge c M_f M_h > 0$, the relative error is at most

$$C' \big[(d/H)^2 + (d/H)^4 \big].$$

Proof. Since $P_s \le P$, D_s is an orthogonal projection. Translation averaging gives

$$\mathbb{E}_s \operatorname{tr} D_s(x,x) = \frac{\operatorname{rank} D_0}{N_s} \le C(a/H)^4.$$

The identity $D_s^2 = D_s$ gives

$$\sum_y \|D_s(x,y)\|_{\mathrm{HS}}^2 = \operatorname{tr} D_s(x,x).$$

Use $f_x \le M_f$, $h_y \le M_h$ and the support count to obtain (4). Expand $P_s = P - D_s$ in the squared Hilbert–Schmidt norm. Cauchy–Schwarz on the joint measure of (x,y,s) bounds the cross term by $\sqrt{2V_0R_D}$; the squared correction contributes $R_D/2$. Division by the assumed lower bound proves the corollary. □

For a centered Gaussian curvature F with covariance P, V_0 is exactly the covariance of $Q_f = \frac{1}{2}\sum_x f_x|F_x|^2$ and Q_h. In this convention the physical Riemann factor a^4 cancels the square of the physical-curvature rescaling a^{-2}. For normalized bumps $f_d = d^{-4}f(\cdot/d)$ and $h_d = d^{-4}h(\cdot/d)$, M_fM_h is of order d^{-8}.

For linear blocking, a fixed number of constraints per coarse block gives the rank hypothesis. Exact covariance under shifting the block origin must still be checked. If every origin gives an exact representation of the same Wilson expectation, averaging the representation changes no observable. Applying this lemma to a background-dependent, nonlinear covariance requires a further comparison; it is not a projection merely because its zero-background approximation is one.

4.2 The free separated signal and its normalization

With q independent colors, use the canonical free potential covariance

$$\mathbb{E}A_\mu^a(x)A_\nu^b(y) = \delta^{ab}\delta_{\mu\nu}G(x-y), \qquad G(z) = \frac{1}{4\pi^2|z|^2}.$$

For $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$, differentiation gives

$$K_{\mu\nu,\rho\sigma} = -\delta_{\nu\sigma}\partial_\mu\partial_\rho G + \delta_{\nu\rho}\partial_\mu\partial_\sigma G + \delta_{\mu\sigma}\partial_\nu\partial_\rho G - \delta_{\mu\rho}\partial_\nu\partial_\sigma G.$$

At $z = |z|e_0$, the six-by-six matrix on pairs $\mu < \nu$ is diagonal, with three entries of each sign $\pm 1/(\pi^2|z|^4)$. Thus

$$\begin{gathered}\|K(z)\|_{\mathrm{HS}}^2 = \frac{6q}{\pi^4|z|^8}, \qquad \operatorname{Cov}(Q(x),Q(y)) = \frac{3q}{\pi^4|x-y|^8}, \\ Q = \frac{1}{2}\sum_{a,\mu<\nu} :(F_{\mu\nu}^a)^2: .\end{gathered} \tag{6}$$

This establishes a nonconstant scalar observable in the reference theory. It does not establish interacting gauge dynamics: the reference theory is free.

5 Coarse fields need not be Gaussian

There is another useful comparison that explicitly retains the coarse mean. It has stronger covariance assumptions than the rank-averaging theorem but needs no Gaussian law for that mean.

Let $P(k)$ be the exact-two-form projection, repeated for q colors. Let $0 \le \chi \le 1$ be even and supported in $|k| \le 1$. Let X be a centered stationary Gaussian field with Fourier covariance

$$C_{a,H}(k) = e^{-a^2|k|^2}(1 - \chi(Hk))P(k).$$

Let m be independent of X and satisfy

$$\sup_x \mathbb{E}|m(x)|^4 \le B^4 H^{-8}.$$

For $F = X + m$ put $Q_f = \frac{1}{2}\int f|F|^2$. Suppose $f, h \ge 0$ and every pair in their supports obeys $d \le |x - y| \le Rd$, with $a \le d/4$ and $d \le H$. Define

$$V_0(f,h) = \frac{3q}{\pi^4} \iint f(x)h(y)|x - y|^{-8}\,\mathrm{d}x\,\mathrm{d}y.$$

Proposition 5.1 (Arbitrary coarse completion)**.** *For positive test masses,*

$$\frac{|\operatorname{Cov}(Q_f, Q_h) - V_0(f,h)|}{V_0(f,h)} \le C_{q,R}\left[\varepsilon_{\mathrm{uv}}(a/d) + (1 + B^2)(d/H)^4 + (1 + B^4)(d/H)^8\right], \quad (7)$$

where one may take $\varepsilon_{\mathrm{uv}}(u) = C(1 + u^{-4})e^{-1/(8u^2)}$ *for* $0 < u \le 1/4$.

Proof. The removed low covariance has norm at most $C_q H^{-4}$, since its Fourier support lies in $|k| \le H^{-1}$ and $\|P(k)\| \le 1$. The heat-regularized scalar propagator is explicit:

$$G_a(z) = \frac{1 - e^{-|z|^2/(4a^2)}}{4\pi^2|z|^2}.$$

Two derivatives show $C_{a,H}(z) = K(z) + D(z)$ on the support separation range, with

$$\|D(z)\|_{\mathrm{HS}} \le C_q[d^{-4}\varepsilon_{\mathrm{uv}}(a/d) + H^{-4}].$$

The exact noncentral Gaussian contraction and total covariance formula give

$$\operatorname{Cov}(Q(x), Q(y)) = \tfrac{1}{2}\|C_{a,H}(x - y)\|_{\mathrm{HS}}^2 + \mathbb{E}[m(x)^T C_{a,H}(x - y)m(y)] + \tfrac{1}{4}\operatorname{Cov}(|m(x)|^2, |m(y)|^2).$$

The last two terms are bounded in absolute value by $B^2H^{-4}\|C_{a,H}\|$ and $B^4H^{-8}/4$. Also

$$\left|\|K + D\|_{\mathrm{HS}}^2 - \|K\|_{\mathrm{HS}}^2\right| \le 2\|K\|_{\mathrm{HS}}\|D\|_{\mathrm{HS}} + \|D\|_{\mathrm{HS}}^2.$$

Integrate against $f(x)h(y)$. Its leading term is bounded below by $3q(\int f)(\int h)/(\pi^4 R^8 d^8)$, yielding (7). □

This theorem does not identify an actual constrained-block covariance with a Fourier high-pass covariance. Nor does it prove the independence or coarse moment hypotheses for an interacting theory. These distinctions remain necessary when composing the proof modules.

6 A finite-window comparison with compatible losses

Lemma 6.1 (Normalized density comparison)**.** *Let* $\mathrm{d}\mu/\,\mathrm{d}\nu = e^{-V}/\nu(e^{-V})$*, where* ν *is a probability measure and* $|V - c| \le \delta$ *for some constant* c*. Put* $\kappa = e^{2\delta} - 1$*. For* $A, B \in L^2(\nu)$*,*

$$|\operatorname{Cov}_\mu(A,B) - \operatorname{Cov}_\nu(A,B)| \le (\kappa + \kappa^2)\sqrt{\operatorname{Var}_\nu(A)\operatorname{Var}_\nu(B)}.$$

Proof. Write $w = \mathrm{d}\mu/\,\mathrm{d}\nu$, $A_0 = A - \nu A$, $B_0 = B - \nu B$. Since $|w - 1| \le \kappa$,

$$\operatorname{Cov}_\mu(A,B) - \operatorname{Cov}_\nu(A,B) = \nu[(w-1)A_0B_0] - \nu[(w-1)A_0]\nu[(w-1)B_0].$$

Two applications of Cauchy–Schwarz give the result. □

For Gaussian curvature with Fourier covariance bounded by $e^{-a^2|k|^2}P(k)$, Parseval gives

$$\operatorname{Var}(Q_f) \le C_q a^{-4}\,\|f\|_2^2\,.$$

For the reference $X + m$ from the preceding section, the corresponding bound is

$$\operatorname{Var}(Q_f) \le C_q a^{-4}\,\|f\|_2^2 + B^2H^{-4}\,\|f\|_2^2 + \tfrac{1}{4}B^4H^{-8}\,\|f\|_1^2\,.$$

For fixed-shape positive bumps of diameter and separation d, their free cross covariance is of order d^{-8}, while the geometric mean of their self-variances is of order $a^{-4}d^{-4}$. Thus normalized density comparison loses the relative factor $(d/a)^4$. An observable supremum estimate would introduce unnecessary further losses.

7.1 One smooth fiber

Let $C > 0$ on $\mathbb{R}^D$ and suppose $R \in C^2$ obeys

$$\sup_x \left\| C^{1/2} D^2 R(x) C^{1/2} \right\| \le \epsilon < 1.$$

Consider the density proportional to $\exp[-x^T C^{-1} x/2 - R(x) + h^T x]$. In whitened coordinates $y = C^{-1/2} x$, $r(y) = R(C^{1/2} y)$, the full minimizer is the unique solution

$$y_* = C^{1/2} h - \nabla r(y_*).$$

The right side is a contraction. It also gives

$$|y_*| \le \frac{|C^{1/2} h - \nabla r(0)|}{1 - \epsilon}.$$

After $y = y_* + z$, the potential is $|z|^2/2 + \widetilde{R}(z)$ up to a constant, with $\widetilde{R}(0) = \nabla \widetilde{R}(0) = 0$ and $|\widetilde{R}(z)| \le \epsilon |z|^2/2$.

The covariance satisfies

$$(1+\epsilon)^{-1} C \;\le\; \mathrm{Cov}(x) \;\le\; (1-\epsilon)^{-1} C.$$

The upper bound is the Brascamp–Lieb variance inequality [21]. For the lower bound, with centered z and whitened potential V, integration by parts gives

$$\mathbb{E}[(v \cdot (z - \mathbb{E}z))(v \cdot \nabla V)] = |v|^2, \quad \mathbb{E}(v \cdot \nabla V)^2 = \mathbb{E}[v^T D^2 V v] \le (1+\epsilon)|v|^2.$$

Cauchy–Schwarz proves the stated lower bound. Gaussian tails justify the integration by parts through compact cutoffs.

This is a valid finite-dimensional statement. A constrained nonlinear fiber additionally requires a legitimate chart, its Jacobian, and a domain margin. If there is a sharp cutoff, shifting the minimizer shifts its boundary; the displayed whole-space proof does not apply without boundary estimates.

7.2 Analyticity can bound the linear force

Suppose P is holomorphic on the actual coordinate polydisc $\|z\|_\infty < R$, with $|P| \le M$. For $\|y\|_\infty \le B \le R/4$, restrict P to the complex line $z = ty/B$. One-variable Cauchy bounds give

$$|P(y) - P(0)| \le \frac{MB}{R - B} \le \frac{2MB}{R}.$$

There is no extra coordinate-count factor beyond the bound M itself. For a localized polymer expansion with sufficient exponential decay, the same Cauchy argument and polymer counting give local gradient

and Hessian row-sum bounds. Thus linear-force control can follow from a sufficiently strong common analytic-domain theorem.

The qualifier "actual" matters: a bound on a formal extension does not prove that all retained fields, constraints and histories lie in that extension. That model-specific domain theorem remains unproved here.

7.3 The sum over histories defeats a global convexity inference

For a positive sum of smooth branches

$$\rho(x) = \sum_h w_h e^{-V_h(x)}, \qquad V_{\text{eff}} = -\log \rho,$$

direct differentiation gives, for finite sums or when differentiation under the sum is justified,

$$D^2 V_{\text{eff}} = \mathbb{E}_x[D^2 V_h] - \operatorname{Cov}_x(\nabla V_h), \tag{9}$$

where $\mathbb{E}_x$ uses the posterior branch weights. The second term is positive semidefinite and is subtracted.

For equal Gaussians centered at $\pm M$,

$$V_{\text{eff}}(x) = \frac{x^2}{2} - \log\cosh(Mx) + \text{constant}, \qquad V''_{\text{eff}}(0) = 1 - M^2.$$

Every branch has Hessian one, but the mixture is not convex near zero when $M > 1$. Furthermore, a linear perturbation has zero Hessian yet can move the mean arbitrarily far. For a quadratic field observable that mean is part of the observable and cannot be removed by renaming the fluctuation variable.

This is the exact failure of the attempted global repair. The true RG representation also includes sharp tests and retained boundary variables. A successful argument still needs estimates on the actual branch weights, their common analytic domains, background moments and source transport. Centering each branch does not prove those estimates.

8 Passing estimates to local fields

Let $D_n = \{(x_1, \dots, x_n) : x_i \neq x_j\}$ and let $\mathcal{S}_{\Delta,n}$ consist of Schwartz functions flat to every order on its complement. Put $d(x) = \min_{i<j} |x_i - x_j|$ for $n \geq 2$ and define

$$p_k(f) = \max_{|\alpha| \leq k} \sup_x (1 + |x|)^k |\partial^\alpha f(x)|.$$

Proposition 8.1 (A sufficient all-order extraction estimate)**.** *Suppose smooth off-diagonal kernels $K_{n,j}$ satisfy, uniformly in cutoff j,*

$$|K_{n,j}(x)| \leq A B^n (n!)^p (1 + |x|)^{mn} \min(1, d(x))^{-qn}, \tag{10}$$

with fixed finite nonnegative m, q, p. For $n = 1$ omit the distance factor. Then some fixed integer k and B' give

$$\left| \int K_{n,j}(x) f(x)\, \mathrm{d}x \right| \leq A (B')^n (n!)^p p_{kn}(f), \qquad f \in \mathcal{S}_{\Delta,n}. \tag{11}$$

There is one subsequence on which these functionals converge for every n and every such test.

Proof. For $d(x) < 1$, project the nearest pair to its coincidence diagonal. The displacement affects only eight coordinates. Taylor's formula and flatness, with $N = \lceil qn \rceil$, give

$$|f(x)| \le \frac{(Cd(x))^N}{N!}\, 2^{kn} p_{kn}(f)(1+|x|)^{-kn}.$$

For $d(x) \ge 1$ use the direct weighted bound. If $q = 0$ or $n = 1$, Taylor's formula is unnecessary. Choose an integer $k > m + q + 5$. Multiplication by (10) cancels the short-distance singularity and leaves an integrable weight bounded by $(1+|x|)^{-5n}$. Since

$$(1+|x|)^{-5n} \le \prod_{i=1}^{n} (1+|x_i|)^{-5},$$

its integral is bounded by a constant to the nth power. Absorb the remaining $C^N 2^{kn}/N!$ into $(B')^n$.

Choose a countable dense set in each flat Schwartz space. Boundedness and diagonal subsequence extraction give convergence on all these tests simultaneously. Approximation using the same uniform seminorm bound extends convergence to every test in each space. Countably many field labels can be included in the diagonal enumeration. □

The hypothesis (10) is not proved for interacting Yang–Mills here. Lattice distributions require a suitable interpolated or directly distributional version; that step cannot be replaced by calling a lattice sum a smooth kernel.

Finite reflection-positive Gram matrices remain positive under entrywise convergence. Symmetry and covariance pass when their testwise defects vanish. The growth bound (11), full Euclidean covariance and the required vacuum/clustering assumptions form a route to the corrected OS reconstruction theorem [12]. A lower bound on a reflected centered two-point norm then gives a non-vacuum vector. These are conditional implications, not established properties of a constructed Yang–Mills limit.

There is an interesting conditional route from diagonal lattice reflection positivity and sufficiently strong scalar insertion estimates to full scalar rotational covariance in the public candidate [20]. Selected algebraic and spectral steps were checked during this attempt. The needed insertion and ultraviolet estimates were not proved, and the argument does not itself construct all tensor observables. We do not use it as a completed theorem here.

10 Actual Wilson exponential moments and large-plaquette sets

This section concerns the actual Wilson measure, without assuming a renormalization-group theorem. We first take $G = \mathrm{SU}(2)$ and fix the normalization

$$a(U) = 1 - \tfrac{1}{2}\Re\,\mathrm{tr}\,U, \qquad A(U) = \sum_p a(U_p), \qquad \mathrm{d}\mu_{n,\beta} = Z_{n,\beta}^{-1} e^{-\beta A(U)} \prod_b \mathrm{d}U_b$$

on the periodic lattice $(\mathbb{Z}/n\mathbb{Z})^4$, with normalized Haar measure. The physical spacing is denoted by a_{lat} when needed. It does not enter these dimensionless inequalities. The only standard non-elementary input in the local-moment argument is Wilson reflection positivity and its chessboard consequence [22, 23]. Related partition-factorization methods appear in [24]; all estimates for the action observable used here are derived explicitly.

10.1 Matching partition powers

Lemma 10.1 (Elementary partition bounds)**.** *Put*

$$Q_n = 3n^4 + 1, \quad R_n = 3n^4 - 4n^3 + 1, \quad c_l = \frac{e^{-1}}{24\pi^3}, \quad c_u = (\pi/2)^{5/2}.$$

For $\beta \geq 1$,

$$(c_l \beta^{-3/2})^{Q_n} \leq Z_{n,\beta} \leq (c_u \beta^{-3/2})^{R_n}. \tag{12}$$

The upper bound holds for every $\beta > 0$.

Proof. Gauge-fix a spanning tree, leaving $4n^4 - (n^4 - 1) = Q_n$ independent Haar links. The eliminated gauge coordinates integrate to one by Haar invariance. In the parametrization $U = \cos\theta + i \sin\theta\, \mathbf{n} \cdot \boldsymbol{\sigma}$, radial Haar measure is $(2/\pi)\sin^2\theta\,\mathrm{d}\theta$. Hence, for $r \leq 1$,

$$H\{\theta \leq r\} \geq \frac{8r^3}{3\pi^3}.$$

Restrict every retained link to $\theta \leq (4\sqrt{\beta})^{-1}$. The angle of a product is bounded by the sum of the angles, so every plaquette has $\beta a(U_p) \leq 1/2$. There are $6n^4$ plaquettes. The restricted integral is at least

$$e^{-3n^4} \left(\frac{\beta^{-3/2}}{24\pi^3} \right)^{Q_n} \geq (c_l \beta^{-3/2})^{Q_n}.$$

For the upper bound, first discard nonnegative actions of plaquettes crossing the periodic boundary, leaving a free box. Gauge-fix all temporal links and a recursively chosen tree on its bottom face. Keep all temporal plaquettes and the bottom-face plaquettes, and discard the remaining nonnegative actions. Integrating each spatial-link chain from top to bottom supplies one factor

$$z(\beta) = \frac{2}{\pi} \int_0^\pi e^{-\beta(1-\cos\theta)} \sin^2\theta\,\mathrm{d}\theta \leq \frac{2}{\pi} \int_0^\infty \theta^2 e^{-2\beta\theta^2/\pi^2}\,\mathrm{d}\theta = c_u \beta^{-3/2}.$$

In dimension d the number of factors obeys

$$R_d(n) = (d-1)(n-1)^2 n^{d-2} + R_{d-1}(n), \qquad R_1(n) = 0.$$

Induction gives $R_d(n) = (d-1)n^d - dn^{d-1} + 1$. At $d = 4$ this is R_n, proving the claim. The difference $Q_n - R_n = 4n^3$ is a boundary term. □

10.2 From a partition bound to a local moment

Choose a fixed compatible chessboard cell of side b and volume $v = b^4$, with the observed plaquette strictly inside. For example $b = 4$ suffices. Use compatible dyadic tori and put $M = n^4/v$.

Theorem 10.2 (Uniform local Wilson exponential moment)**.** *For $X_p = \beta a(U_p)$ and $0 \le t < 1$,*

$$\mathbb{E}_{n,\beta} e^{tX_p} \le c_u^{R_n/M} c_l^{-Q_n/M} (1-t)^{-3R_n/(2M)} \beta^{6v/n}. \tag{13}$$

Every thermodynamic limit of these periodic states therefore satisfies

$$\boxed{\mathbb{E}_\beta e^{t\beta a(U_p)} \le A_v (1-t)^{-\gamma_v}, \qquad A_v = (c_u/c_l)^{3v}, \quad \gamma_v = 9v/2.} \tag{14}$$

The constants are independent of $\beta \ge 1$ and physical spacing.

Proof. Chessboard gives

$$\mathbb{E} e^{tX_p} \le \left[\mathbb{E} \prod_{c=1}^{M} \theta_c e^{tX_p} \right]^{1/M}.$$

The disseminated plaquettes are distinct. Their action sum A_* satisfies $0 \le A_* \le A$, so

$$\mathbb{E} \prod_c \theta_c e^{tX_p} = Z_{n,\beta}^{-1} \int e^{-\beta A + t\beta A_*} \, \mathrm{d}U \le \frac{Z_{n,(1-t)\beta}}{Z_{n,\beta}}.$$

It is not necessary for the disseminated plaquettes to exhaust all orientations. Apply (12) and use $(Q_n - R_n)/M = 4v/n$ to obtain (13). At fixed β, the cylinder observable is bounded and continuous, so it passes to every weak thermodynamic limit. Taking $n \to \infty$ gives (14). □

The finite-size factor is $\exp(6v \log \beta/n)$. It is uniformly bounded if $n \ge c \log \beta$, and tends to one if $\log \beta/n \to 0$. In particular, it is harmless in the simultaneous regime $n \asymp \ell/a_{\mathrm{lat}}$ and $\beta = O(\log(1/a_{\mathrm{lat}}))$. A bound uniform in β at fixed n has not been proved by this argument.

Generalized chessboard yields the corresponding product estimate for one marked plaquette in each of finitely many distinct cells. Let $B_v = A_v 2^{\gamma_v}$. Since

$$x^k \mathbf{1}_{x \ge T} \le k! 4^k e^{-T/4} e^{x/2} \quad (x \ge 0),$$

we obtain the actual insertion-weighted bound

$$\mathbb{E} \prod_{i=1}^{r} \left[X_{p_i}^{k_i} \mathbf{1}_{X_{p_i} \ge T_i} \right] \le B_v^r \prod_{i=1}^{r} k_i! 4^{k_i} e^{-T_i/4}. \tag{15}$$

This is a chessboard estimate, not an independence statement. Ordinary exponential moments also give factorial moments without the indicators.

For compact connected G of dimension m with a faithful unitary representation ρ, replace $a(g)$ by $1 - \Re \operatorname{tr} \rho(g)/\dim \rho$. It has a unique nondegenerate minimum at the identity. Local exponential coordinates and compactness give $z(\beta) \le C_G \beta^{-m/2}$ and the corresponding Haar-ball lower bound. The same tree and chain proof gives (12) with exponent $m/2$. Wilson reflection positivity follows from the positive character expansion of $\exp(\beta \Re \chi_\rho/\dim \rho)$. Consequently

$$\gamma_v = 3mv/2, \qquad A_v = (c_u/c_l)^{3v}, \qquad \text{finite-size factor } \beta^{2mv/n},$$

with group-dependent positive constants c_l, c_u. Thus the qualitative result is not specific to SU(2).

10.3 Connected bad-plaquette sets

Call p bad when $X_p \geq T$. Color plaquettes by their orientation and base point modulo b, using $C_{\mathrm{col}} = 6b^4$ colors. Each color admits a translated board placing at most one of its plaquettes in the interior of each cell. This also covers plaquettes crossing the faces of an initially chosen board. Let

$$q = B_v e^{-T/2} \leq 1.$$

For any finite plaquette set S, one color contains at least $|S|/C_{\mathrm{col}}$ of its plaquettes. Chessboard and exponential Markov therefore give

$$\Pr(S \text{ is entirely bad}) \leq q^{|S|/C_{\mathrm{col}}}. \tag{16}$$

Give the plaquette adjacency graph any fixed local rule, with maximum degree $D \geq 2$. A connected set of m plaquettes containing a specified plaquette can be encoded by a depth-first traversal of a deterministic spanning tree. There are at most $D^{2(m-1)}$ such traversals. If $\mathcal{C}(p)$ is the bad component of p,

$$\Pr(|\mathcal{C}(p)| \geq m) \leq D^{2(m-1)} q^{m/C_{\mathrm{col}}}.$$

In particular, if

$$T \geq 2 \log B_v + 2C_{\mathrm{col}}(2 \log D + c),$$

this is at most $D^{-2}e^{-cm}$. For a bounded physical region, a union bound gives

$$\Pr(\text{a bad component meeting it has diameter} \geq r) \leq C a_{\mathrm{lat}}^{-4} e^{-c'r/a_{\mathrm{lat}}}.$$

These are statements about bare Wilson plaquettes. They do not imply decay of physical Yang–Mills correlations, a mass gap, or equivalent bounds for every test used in a multiscale history.

11 Arbitrary-exterior control of excess action

There is no uniform absolute plaquette moment bound under every exterior condition. For SU(2), conditioning on all links except one gives a unit quaternion q with density proportional to

$$\exp(\beta q \cdot S), \qquad S = \sum_{i=1}^{6} v_i,$$

where the six v_i are the staples. If $v_1 = v$ and $v_2 = \cdots = v_6 = -v$, then q concentrates near $-v$ and the target plaquette action $1 - q \cdot v$ tends to 2. Its conditional exponential moment grows like $e^{2t\beta}$. Such a frustrated exterior is realizable by choosing the distinct exterior staple paths. Uniformity over arbitrary boundaries must instead retain an exterior cost or use excess energy above the compatible minimum.

Theorem 11.1 (Conditional excess-action bound)**.** *Let B contain N free bonds; fix all other bonds arbitrarily. For compact connected G of dimension m, use the faithful-representation action above and let S_e be the sum of plaquette actions depending on a free bond. Put*

$$S_{\min}(e) = \min S_e, \qquad \Delta_e = S_e - S_{\min}(e) \geq 0.$$

There is a constant $c_G \in (0,1]$, depending only on the group, representation and metric, such that for $\beta \geq 1$,

$$e^{\beta S_{\min}(e)} Z_e(\beta) \geq (c_G \beta^{-m/2})^N, \tag{17}$$

$$K_{e,\beta}\big[e^{t\beta\Delta_e}\big] \leq c_G^{-N} \beta^{mN/2}, \qquad 0 \leq t \leq 1. \tag{18}$$

For $u, w > 0$, $t \geq 0$, $t + u + w \leq 1$, and integer $k \geq 0$,

$$K_{e,\beta}\Big[(\beta\Delta_e)^k e^{t\beta\Delta_e} \mathbf{1}_{\beta\Delta_e \geq T}\Big] \leq k! w^{-k} \exp\Big[-uT + N\log(1/c_G) + \tfrac{mN}{2}\log\beta\Big]. \tag{19}$$

The constants are independent of the exterior and total lattice volume.

Proof. Choose a minimizing configuration U^*, which exists by compactness, and product coordinates $U_b = U_b^* \exp X_b$. Set $c_\rho = \sup_{|Y|=1} \| \mathrm{d}\rho(Y)\|_{\mathrm{op}}$. Unitarity and Duhamel's formula bound first exponential derivatives by $c_\rho|Y|$ and second derivatives by $c_\rho^2|Y||Z|$. A plaquette has four factors and a bond has six incident plaquettes. Thus the coordinate Hessian H of S_e satisfies

$$\sup_b \sum_c \|H_{bc}\|_{\mathrm{op}} \leq 24 c_\rho^2.$$

This bound is uniform in the exterior; mixed blocks vanish unless the bonds share a plaquette. Since $DS_e(U^*) = 0$, Taylor's formula gives

$$0 \leq \Delta_e(U^* e^X) \leq 12 c_\rho^2 \sum_b |X_b|^2.$$

Choose r_0 within an injectivity neighborhood and $h_G > 0$ such that a Haar ball of radius $r \leq r_0$ has measure at least $h_G r^m$. Restrict every $|X_b| \leq r_0/\sqrt{\beta}$. Its product measure is at least $(h_G r_0^m \beta^{-m/2})^N$, and $\beta\Delta_e \leq 12 c_\rho^2 r_0^2 N$. This proves (17) with

$$c_G = \min\{1, h_G r_0^m e^{-12 c_\rho^2 r_0^2}\}.$$

The numerator $\int e^{-(1-t)\beta\Delta_e}\, \mathrm{d}U_B$ is at most one for $0 \leq t \leq 1$, proving (18). Finally use $x^k \leq k! w^{-k} e^{wx}$ and $\mathbf{1}_{x \geq T} \leq e^{u(x-T)}$. □

If no plaquette meets two different free-bond blocks, conditioning on their complement factors the Wilson law exactly over those blocks. Products of the bounds (19) then hold in every Wilson Gibbs state, without reflection positivity or a good-exterior assumption. The exterior-dependent minima are fixed during the conditional integrations.

This is concentration of excess action, not distance from a unique minimum. Several minima and gauge degeneracy are allowed. A frustrated configuration may have a large minimum but small excess. Furthermore a nonlinear block-average fiber is not a product-Haar conditional space; its reference volume and allowed tangent geometry need a separate proof.

11.1 Centered constant-source moments

Put $H = N\log(1/c_G) + (mN/2)\log\beta$ and $Y = \beta\Delta_e$. Equation (18) at $t = 1/2$ gives

$$\Pr(Y \geq s) \leq \min\{1, e^{H - s/2}\}.$$

Thus Y is stochastically dominated by $2H + E$, where E is exponential with rate $1/2$. Since $\|E\|_p = 2(p!)^{1/p} \le 2p$ for integer $p \ge 1$,

$$\|S_e - \mathbb{E}_e S_e\|_{L^p(K_{e,\beta})} \le \frac{4(H+p)}{\beta}. \tag{20}$$

This is uniform in an arbitrarily frustrated exterior. If A_f sums the at most $6N$ plaquettes touching the block and $|f_p - f_0| \le \ell \operatorname{Lip}(f)$ there, then $0 \le a_p \le 2$ also gives

$$\|A_f - \mathbb{E}_e A_f\|_p \le \frac{4|f_0|(H+p)}{\beta} + 24N\ell \operatorname{Lip}(f).$$

The logarithmic loss does not imply a divergent separated correlation: before source supports coalesce, local repeated source terms can be removed in the finite-source algebra. Nevertheless these are estimates for fixed-exterior free-link blocks, not for every coarse conditional measure.

11.2 An exact one-link Gamma bound

For SU(2) and $\beta > 0$, let $r = |S|$, $\kappa = \beta r$, and $F = 6 - r$, the minimum of the six-plaquette star action. The excess variable

$$X = \beta\big[(6 - q \cdot S) - F\big] = \kappa(1 - q \cdot S/r)$$

is zero if $r = 0$. Otherwise its density relative to a Gamma$(3/2, 1)$ density is proportional to

$$\sqrt{2 - x/\kappa}\, \mathbf{1}_{0 \le x \le 2\kappa},$$

which is decreasing. For an increasing function φ and a decreasing nonnegative function h, the identity

$$2 \operatorname{Cov}(\varphi(X), h(X)) = \mathbb{E}[(\varphi(X) - \varphi(X'))(h(X) - h(X'))] \le 0$$

shows stochastic domination by that Gamma variable. Therefore, for every exterior and every β,

$$K_e[e^{tX}] \le (1-t)^{-3/2}, \qquad 0 \le t < 1. \tag{21}$$

The absolute star action is $F + X/\beta$; its exterior frustration floor F is not bounded by this assertion.

11.3 Force moments without convexity

Proposition 11.2 (Compact-space force bound)**.** *Let* $\mathrm{d}\nu_\beta \propto e^{-\beta S}\,\mathrm{d}v$ *on a compact Riemannian manifold without boundary, with its Riemannian volume, and suppose* $\operatorname{Hess} S \le LI$ *and* $\Delta S \le K$ *for* $L, K > 0$*. Then*

$$\mathbb{E}_{\nu_\beta} e^{\lambda\beta|\nabla S|^2} \le (1 - 2L\lambda)^{-K/(2L)}, \qquad 0 \le \lambda < (2L)^{-1}. \tag{22}$$

No lower Hessian bound is required.

Proof. Integrate the divergence of $e^{-\beta S}|\nabla S|^{2k-2}\nabla S$. Its integral vanishes, and consequently

$$\beta\mathbb{E}|\nabla S|^{2k} = \mathbb{E}\Big[|\nabla S|^{2k-2}\Delta S + (2k-2)|\nabla S|^{2k-4} \operatorname{Hess} S(\nabla S, \nabla S)\Big] \le [K + 2L(k-1)]\mathbb{E}|\nabla S|^{2k-2}.$$

Induction gives the moments of a Gamma upper majorant, $\mathbb{E}(\beta|\nabla S|^2)^k \le (2L)^k (K/(2L))_k$. Sum the nonnegative exponential series to obtain (22). □

For the actual Wilson action on N free links, use a bi-invariant product metric. The Hessian proof above gives $L = 24c_\rho^2$. Writing $C_\rho = -\sum_a \mathrm{d}\rho(T_a)^2$, the Laplacian satisfies $K = 6N \operatorname{tr} C_\rho / \dim \rho$. For SU(2) with unit-quaternion generators, one may take $L = 24$ and $K = 18N$. For a single free link, $L = 6$, $K = 18$, giving $(1 - 12\lambda)^{-3/2}$. These bounds hold for every exterior.

Unlike the lower-Hessian argument, this upper-Hessian estimate can survive a smooth positive mixture. If

$$S_{\mathrm{eff}} = -\beta^{-1} \log \sum_h w_h e^{-\beta S_h},$$

with field-independent $w_h \geq 0$ and common smooth domains, then

$$D^2 S_{\mathrm{eff}} = \mathbb{E}_h D^2 S_h - \beta \operatorname{Cov}_h(\nabla S_h) \leq LI$$

whenever every branch has that upper bound; the analogous Laplacian bound also passes through. This controls the force of the effective action. It does not control spatially separated source covariances, and it does not apply through sharp history tests without an additional argument.

12 Finite-source transport with all insertion terms retained

For N labeled insertions $X_1, \ldots, X_N$, work in the commutative algebra

$$\mathfrak{A}_N = \mathbb{R}[z_1, \ldots, z_N]/(z_1^2, \ldots, z_N^2), \qquad z_A = \prod_{i \in A} z_i.$$

Every zero-constant element is nilpotent of order at most $N + 1$. Exponentials and logarithms with constant coefficient one are finite polynomials. This removes a zero-free complex-neighborhood requirement from fixed-order cumulants. Repeated copies of one observable receive distinct labels and are still allowed.

Proposition 12.1 (Exact conditional transport)**.** *For decreasing sigma-fields $\mathcal{F}_0 \supset \cdots \supset \mathcal{F}_J$, define*

$$M_j(z) = \mathbb{E}\left[\prod_i (1 + z_i X_i) \mid \mathcal{F}_j\right], \qquad K_j(z) = \log M_j(z) = \sum_{A \neq \varnothing} z_A K_j(A).$$

Then

$$K_{j+1}(A) = \sum_{\pi \in \Pi(A)} \kappa\big(K_j(B) : B \in \pi \mid \mathcal{F}_{j+1}\big), \tag{23}$$

where $\Pi(A)$ is the set of partitions of A and κ denotes conditional joint cumulants.

Proof. Coefficient extraction from the finite logarithm gives

$$K_j(A) = \sum_{\pi \in \Pi(A)} (-1)^{|\pi|-1} (|\pi| - 1)! \prod_{B \in \pi} \mathbb{E}\left[\prod_{i \in B} X_i \mid \mathcal{F}_j\right].$$

The tower property yields $e^{K_{j+1}} = \mathbb{E}[e^{K_j} \mid \mathcal{F}_{j+1}]$. Expanding both finite polynomials and collecting the coefficient z_A proves (23). It is the finite-source version of the law of total cumulance [25]. □

For randomized exact RG kernels, use the extended Markov chain $U_0 \to \cdots \to U_J$ and the reverse filtration $\mathcal{F}_j = \sigma(U_j, \ldots, U_J)$. The Markov property makes a conditional insertion given $\mathcal{F}_j$ equal its

conditional expectation given U_j. Coarse variables therefore need not be deterministic functions of the original field.

At order two, (23) reads

$$K_{j+1}(il) = \mathbb{E}[K_j(il) \mid \mathcal{F}_{j+1}] + \mathrm{Cov}(K_j(i), K_j(l) \mid \mathcal{F}_{j+1}).$$

Assume now that $\mathcal{F}_0$ contains the original insertions. Writing $D_j^i = \mathbb{E}[X_i \mid \mathcal{F}_j] - \mathbb{E}[X_i \mid \mathcal{F}_{j+1}]$ gives the exact bracket decomposition

$$\mathrm{Cov}(X_i, X_l) = \mathrm{Cov}(\mathbb{E}[X_i \mid \mathcal{F}_J], \mathbb{E}[X_l \mid \mathcal{F}_J]) + \sum_{j<J} \mathbb{E}[D_j^i D_j^l].$$

At every higher order all partitions in (23) must likewise be retained. If $\mathcal{F}_0$ contains the original variables, define

$$\Gamma_j(A) = \sum_{\substack{\pi \in \Pi(A) \\ |\pi| \geq 2}} \kappa(K_j(B) : B \in \pi \mid \mathcal{F}_{j+1}).$$

For $|A| \geq 2$ the full cumulant is

$$\kappa(X_i : i \in A) = \sum_{j<J} \mathbb{E}\Gamma_j(A) + \sum_{\substack{\pi \in \Pi(A) \\ |\pi| \geq 2}} \kappa(K_J(B) : B \in \pi). \tag{24}$$

The terminal coarse-field terms are part of the identity.

For positive source weights $r_A = \prod_{i \in A} r_i$, the graded norm

$$\|F\|_{N,r} = \|F_\varnothing\|_\infty + \sum_{A \neq \varnothing} r_A \, \|F_A\|_{L^{N/|A|}}$$

is submultiplicative by Hölder, and conditional expectation contracts it. This is an averaged coefficient framework. It does not make the norm of a bare dimension-four insertion cutoff independent.

12.1 Locality, contact terms and a real integral representation

In a fixed separator decomposition of a finite-range Gibbs measure, conditioning on the separator makes the remaining components independent. If $X_i = X_i^S + \sum_B X_{i,B}$, then

$$K_S(z) = \sum_i z_i X_i^S + \sum_B \log \mathbb{E}_B \exp\left(\sum_i z_i X_{i,B} \right).$$

Thus a mixed coefficient vanishes unless one component meets all its source supports. This is exact locality before coalescence. The retained separator still carries fine-scale boundary fluctuations; the identity alone gives no ultraviolet norm bound.

For disjoint sources $w(x) = \sum_i z_i f_i(x)$, one has $w(x)^2 = 0$. The same holds for products of source jets at one point. Every deterministic point-local counterterm therefore becomes linear in the labels and drops out of separated connected coefficients of order at least two. In particular, for an ordinary local function b,

$$b(w(x)) - b(0) = b'(0) w(x).$$

Once a block intersects several supports, its nonlinear source responses are genuine multi-support terms. Averaging the source profile does not turn them into point-local contacts.

For a fixed real common history representation with $w_h \geq 0$, finite coefficient extraction gives

$$\left| [z_A] \int \sum_h w_h e^{\Phi_h(z)} \right| \leq \int \sum_h w_h \left| [z_A] e^{\Phi_h(z)} \right|, \tag{25}$$

provided the absolute coefficient moments are integrable. For general real or complex weights use $|w_h|$ on the right. There is no supremum over a complex source family inside the integral. Common deterministic vacuum terms must first be extracted. History-dependent linear terms $\sum_i z_i c_i(h)$ produce genuine history cumulants after summing h and taking a logarithm; they cannot be cancelled as deterministic contacts.

13 The smooth-source gain for the actual Wilson current

Use the bi-invariant metric identifying SU(2) with the unit three-sphere. If p contains b,

$$\|\nabla_b a_p\|^2 = \sin^2\theta_p \le 2a_p.$$

Let $f_p = f(x_p)$ at the plaquette center, $f_b = f(x_b)$ at the bond midpoint, and $A_f = \sum_p f_p a_p$. Exactly,

$$\nabla_b A_f = f_b \nabla_b A + R_b(f), \qquad R_b(f) = \sum_{p \ni b} (f_p - f_b)\nabla_b a_p. \tag{26}$$

Six plaquettes meet a bond and $|x_p - x_b| = a_{\mathrm{lat}}/2$. Cauchy–Schwarz therefore gives

$$\|R_b(f)\|^2 \le 3a_{\mathrm{lat}}^2 \,\mathrm{Lip}(f)^2 \sum_{p \ni b} a_p.$$

Combining ordinary Hölder on these six plaquettes with (14) proves

$$\mathbb{E}\exp\left[\frac{s\beta\,\|R_b(f)\|^2}{a_{\mathrm{lat}}^2\,\mathrm{Lip}(f)^2}\right] \le A_v(1-18s)^{-\gamma_v}, \qquad 0 \le s < 1/18. \tag{27}$$

When $\mathrm{Lip}(f) = 0$, $R_b(f) = 0$. In particular,

$$\mathbb{E}\,\|R_b(f)\|^{2k} \le B_v k! \left[\frac{36a_{\mathrm{lat}}^2\,\mathrm{Lip}(f)^2}{\beta}\right]^k, \tag{28}$$

$$\beta\mathbb{E}\,\|R_b(f)\|^2 \le 36(\log A_v + \gamma_v \log 2)a_{\mathrm{lat}}^2\,\mathrm{Lip}(f)^2. \tag{29}$$

This is an actual Gibbs expectation estimate, without a small-field conditioning assumption. It verifies a one-power-of-spacing gain in the current's root-mean-square size.

At a constrained Wilson saddle, let Q be the linearized constraint, let $QB = 0$, and write $J = Q^*\lambda$. Multiplication by f gives

$$\langle J, M_f B\rangle = \langle \lambda, (QM_f - M_{f^c}Q)B\rangle.$$

If Q has a localized kernel of range $O(\ell)$ and a uniform Schur bound, then

$$\|QM_f - M_{f^c}Q\| \le C\ell\,\mathrm{Lip}(f).$$

A bounded right inverse of Q also controls λ. This is the constrained form of local freezing. For a full effective saddle of $xA + K$, however, the fiber gradient satisfies $\nabla A = -x^{-1}\nabla K$. A constant source leaves this additional remainder term; it has no source-gradient gain. Nonlinear constraint charts also contribute their second derivatives. Those terms must remain in the physical insertion.

14 A clean one-step source expansion on its nonlinear fiber

This section verifies the primitive source input for a fixed clean step. Its covariance and its existing unmarked cluster majorant remain inputs; no uniform multilayer source theorem is assumed.

14.1 Local constraint coordinates

For definiteness use the explicit one-step path-log average in [20]:

$$\Pi_x(U) = U(c_-, x)U([x, x'])U(x', c_+), \qquad x' = x + L\eta e_\mu,$$

$$W_x(U) = \Pi_x(U)U[c]^{-1}, \qquad \mathcal{M}(U)(c) = \exp\left(L^{-4} \sum_{x \in B(c_-)} \log W_x(U) \right) U[c].$$

Internal contours are means of paths contained in their individual blocks. At a regular real field all closed transports lie in one fixed logarithmic neighborhood. Comparing two such paths uses at most C_L plaquettes. The implicit mean equation has derivative $-I$ when its inputs coincide, so the analytic implicit-function theorem gives a common local radius and bounded derivatives depending on L, not volume.

Let Q linearize this average. For each coarse bond c, use the L^3 fine bonds crossing its separating face. A path in another independent row crosses its own face; internal contours cross no block face. Vary the face bonds by one Lie-algebra vector, transported from c_-. These seeds have disjoint supports and

$$Q_c R_c^0 = I + O(C_L \epsilon)$$

when the determining plaquette deviations are at most ϵ. For $C_L \epsilon < 1/2$, set $R_c = R_c^0 (Q_c R_c^0)^{-1}$. Then $QR = I$, R is local and $\|R\| \le C_L$.

In compatible Lie-algebra coordinates let $q(A) = 0$ be the nonlinear constraint. On $\ker Q$, solve $q(B + Rr) = 0$ row by row. The map

$$r_c \longmapsto -\big(q_c(B + R_c r_c) - r_c\big)$$

has zero derivative in r_c at the origin, and its value at zero is $O(C_L \|B\|_\infty^2)$. On a sufficiently small fixed box it is a contraction. This supplies a local holomorphic chart

$$\Phi(B) = B + Rr(B), \quad q(\Phi(B)) = 0, \quad \|r(B)\|_\infty \le C_L \|B\|_\infty^2 .$$

The local projection $P = I - RQ$ gives a full independent-coordinate extension $\Phi(PB)$. If an internal tree gauge is imposed, this box consists of the remaining tree-free coordinates; R has no tree support and preserves that space. The projection has bounded range and agrees with Φ on $\ker Q$. This construction is for the displayed one-step map, not every nonuniform seam operation.

The constrained source Hessian is exactly

$$D^2(A_f \circ \Phi)(0)[B, C] = D^2 A_f[B, C] - \langle R^* J_f, D^2 q(0)[B, C] \rangle. \tag{31}$$

For a constant source at a Wilson saddle, the first derivative vanishes but this Lagrangian Hessian remains. It must not be replaced by the ambient Hessian.

14.3 Exact Gaussian cycles and paths

Let C be the fixed covariance on the actual gauge-fixed tangent space. With

$$j(z) = g\sum_i z_i l_i, \qquad T(z) = g^2 \sum_i z_i H_i,$$

where l_i, H_i are the actual constrained derivatives, Gaussian integration gives the finite-jet identity

$$\log \mathbb{E}_C e^{j\cdot b + \frac{1}{2} b^T T b} = -\tfrac{1}{2}\operatorname{tr}\log(I - CT) + \tfrac{1}{2} j^T (C^{-1} - T)^{-1} j. \tag{35}$$

All inverses are on the tangent space and every source series terminates. The identity follows either by Wick's formula or by taking the finite Taylor jet of the ordinary Gaussian identity near zero. Its distinct-label quadratic coefficient is

$$\frac{g^{2n}}{2n} \sum_{\pi \in S_n} \operatorname{tr}(C H_{\pi(1)} \cdots C H_{\pi(n)}).$$

The other terms are paths with two linear endpoints and $n-2$ Hessian insertions. Their prefactor is g^{2n-2}; at a regular Wilson saddle the two endpoint currents supply the additional curvature and source-gradient factors. Constant-source endpoints vanish.

Use local ambient representatives of (31), or their extension by the local $P = I - RQ$. Since $PC = C$ and $CP^* = C$, the Gaussian contractions are unchanged. An orthogonal projection with long range must not be inserted and then called local.

Define $Z_{\mathrm{quad}} = \mathbb{E}_C \exp(\frac{1}{2} b^T T b)$, excluding the linear source. Suppose the actual covariance satisfies the exponential Schur bound

$$\sup_b \sum_c e^{\mu |x_b - x_c|/\ell} \, \|C_{bc}\| \le C_0$$

and its reverse, and the Hessian representatives have range at most $r_0 \ell$ with row/column norms at most $h \|f_i\|_\infty$. Weight every edge of a cycle by its exponential distance. A cycle visiting all enlarged source supports spans them. Summing successively with the Schur bounds and rooting the trace at label 1 yields

$$|[z_1 \cdots z_n] \log Z_{\mathrm{quad}}| \le \tfrac{1}{2}(n-1)! N_1 (C g^2)^n \prod_i \|f_i\|_\infty \, e^{-\mu \mathcal{T}(S_1,\dots,S_n)/\ell}. \tag{36}$$

Here N_1 counts coordinates near the first support and $\mathcal{T}$ is the minimum length of a connected network visiting all supports. Finite-range Hessian edges cost only C^n. Paths have the corresponding endpoint bound. The exponential suppresses every fixed polynomial cutoff loss while supports remain separated. This proves localization for the Gaussian part of the fixed clean step, using its actual covariance; it does not replace a background-dependent covariance by a flat one.

15.1 An actual averaged source-derivative bound

There is a useful normalized source bound even before a convex-fiber theorem is applied. Split the actual compact Wilson variables into x, a block of N links, and retained variables y, including a link b. Write the full joint action as $S(x, y)$, including terms independent of x, and let

$$M(y) = \mathbb{E}[S(x, y) \mid y], \qquad H = N \log(1/c_G) + (mN/2) \log \beta.$$

For SU(2), the following estimate is uniform in any fixed exterior of the joint system:

$$\|D_b M\|_{L^2(\mu_y)} \le (4H+9)\sqrt{18/\beta}. \tag{37}$$

To prove it, put $J_b = D_b S$. Differentiating the actual normalized conditional integral gives

$$D_b M = \mathbb{E}[J_b \mid y] - \beta \operatorname{Cov}_x(S, J_b \mid y).$$

The conditional variance bound (20) gives $\operatorname{Var}_x(S \mid y)^{1/2} \le 4(H+2)/\beta$ uniformly in y. Conditional Cauchy–Schwarz, followed by integration over y, gives

$$\|\operatorname{Cov}_x(S, J_b \mid y)\|_2 \le \frac{4(H+2)}{\beta}\, \|J_b\|_{L^2(\mu)}\,.$$

Integrating by parts in the one retained link yields $\beta\mathbb{E}|J_b|^2 = \mathbb{E}\Delta_b S \le 18$; alternatively use (22). Conditional expectation contracts L^2. Combining these facts proves (37). No independence of internal histories or lower Hessian bound was used.

In the canonical direction gD_b, $g = \beta^{-1/2}$, the bound is

$$g\, \|D_b M\|_2 \le \sqrt{18}\,(4H+9)/\beta.$$

For a smooth source A_f, use the centered bound following (20) and the exact current identity (26). If $|f_p - f_0| \le \ell \operatorname{Lip}(f)$ on plaquettes touching the hidden block, then

$$g\, \|D_b \mathbb{E}[A_f \mid y]\|_2 \le C\left[\frac{|f_b| + |f_0|(H+2)}{\beta} + N\ell \operatorname{Lip}(f)\right]. \tag{38}$$

Here the pointwise bound on the current remainder contributes at most $C a_{\mathrm{lat}} \operatorname{Lip}(f)/\sqrt{\beta}$, absorbed when $\ell \ge a_{\mathrm{lat}}$. The constant-source part must include the outside action: replacing the total force by only a selected subset of staples would lose its cancellations.

These are actual one-block estimates in an averaged derivative norm. They do not bound a supremum over all coarse fields or the derivative of every separately extracted activity. On straight path-product coarse coordinates, a product-Haar change of variables makes an analogous estimate possible; a retained coarse link enters through a single fine pivot. The hidden-variable chart can use radius β^{-1} instead of $\beta^{-1/2}$, replacing H by $N\log(1/c'_G) + mN\log\beta$. This observation uses the bare β and does not establish a physical running-coupling trajectory or a smooth block-average construction.

15.2 All finite source orders on the same block

The preceding argument also controls the source/field derivative of conditional cumulants. Suppose that, for $i = 1, \dots, n$,

$$\|A_i - \mathbb{E}_x A_i\|_{L^{2n}(x|y)} \le V_i$$

uniformly in y. Let $c_r = \sum_{\pi \in \Pi(\{1,\dots,r\})} (|\pi| - 1)!$. Then, for every $p \ge 2$,

$$\|\kappa_x(A_1, \dots, A_n, J_b)\|_{L^p(\mu_y)} \le c_{n+1} \prod_i V_i\ \|J_b\|_{L^p(\mu)}\,. \tag{39}$$

To prove it, conditionally center the A_i, which leaves this cumulant unchanged, and use the partition formula. In each term the unique block containing J_b is bounded by conditional Cauchy–Schwarz for J_b and Hölder for its A_i factors. The other blocks use Hölder alone. The resulting pointwise bound is

$$c_{n+1} \prod_i V_i \left(\mathbb{E}_x[|J_b|^2 \mid y]\right)^{1/2}.$$

Conditional Jensen gives (39). No conditional force bound at a fixed y_b is being assumed: its norm is taken under the full joint Gibbs law.

For constant action sources, $A_i = S$, equation (20) supplies $V = 4(H + 2n)/\beta$. Differentiating the normalized finite generating polynomial gives the exact identity

$$D_b\kappa_n(S \mid y) = n\kappa_n(\underbrace{S, \dots, S}_{n-1}, J_b \mid y) - \beta\kappa_{n+1}(\underbrace{S, \dots, S}_{n}, J_b \mid y).$$

At $n = 1$ the first term is $\mathbb{E}[J_b \mid y]$. The force estimate gives $\|J_b\|_p \le C_p\beta^{-1/2}$. For $g = \beta^{-1/2}$ and $n \ge 2$, it follows that

$$g\,\|D_b\kappa_n(S \mid y)\|_p \le C_p\left[nc_n\beta^{-1}V^{n-1} + c_{n+1}V^n\right] \tag{40}$$

$$\le C_{n,p}\frac{(H + 2n)^n}{\beta^n}. \tag{41}$$

The $n = 1$ estimate was proved above. The centered smooth-source bounds give corresponding estimates for distinct labels. This controls all fixed source orders through the complete positive hidden integral of the stated block, including any decomposition of it into histories. Its constants depend on the hidden block size. It supplies neither a spatial tree factor nor an induction bound for previously transported observables over arbitrarily many renormalization steps.

15.3 Spatial decay on a controlled real fiber

The following theorem supplies the desired exponential separation factor on a genuinely controlled fiber. Its hypotheses are not asserted uniformly for the full history mixture.

Theorem 15.1 (Weighted conditional covariance)**.** *Let $\mathrm{d}\nu = Z^{-1}e^{-V(z)}\,\mathrm{d}z$ on a finite-dimensional real space with coordinates indexed by spatial blocks. Assume sufficient smoothness, $D^2V \ge m_0^2 I$, and*

$$K_2(\alpha) = \sup_{z,i}\sum_j \left\|V''_{ij}(z)\right\| d(i,j)^2 e^{\alpha d(i,j)} < \infty, \qquad \alpha^2 K_2(\alpha) \le m_0^2.$$

If ∇F and ∇G are supported in A and B, respectively, then

$$|\operatorname{Cov}_\nu(F, G)| \le \frac{2}{m_0^2}e^{-\alpha d(A,B)}\,\|\nabla F\|_{L^2(\nu)}\,\|\nabla G\|_{L^2(\nu)}\,. \tag{42}$$

For exponentially localized gradients the same statement holds with the corresponding weighted gradient norms.

Proof. Let $\mathcal{L} = -\Delta + \nabla V \cdot \nabla$ on $L^2(\nu)$ and solve $\mathcal{L}u = G - \mathbb{E}G$. Uniform convexity supplies the spectral gap. Integration by parts and differentiation yield

$$\operatorname{Cov}(F, G) = \mathbb{E}[\nabla F \cdot \nabla u], \qquad (\mathcal{L} \otimes I + D^2V)\nabla u = \nabla G.$$

Put $R = (\mathcal{L} \otimes I + D^2V)^{-1}$ and let W multiply coordinate i by $e^{\alpha d(i,A)}$. It commutes with $\mathcal{L}$. The symmetric part of $WD^2VW^{-1} - D^2V$ has block entries

$$V''_{ij}\,[\cosh(\log W_i - \log W_j) - 1].$$

The distance function is 1-Lipschitz, and $\cosh t - 1 \le (t^2/2)e^{|t|}$. The Schur norm is therefore at most $\alpha^2 K_2(\alpha)/2$. The real part of the weighted operator is at least $m_0^2/2$, as is that of the operator with W and W^{-1} interchanged. Hence $\|W^{-1}RW\| \le 2/m_0^2$. In

$$\langle \nabla F, R\nabla G\rangle = \langle W\nabla F, (W^{-1}RW)W^{-1}\nabla G\rangle,$$

the first weight is one on A and the second is at most $e^{-\alpha d(A,B)}$ on B. This proves (42). Approximation proves it for the stated Sobolev class. For gradient tails use $d(i,A) \ge d(A,B) - d(i,B)$. □

The quadratic $\cosh -1$ loss is essential. For a massive discrete elliptic operator, the second spatial moment of its off-diagonal Hessian stays $O(1)$ while $m_0^2 \asymp \ell^{-2}$, allowing $\alpha \asymp \ell^{-1}$ even when $a_{\mathrm{lat}} \ll \ell$.

On a common fixed reference domain, differentiation of a normalized conditional integral gives the exact score identity

$$\partial_W \mathbb{E}_W F_W = \mathbb{E}_W(\partial_W F_W) - \mathrm{Cov}_W(F_W, \partial_W S_W). \tag{43}$$

Equation (42) bounds this covariance when the observable and score gradients satisfy its hypotheses. Moving domains, sharp tests, Jacobians and nonlinear constraints require their own terms or a fixed-reference reformulation.

For a positive history mixture $\nu = \sum_h p_h \nu_h$, the exact decomposition is

$$\mathrm{Cov}_\nu(F,G) = \sum_h p_h \,\mathrm{Cov}_{\nu_h}(F,G) + \mathrm{Cov}_p(\mathbb{E}_h F, \mathbb{E}_h G). \tag{44}$$

The theorem controls the first term on suitable fibers. It does not control the second. For Gaussian branches with identical massive local covariance and means $\pm M$ at all coordinates, the mixture adds M^2 covariance at arbitrary separation. The missing history estimate must therefore establish spatial connection of the inserted activities, not only small total history masses.

18 Clean averaging maps and Wilson precision estimates

All norms in this section use physical lattice volume. For spacing η, $\|B\|_2^2 = \eta^4 \sum_b |B_b|^2$ and $W_\eta = (-\Delta_\eta + 1)^{1/2}$.

Proposition 18.1 (One-sided precision bound)**.** *In a fixed local gauge let $U_b = e^{\eta A_b}$, with*

$$\|A\|_{1,\infty} = \max\{\|A\|_\infty, \|\nabla_\eta A\|_\infty\} < h.$$

Let $K_\eta(A)$ be the holomorphic Wilson Hessian in multiplicative link coordinates plus the full covariant Feynman-gauge term $D_A D_A^\mathsf{T}$. On a fixed small complex ball,

$$\left\|(K_\eta(A) - K_\eta(0))W_\eta^{-1}\right\| + \left\|W_\eta^{-1}(K_\eta(A) - K_\eta(0))\right\| \le C(h + h^2), \tag{53}$$

uniformly in η and volume. Fixed background jets have the corresponding bounds by Cauchy's formula. The same assertion holds for the covariant site Laplacian.

Proof. Transport the four oriented link variations of a plaquette to one endpoint and write them $X_i = \eta\,\mathrm{Ad}(P_i)(\epsilon_i B_i)$. At plaquette holonomy one, the symmetric second trace variation is the squared sum of the X_i, the covariant plaquette curl. The difference at general holonomy is bounded by $C\,|U_p - 1|\,\eta^2 \sum_{b,c\in p} |B_b|\,|C_c|$. The physical first-difference bound gives $|U_p - 1| \le C\eta^2(h + h^2)$. Summation with finite plaquette overlap therefore leaves a bounded operator in physical L^2, with no inverse power of η.

The transported curl is $d_\eta + C_A$, where C_A is a finite-range multiplication/shift operator. Its norm is $O(h)$ and its first coefficient differences are $O(h + h^2)$. Expand the square. In $d_\eta^\mathsf{T} C_A W_\eta^{-1}$, commute the flat difference through the coefficient. The discrete product rule and $\left\|\nabla_\eta W_\eta^{-1}\right\| \le C$ give the bound; the transposed term and $C_A^\mathsf{T} C_A$ are handled directly. The covariant gauge term has the same structure. Holomorphic Cauchy bounds on the stated physical norm ball give the fixed jets. □

The current/multiplier term of the actual constrained Lagrangian is not part of K_η here. Its coefficient must be retained or separately cancelled by the exact current Ward identities. An amplitude-only ball would also be insufficient for (53).

Lemma 18.2 (Stable iteration of a fixed local analytic average)**.** *At spacing ξ, suppose the straightened physical potential map has, on every coarse component, the translated finite-patch form*

$$T_\xi(A) = (L\xi)^{-1} F(\xi A|_{\mathrm{patch}}), \qquad F(0) = 0.$$

Its analytic radius depends on the fixed L only. Suppose its flat derivative Q^0 is the normalized straight average, hence a contraction in physical L^2, in the sup norm, and from fine to coarse first differences. Then

$$\|T_\xi(A)\|_{1,\infty} \le \|A\|_{1,\infty} + C_L\xi \,\|A\|_{1,\infty}^2 , \tag{54}$$

$$\left\|DT_\xi(A) - Q^0\right\|_{2\to 2} \le C_L\xi \,\|A\|_\infty . \tag{55}$$

For $\xi_i = L^{i-k}$ and small enough initial h, all iterated potentials stay within $2h$, and the full derivative has norm at most $\exp(2C_Lh/(L-1))$, independently of k.

Proof. For $R_\xi = T_\xi - Q^0$, finite-patch analyticity gives $|R_\xi| \le C_L\xi\,\|A\|_\infty^2$ and $\|DR_\xi\| \le C_L\xi\,\|A\|_\infty$. Adjacent coarse patches differ by a translation $L\xi$; their input difference is at most $L\xi\,\|\nabla_\xi A\|_\infty$. Divide the output difference by $L\xi$ to obtain the first-difference bound. The finite patch and physical volume weights give the L^2 Schur bound. Now $h_{i+1} \le h_i + C_L\xi_i h_i^2$ and $\sum_i \xi_i \le 1/(L-1)$ imply $h_i \le 2h$ by induction. Multiply the derivative bounds to conclude. □

The straightened one-step path-log/Federbush construction has this finite local analytic form and the indicated flat derivative. Its site average requires the analogous analytic perturbation estimate. To use the lemma at every depth of an actual determinant, the same straightening and its accompanying gauge factors must be identified consistently. The plain holonomy average samples gauge parameters at block centers and does not have the same bounded derivative.

For momentum conjugation, weighted Schur/Jensen bounds on the positive flat average, together with the Schur bound on its perturbation, cost at most $e^{C_L|\Im p|\xi_i}$ per step. Their product remains bounded on a fixed momentum strip. An unweighted operator norm alone would not justify this conclusion.

18.1 Coarse constraint factors

Let $Q(A)$ and $R(A)$ be the form and site averages on this clean common core. The site map has $R(0)R(0)^{\mathsf T} = I$. Small perturbation makes $RR^{\mathsf T}$ uniformly invertible, so

$$1 - P = R^{\mathsf T}(RR^{\mathsf T})^{-1}R, \qquad E = Q^{\mathsf T}Q - D_A(1-P)D_A^{\mathsf T}$$

have uniformly bounded energy-sandwiched jets. Factor through QW^{-1}, $RD_A^{\mathsf T}W^{-1}$, their transposes, and $(RR^{\mathsf T})^{-1}$ before differentiating. All such terms factor through a coarse space of fixed dimension per physical cell.

There is also an explicit clean right inverse. For $N_0 = L^k$, $\eta = N_0^{-1}$, the unaliased flat straight-average multiplier, up to a phase, is

$$q_\mu(p) = a_{N_0}(p_\mu)^2 \prod_{\nu\neq\mu} a_{N_0}(p_\nu), \qquad a_{N_0}(t) = \frac{\sin(t/2)}{N_0 \sin(t/(2N_0))}.$$

On the coarse Brillouin zone, $|q_\mu| \ge (2/\pi)^5$. Embed a coarse Fourier mode in this fine alias and divide by q_μ. The resulting E_0 satisfies $Q_0\mathsf{E}_0 = I$ and $\|W\mathsf{E}_0\| \le C$, since the fine physical difference multiplier is bounded by $|p|$. The right inverse $\mathsf{E}(A) = \mathsf{E}_0(Q(A)\mathsf{E}_0)^{-1}$ persists on the small complex core. For a positive real precision H comparable to W^2, Cauchy–Schwarz gives

$$cI \le QH^{-1}Q^* \le CI.$$

Indeed $\|v\|^2 = \langle Q\mathsf{E}v, v\rangle \le \left\|H^{1/2}\mathsf{E}v\right\| \left\|H^{-1/2}Q^*v\right\|$ proves the lower bound. Complex invertibility follows by perturbation. This argument concerns the normalized straightened clean map, not arbitrary layered seam constraints.

19.1 The actual Wilson lattice coefficient

The missing numerical magnetic numerator calculation can be replaced by an existing *lattice* calculation. In background Feynman gauge, Lüscher–Weisz [26], equation (5.7), compute

$$\nu^{(1)}(p) = \frac{11N}{48\pi^2}\log(1/p^2) + O(1).$$

Their Wilson action, multiplicative link coordinates, gauge fixing and ghost precision match the Hessian and covariant site Laplacian used here [27]. Only this finite Gaussian-order calculation is used; convergence of a perturbation series is not assumed.

Let

$$\mathcal{D}_z(B) = \log\det(\Delta_B + z) - \tfrac{1}{2}\log\det(K_B + z) - (B = 0), \qquad z > 0.$$

Take field derivatives at finite volume, then the thermodynamic limit at fixed z. Denote by $b_m(z)$ its quadratic coefficient in the normalized Wilson action $A = \sum_p (1 - N^{-1}\Re\,\mathrm{tr}\,U_p)$.

Proposition 19.1 (Mass-regulated Wilson coefficient)**.** *For $0 < z \le z_0$,*

$$b_m(z) = \frac{11N^2}{24\pi^2}\log(1/z) + O(1), \qquad |z b_m'(z)| + \left|z^2 b_m''(z)\right| \le C. \tag{56}$$

For SU(2) *the logarithmic coefficient is* $11/(6\pi^2)$.

Proof. At zero background put $\psi(q) = \sum_\mu 4\sin^2(q_\mu/2)$ and $G_z = (\psi + z)^{-1}$. The determinant Hessian is a finite sum of tadpole and bubble integrands

$$I(q,p,z) = \frac{T(q,p)}{\psi(q)+z} + \frac{N(q,p)}{(\psi(q)+z)(\psi(q+p)+z)}.$$

The finite-range vertices are trigonometric polynomials. The first background vertex is $O(|q| + |p|)$, because constant background and constant quantum field have no quadratic term linear in the background. Thus $N = O((|q| + |p|)^2)$, $\partial_p N = O(|q| + |p|)$ and $\partial_p^2 N = O(1)$; the second vertex is bounded.

Let $\Phi_z(p)$ be the integrated color-normalized spatial trace, with measure $(2\pi)^{-4}\,\mathrm{d}^4q$. Background gauge covariance, preserved by the scalar mass, and hypercubic symmetry imply $\Phi_z(te) = 3c(z)t^2 + O_z(t^4)$ for a unit vector e. The full massless sum has $\Phi_0(0) = 0$ and zero first derivative. These cancellations are used *before* splitting momentum space: replace each integrand by

$$J(q,p) = I(q,p,0) - I(q,0,0) - p\cdot\nabla_p I(q,0,0).$$

Its integral is $\Phi_0(p)$ after summing diagrams.

Set $m = \sqrt{z}$ and $p = me$. On $|q| \le 4m$, the massive second momentum derivative is bounded by $C(|q|^2 + m^2)^{-2}$, whose integral is $O(1)$. The massless bubble and the subtractions have bounds

$$Cm^2\int_{|q|\le 4m} \frac{\mathrm{d}^4q}{|q|^2\,|q+me|^2} = O(m^2),$$
$$\int_{|q|\le 4m} |q|^{-2}\,\mathrm{d}^4q + m\int_{|q|\le 4m} |q|^{-3}\,\mathrm{d}^4q = O(m^2).$$

On $|q| \geq 4m$, the third momentum derivative is $O(|q|^{-5})$. The Taylor remainder divided by m^2 therefore integrates to $O(m \int_{4m}^{1} r^{-2}\,\mathrm{d}r) = O(1)$. Changing the mass from zero to m^2 in the second Taylor coefficient costs $O(m^2 \int_{4m}^{1} r^{-3}\,\mathrm{d}r) = O(1)$. The rest of the Brillouin zone is uniformly regular. Consequently

$$c(m^2) = \frac{\Phi_0(me)}{3m^2} + O(1) = \nu^{(1)}(me) + O(1).$$

The normalized Wilson action has transverse Hessian $P/(2N)$ in the conventions of [26]; the Gaussian log integral has coefficient $+\nu^{(1)}P$. Thus $b_m = 2Nc$, including its positive sign.

Finally, direct differentiation of these finite integrals gives

$$\left|\partial_z^j \partial_p^2 I(q,0,z)\right| \leq C_j(\psi(q)+z)^{-2-j}, \qquad j = 1,2.$$

Its four-dimensional integral is $O(z^{-j})$. This proves the weighted derivative bounds; they do not follow by differentiating an unspecified $O(1)$ remainder. □

At positive mass, compute the gauge-invariant diagonal trace density in a gauge centered at its observation site. A background gauge transformation removes the constant part of an affine Abelian potential there. Field derivatives then involve polynomial factors in relative distances, controlled by the massive decay, and the boundary contribution per volume tends to zero. This gives the constant magnetic-field second derivative as the same F^2 coefficient; a sharp cutoff of an uncentered affine potential would not justify the argument. For the SU(2) convention $U_p = e^{if\sigma_3}$, $a_p = 1 - \cos f$, so no further factor is needed. The order of limits is field derivative, volume, then mass tending to zero.

19.2 An exact trace remainder

Lemma 19.2 (Finite-rank constraint versus mass)**.** *Let $W \geq 1$, $B = K + I$, $A = K + F$, $A_t = B + t(F - I)$. Suppose on a common parameter domain*

$$\left\|W A_t^{-1} W\right\| \leq C, \qquad \left\|W^{-1} F W^{-1}\right\| \leq C, \qquad \operatorname{rank} F \leq r.$$

With $G = B^{-1}$ and coherent logarithms,

$$\log\det A = \log\det B - \operatorname{tr} G - \tfrac{1}{2}\operatorname{tr} G^2 + \mathcal{R}, \qquad |\mathcal{R}| \leq C'(r + \operatorname{tr} W^{-6}). \tag{57}$$

Proof. Set $E = F - I$. The exact resolvent identity, stopped after two substitutions, is

$$A_t^{-1} = G - tGEG + t^2 GEGEA_t^{-1}.$$

Integrate $\operatorname{tr}(EA_t^{-1})$ in t. The terms without F in the first two traces are $-\operatorname{tr} G - \frac{1}{2}\operatorname{tr} G^2$. Every other such term has bounded energy-sandwiched factors and one rank-r factor. Expand the three E factors in the remaining integral $\int_0^1 t^2 \operatorname{tr}(EGEGEA_t^{-1})\,\mathrm{d}t$. Each term containing F is $O(r)$, by inserting W, W^{-1} cyclically. The term without F is $-\operatorname{tr}(G^2 A_t^{-1})$, bounded by Schatten Hölder:

$$\left|\operatorname{tr}(G^2 A_t^{-1})\right| \leq \|G\|_{\mathfrak{S}^3}^2 \left\|A_t^{-1}\right\|_{\mathfrak{S}^3} \leq C \operatorname{tr} W^{-6}.$$

The last step writes every inverse as $W^{-1}CW^{-1}$ with bounded C. This is a finite identity, not an infinite formal resolvent expansion. □

In four dimensions, for $W = (-\Delta_\eta + 1)^{1/2}$, the trace $\operatorname{tr} W^{-6}$ per physical unit volume is bounded uniformly in η, by the convergent Fourier sum of $(1 + |q|^2)^{-3}$. Coarse-rank constraint terms have r proportional to coarse volume. Thus (57) has a cutoff-independent remainder per coarse cell. A saddle determinant has an additional coarse Schur determinant; the preceding right-inverse estimate controls it on the clean class.

For the determinant pair the explicit mass terms combine as $\mathcal{L}(1) - \mathcal{L}'(1) + \frac{1}{2}\mathcal{L}''(1)$. A uniform bound on the momentum-quadratic coefficient of the remainder additionally needs an exponentially summable Hessian kernel or a common external momentum strip. A per-volume amplitude bound alone is insufficient. Nor does the finite identity identify the exact gauge and constraint factors of a layered history.

If those actual identifications give

$$\sum_{l=1}^{k} b_l^0 = b_m(\eta_k^2) - \eta_k^2 b_m'(\eta_k^2) + \tfrac{1}{2}\eta_k^4 b_m''(\eta_k^2) + O(1), \qquad \eta_k = L^{-k},$$

with a uniform extracted-coefficient remainder, Proposition 19.1 yields

$$\sum_{l=1}^{k} b_l^0 = k\,\frac{11N^2}{12\pi^2}\log L + O(1).$$

This cumulative estimate suffices for the real ultraviolet arithmetic. Indeed, if $s_j = \sum_{l\le j}(b_l^0 - c_0)$ is bounded and $x_{j+1} = x_j - b_{j+1}^0 + r_j$, then $y_j = x_j + s_j$ obeys $y_{j+1} = y_j - c_0 + r_j$ exactly. Since s_j is bare-coupling independent, the tangent is unchanged. Geometric convergence of the individual b_j^0 is unnecessary for this reduction.

20.1 Local normalization quotients

Lemma 20.1 (Finite-source denominator)**.** *Let ν be a probability measure and Ψ nilpotent. If $\sup_\xi \|\Psi(\xi,z)\|_r \le q < \log 2$, then $D(z) = \mathbb{E}_\nu e^\Psi$ satisfies*

$$\left\|D^{-1}\right\|_r \le (2-e^q)^{-1}, \qquad \|\log D\|_r \le -\log(2-e^q).$$

For $q \le 1/4$ these are at most e^{2q} and $2q$. If the normalized reference measure is complex with total variation at most L, replace $e^q - 1$ by $L(e^q-1)$ in the geometric and logarithmic bounds.

Proof. Write $D = 1 + H$. Positivity gives $\|H\|_r \le e^q - 1 < 1$. The inverse and logarithm are finite algebra series, bounded by their positive geometric and logarithmic series. Total variation gives the complex-reference version. □

Here is the primitive Wilson input. In a fixed regular normalization cell Λ, suppose the actual reconstructed plaquettes satisfy

$$|1 - U_{0,p}| \le C\alpha\eta^2, \qquad |U_p(B) - U_{0,p}| \le Ct\eta^2, \qquad \#p \le C\left|\Lambda^+\right|\eta^{-4}.$$

The Wilson action vanishes quadratically at the identity, so $|a_p(U(B)) - a_p(U_0)| \le C\eta^4(\alpha t + t^2)$, also on a fixed small complex neighborhood. For $w = \sum_i z_i f_i$ and $\rho = \sup_p \sum_i r_i\,|f_i(p)|$,

$$\sup_B \|A(w,U(B)) - A(w,U_0)\|_r \le C\rho\left|\Lambda^+\right|(\alpha t + t^2). \tag{59}$$

The fine spacing cancels. With $t = gP$ and the stated regular cutoffs, this is small on a fixed cell. An entire component can contain arbitrarily many cells; invert its local normalization factors separately and pay their extensive costs with the component reserve.

The factorization of a denominator is exactly

$$D_\Lambda(w) = e^{A(w,U_0)} D_\Lambda(0) \mathbb{E}_{\nu_0} e^{A(w,U(B)) - A(w,U_0)}.$$

The factor $e^{-A(w,U_0)}$ in its inverse is field dependent and must remain in the source/action bookkeeping. It is not an arbitrary vacuum subtraction.

Under the original coefficientwise component reserve, bounded overlap of charged cells, and an actual marked bridge majorant, these estimates close the primitive source loads and normalization algebra through one completed **R** step. They do not construct the marked bridge family on the extended paired-field domains. Also, the rough load in (59) can be of order $x^{-1}\log^a x$, which is not summable over all scales; its role is to pay local reserves. The normalized constant-source cancellation must still be used for multiscale tangent control.

21 A common analytic envelope with controlled parameter size

This section gives an alternative way to control weak cross-component background dependence. Its application requires the indicated common domain; exponential operator-norm locality alone is not enough.

Lemma 21.1 (Nuclear image envelope)**.** *Let $K : B_E(p_0, R) \to L^1(\mu)$ be holomorphic with norm at most M. If $T : X \to E$ has nuclear norm $s < R/e$, there is a common measurable envelope for canonical representatives of $K(p_0 + Tx)$, $\|x\| \le 1$, whose integral is at most*

$$\frac{M}{1 - es/R}.$$

Proof. Write $Tx = \sum_a \lambda_a \phi_a(x) u_a$ with $\|\phi_a\|, \|u_a\| \le 1$ and $\sum_a |\lambda_a|$ arbitrarily close to s. Multivariable Cauchy on radii R/m gives $\|D^m K[u_1, \dots, u_m]\|_1 \le M m^m R^{-m}$. Expand the Taylor series after substituting the nuclear decomposition, take absolute values of every coefficient, and use Tonelli. Its integral is bounded by $M \sum_{m \ge 0} (m^m/m!)(s/R)^m \le M \sum_{m \ge 0} (es/R)^m$. The convergent coefficient sum gives common almost-everywhere representatives and the envelope. □

There is an extensive version that avoids requiring the *total* influence on a large component to be small. Suppose $E = \bigoplus_b^\infty E_b$ and K is holomorphic on the product ball of radius R. If each output-block map T_b has nuclear norm $s_b < R/e$, the same proof by multidegrees gives

$$\int \text{envelope} \le M \prod_b (1 - e s_b/R)^{-1} \le M \exp\left(\frac{2e}{R} \sum_b s_b\right) \quad \text{if } s_b \le R/(2e). \tag{60}$$

Indeed the mixed derivative bound is $M \prod_b (n_b/R)^{n_b}$ and its Taylor denominator is $\prod_b n_b!$. If bounded-overlap smooth local variations satisfy $\|\sum_b v_b\|_E \le C_{\mathrm{ov}} \sup_b \|v_b\|$, pull back K to their product ball of radius R/C_{ov}. This handles first-difference norms without introducing artificial jumps across patch boundaries.

For a bounded physical forcing, a separated Green operator provides a concrete nuclear decomposition:

$$(Tf)(x) = \eta^4 \sum_{y \in Y} G(x, y) f(y), \qquad \|T\|_{\mathrm{nuc}} \le \eta^4 \sum_{y \in Y} \|G(\cdot, y)\|_{C^1(X)}.$$

A cutoff-uniform separated Green/gradient bound makes this exponentially small in the buffer width. The product version then costs $\exp(C\,|X|\,e^{-c\,\mathrm{buffer}})$, compatible with an extensive component reserve. For nonlinear layered reconstructions one still has to identify a common fixed forcing representation and show that the summed perturbations stay in the actual paired-field domain. A response operator depending on the very hidden variables being integrated cannot be substituted into this lemma without that additional argument.

22 Conditional closure of clean source transport

The clean source sector can be obtained by differentiating one actual source-free RG map, provided that map is established on *spatial* product neighborhoods. A disc for a single constant coupling, or a product in scale indices, is insufficient.

Let X_j be the local marginal coefficient field, including all tensor types admitted by the scale map, and K_j its irrelevant state. Fix charts, tests, references, extraction maps, R_j and b_j independently of

the density perturbation. Use physical scales $\ell_j = aL^j$ with fixed $L > 1$. Suppose exact holomorphic maps have the form

$$X_{j+1} = R_j X_j - b_j \mathbf{1} + E_j(X_j, K_j), \qquad K_{j+1} = T_j(X_j, K_j),$$

on enlarged neighborhoods $\|X - x_j\mathbf{1}\|_\infty < \rho x_j$, $\|K - K_j\| < \rho_K x_j^{-p}$. The local average R_j preserves the whole scalar coefficient sector, preserves its constant, has absolute row sum at most one, and range $O(\ell_{j+1})$. Preserving only the constant would leave a possible extra tensor-valued geometric term in the commutator below. Assume $p > 0$, $0 < \epsilon \le p$, and that K_0 is independent of X_0. The centers $(x_j\mathbf{1}, K_j)$ form the actual reference orbit starting at $X_0 = x_0\mathbf{1}$. They satisfy $b/2 \le x_j - x_{j+1} \le 3b/2$ for a fixed $b > 0$, stay above a sufficiently weak floor, and obey $\|K_j\| \le D x_j^{-p}$.

Use the weighted row norm $\|A\|_{\alpha,\ell} = \sup_u \sum_v e^{\alpha d(u,v)/\ell} \|A(u,v)\|$. The required *unmarked* derivative bounds are

$$\begin{aligned} \|D_X E_j\|_{\alpha,\ell_{j+1}} &\le C x_j^{-1-\epsilon}, \quad \|D_K E_j\|_{\alpha,\ell_{j+1}} \le B, \\ \|D_X T_j\|_{\alpha,\ell_{j+1}} &\le C x_j^{-1-p}, \quad \|D_K T_j\|_{\alpha,\ell_{j+1}} \le q_\alpha < 1, \end{aligned} \tag{61}$$

uniformly on those neighborhoods. All constants, the domain radii, and $1 - q_\alpha > 0$ are uniform in scale, cutoff and volume. They are genuine hypotheses about the actual map. Spatial locality and irrelevant contraction do not follow from a scalar partition-function estimate.

Directional Cauchy bounds in a product neighborhood give operator norms. They do not alone bound the sum of norms of a Banach-valued spatial kernel. The weighted kernel bounds in (61) remain explicit locality inputs.

Proposition 22.1 (Clean source closure from the local unmarked map)**.** *Under these assumptions, for $Y_j = D_{X_0} X_j$ and $Z_j = D_{X_0} K_j$,*

$$\|Y_j\|_{\alpha',\ell_j} \le C, \qquad \|Z_j\|_{\alpha',\ell_j} \le C x_j^{-1-p}, \qquad \|Y_j - R_{j:0}\|_{\alpha',\ell_j} \le C x_j^{-\epsilon}.$$

Here $0 < \alpha' \le \alpha$ is fixed and $R_{j:0} = R_{j-1} \cdots R_0$. For scalar f, write $R_j(f e_0) = (R_j^{\mathrm{sc}} f) e_0$ and set $f_j = R_{j:0}^{\mathrm{sc}} f$, $y_j = Y_j[e_0]$, $z_j = Z_j[e_0]$. Then

$$\|Y_j[f e_0] - f_j y_j\| \le C \ell_j \operatorname{Lip}(f) x_j^{-\epsilon}, \tag{62}$$

$$\|Z_j[f e_0] - f_j z_j\| \le C \ell_j \operatorname{Lip}(f) x_j^{-1-p}. \tag{63}$$

For every finite order m,

$$\|D^m K_j[f_1, \dots, f_m]\| \le C^m m! x_j^{-p-m} \prod_i \|f_i\|_\infty, \tag{64}$$

$$\|D^m X_j[f_1, \dots, f_m]\| \le C^m m! x_j^{1-m-\epsilon} \prod_i \|f_i\|_\infty, \quad m \ge 2. \tag{65}$$

Proof. The exact chain rule includes every old insertion:

$$Y_{j+1} = (R_j + D_X E_j) Y_j + (D_K E_j) Z_j, \qquad Z_{j+1} = (D_X T_j) Y_j + (D_K T_j) Z_j.$$

For a fixed final n, use the common spatial weight α'/ℓ_n throughout the prefix. The averaging norm is at most $r_j = e^{C\alpha' \ell_{j+1}/\ell_n}$. Choose $C_* > C/(1 - q_\alpha)$ and set $V_{j+1} = (r_j + C x_j^{-1-\epsilon} + B C_* x_j^{-1-p}) V_j$, $V_0 = 1$. Induction gives $\|Y_j\| \le V_j$ and $\|Z_j\| \le C_* x_j^{-1-p} V_j$. The logarithm of V_n is bounded by a geometric scale sum and the convergent sums of $x_j^{-1-\epsilon}$ and x_j^{-1-p}. Variation of constants for $Y_j - R_{j:0}$ proves its smaller bound.

For a kernel $A_j = Y_j - R_{j:0}$, the first commutator is $\sum_v A_j(u,v)(f(v) - f_j(u))$. The bracket is bounded by $\mathrm{Lip}(f)(d(u,v) + C\ell_j)$. Its weighted row sum gives (63); use Z_j for the second.

The same estimates on smaller complex neighborhoods show that every initial local perturbation of norm less than cx_n stays inside the admitted boxes through level n. At a proposed first exit, the changes in X_j and K_j are bounded by $C\,\|h_0\|$ and $Cx_j^{-1-p}\,\|h_0\|$, strictly within their margins for small c. Holomorphic composition is therefore valid on this ball. Cauchy applied to the first tangent, with its radius divided among the remaining directions, proves (65). The polarization cost is absorbed in $C^m m!$. □

The weighted first derivative and Cauchy also retain a rooted support factor

$$\exp\left[-\frac{\alpha'}{m}\frac{\mathcal{T}_u(\operatorname{supp} f_1, \dots, \operatorname{supp} f_m)}{\ell_j}\right],$$

because the same coefficient is bounded by the exponential distance to any chosen first support, and a tree through the root has length at most m times the largest such distance. The decay rate here is allowed to depend on the fixed source order. Actual local vacuum increments must also be retained; with their corresponding unmarked derivative bounds, their degree-m coefficients are $O(C^m m! x_j^{-m})$ per current block. For separated supports the rooted factor makes their ultraviolet scale sum finite despite the number of blocks.

For the distinguished scalar direction e_0, the same theorem gives $Y_j[e_0] = e_0 + O(x_j^{-\epsilon})$. The absolute value of its scalar projection is therefore at least $1 - Ch^{-\epsilon} > 0$ at a sufficiently weak floor, while other marginal components remain small. Identifying this tangent with the earlier one-coupling scalar recurrence would require preservation of that entire homogeneous family; nonvanishing does not require that extra identification.

This proposition closes the clean higher-source induction *under* the local unmarked map theorem. It does not prove (61) or its spatial product domains for all Wilson histories. For a frozen linear transfer kernel, derivatives of the logarithmic effective density are the conditional cumulants. When its presentation also contains density-dependent normalization quotients, the derivatives are the complete logarithmic source coefficients and every quotient derivative must be included. The full source problem has thus been reduced to concrete stability and locality properties of the same density map, rather than to an independent list of assumed source formulas.

23.1 Rare components and clean convexity do not suffice

The proposed shortcut through a weak Poincaré inequality fails as a general implication. Let H be a fair global bit. Conditional on H, let local marks B_v be independent with probabilities ε or 2ε. Conditional on all marks, let X_v be independent centered Gaussians of variance one for $B_v = 0$ and two for $B_v = 1$. Every fixed-history fiber is uniformly convex with Poincaré constant at most two. Also $\mathbb{P}(B_v = 1 \ \forall v \in A) \le (2\varepsilon)^{|A|}$, so the standard rare connected-component bounds hold.

Nevertheless the local insertion $F(x) = x^4 - x^6/5$ has zero constant and quadratic Taylor jets and zero clean Gaussian mean, while its bad-fiber mean is -12. For distinct sites,

$$\operatorname{Cov}(F(X_v), F(X_w)) = \operatorname{Var}_H(-12 p_H) = 36\varepsilon^2,$$

independently of separation. Deterministic vacuum subtraction leaves this covariance unchanged. Centering separately by the random history would remove part of the physical observable. This is a counterexample to the proposed inference, not a counterexample to Yang–Mills. It shows why the actual local formation and weighting of histories cannot be replaced by their rarity. If H is kept in the retained state, the conditional covariance of distant variables vanishes and the displayed contribution becomes the terminal covariance of their conditional means. This is consistent with the completed-$\mathbf{R}$ factorization at fixed exterior data. The example prohibits discarding that terminal contribution or inferring an unconditional bridge estimate from rarity alone.

23.2 The exact reconstruction requirement

The primary Osterwalder–Schrader reconstruction theorem [12], section IV.1, accepts the linear-order/factorial-growth condition $E0'$ on Schwartz tests flat at coincidences. Its condition $E0''$ on all Schwartz tests is explicitly stronger. Thus a separate arbitrary extension across the diagonal is not mandatory if $E0'$ and the remaining Euclidean, positivity, symmetry and clustering axioms are proved on the correct algebra.

The earlier flat-test estimate supplies an implication toward $E0'$ once its uniform kernel hypotheses hold for the full hierarchy. Those actual bounds, the full gauge-invariant operator family, its covariance and Yang–Mills identification, and the required vacuum properties have not been constructed here. Correcting the reconstruction target removes an unnecessary requirement; it does not supply these missing estimates.

25.2 2. Exact finite-dimensional Gaussian source transport

Let V be the hidden linear space, C a curvature/difference operator on the full variables, and D its restriction to V. Assume $H_0 = D^*D > 0$. For a fixed coarse datum B, let u_B be the homogeneous minimum and $F_B = Cu_B$. Thus

$$D^*F_B = 0.$$

The weighted action is

$$S_\lambda(v;B) = \tfrac{1}{2}\langle F_B + Dv, \Lambda_\lambda(F_B + Dv)\rangle, \qquad \lambda = x + h, \quad x > 0,$$

where Λ denotes diagonal multiplication on curvature components. The inner products are bilinear when parameters are complex; operator norm bounds use their Hermitian counterparts. Put

$$q_B(f) = H_0^{-1/2} D^* \Lambda_f F_B, \qquad T(f) = H_0^{-1/2} D^* \Lambda_f D H_0^{-1/2}.$$

Then

$$\|T(f)\| \le \|f\|_\infty, \qquad \|q_B(f)\| \le \|\Lambda_{f-c} F_B\| \tag{4.2}$$

for any scalar constant c. The second bound uses $D^*F_B = 0$ and the contraction $H_0^{-1/2}D^*$. For $\|h\|_\infty < x$, the hidden precision is invertible and completing the square gives the exact critical energy

$$\begin{aligned} E_{x+h}(B) = \frac{x}{2}\|F_B\|^2 + \frac{1}{2}\langle F_B, \Lambda_h F_B\rangle \\ - \frac{1}{2x}\left\langle q_B(h), (I + x^{-1}T(h))^{-1} q_B(h)\right\rangle. \end{aligned} \tag{4.3}$$

This identity is holomorphic on the entire indicated product ball, and the inverse norm is at most $(1 - \|h\|_\infty/x)^{-1}$. It concerns the energy norm of a linear Gaussian problem; it does not assert a cutoff-uniform physical C^1 norm for the minimizing field.

For $h = \sum_i z_i f_i$, expand the inverse in (4.3). For every $m \ge 2$,

$$D^m E_x(B)[f_1, \dots, f_m] = \frac{(-1)^{m-1}}{2x^{m-1}} \sum_{\sigma \in S_m} \left\langle q_B(f_{\sigma(1)}), T(f_{\sigma(2)}) \cdots T(f_{\sigma(m-1)}) q_B(f_{\sigma(m)})\right\rangle. \tag{4.4}$$

There are exactly m-2 interior T factors. This is a finite coefficient identity, not an assertion that an infinite physical perturbation series converges as the cutoff is removed.

If all curvature components in the block have physical diameter at most C ℓ, set $o(f) = \inf_c \|f - c\|_\infty$ on that block. From (4.2)–(4.4),

$$|D^m E_x(B)[f_1, \dots, f_m]| \le (m-2)!\, x^{1-m} \|F_B\|^2 \sum_{i<j} o(f_i) o(f_j) \prod_{r \neq i,j} \|f_r\|_\infty. \tag{4.5}$$

The coefficient (m-2)! follows because each ordered endpoint pair occurs (m-2)! times, and the two orders cancel the prefactor 1/2. In particular $o(f) \le C\ell \,\mathrm{Lip}(f)$, and a mixed derivative vanishes when all but at most one input direction are constant.

The result is stronger and more accurate than assigning a small quantum factor $x^{-\epsilon}$ to every spatial response. It identifies two oscillation factors and the exact classical power x^{1-m}.

25.3 3. The same Gaussian integral supplies the complete normalization

Up to a λ-independent factor, integration over V gives

$$Z_\lambda(B) = e^{-E_\lambda(B)} \det(xH_0 + D^* \Lambda_h D)^{-1/2}. \tag{4.6}$$

The field-independent Gaussian normalization is not omitted. Its degree-m coupling derivative is

$$D^m \left[-\tfrac{1}{2} \log\det(xH_0 + D^*\Lambda_h D) \right]_{h=0} [f_1, \dots, f_m] = \frac{(-1)^m}{2mx^m} \sum_{\sigma \in S_m} \mathrm{Tr}\, T(f_{\sigma(1)}) \cdots T(f_{\sigma(m)}). \tag{4.7}$$

Consequently its absolute value is at most

$$\frac{(m-1)!}{2x^m} \dim(V) \prod_i \|f_i\|_\infty. \tag{4.8}$$

The projection

$$\Pi = D(D^*D)^{-1} D^*$$

turns each trace in (4.7) into a curvature-space cycle $\mathrm{Tr}(\Lambda_{f_1} \Pi \Lambda_{f_2} \Pi \cdots \Lambda_{f_m} \Pi)$. Formula (4.4) similarly gives chains whose endpoints meet the coarse background. If Π and the harmonic lift have uniform exponentially weighted kernel bounds at physical scale ℓ, these chains and cycles yield connected spatial bounds directly: the sum of their link lengths is at least the minimal connecting-tree length of the marked supports and retained roots. Spare exponential weight absorbs all summations and finite derivative moments. This derives Gaussian locality from the actual hidden covariance. It does not manufacture locality of the random history weights.

The dimension factor in (4.8) becomes a sum over roots, rather than a factor per source. For positively separated source supports the cycle bound is exponentially small at scales below their separation. For overlapping supports it contains genuine local vacuum/contact terms.

25.4 4. A nonlinear classical version of the two-oscillation cancellation

Suppose $E(\lambda; B)$ is the analytic critical value of an action linear in λ, with constraints and coordinates independent of λ. Assume a unique analytic critical point on a local coefficient ball. Then for scalar t

near one,

$$E(t\lambda; B) = tE(\lambda; B). \tag{4.9}$$

Indeed multiplying the entire action by t leaves its stationary equations and selected branch unchanged. Thus D E(x 1;B)[f] is independent of x, and for $m \geq 2$

$$D^m E(x1; B)[f, 1, \ldots, 1] = 0. \tag{4.10}$$

Write each $f_i = c_i + \delta_i$ with c_i a constant. Multilinearity and (4.10) leave only terms containing at least two $delta_i$. If Cauchy gives

$$\|D^m E(x1; B)\| \leq MC_0^m m! x^{1-m},$$

then

$$|D^m E(x1; B)[f_1, \ldots, f_m]| \leq MC_1^m m! x^{1-m} \sum_{i<j} o(f_i) o(f_j) \prod_{r \neq i,j} \|f_r\|_\infty. \tag{4.11}$$

To bound the subset expansion, use $o(f_i) \leq 2\|f_i\|_\infty$ and assign each surviving subset one of its pairs; the remaining choices cost at most 3^{m-2}, absorbed in C_1^m. If the analytic radius is $x/(\log x)^s$ instead, the right hand side acquires $(\log x)^{sm}$.

This is applicable to an actual fixed regular Wilson constraint chart once that chart's analytic critical point and domain are proved. It does not assume a quadratic Wilson action. At nonzero inherited interactions the full action is no longer homogeneous in the Wilson coefficient alone; their differentiated contributions must be retained as perturbative remainders.

25.5 5. Why the oscillation gain is sufficient for a local ultraviolet sum

For geometric physical scales $\ell_j = \ell_n L^{j-n}$ and a weak coupling trajectory satisfying $x_j \geq x_n/2$, $j \leq n$, $m \geq 2$,

$$\sum_{j<n} \ell_j^2 x_j^{1-m} \leq \frac{2^{m-1} \ell_n^2 x_n^{1-m}}{L^2 - 1}. \tag{4.12}$$

The actual trajectory need not be monotone. Its bounded partial-sum error gives $x_j \geq x_n - 2C_s \geq x_n/2$ after the weak floor is fixed. The extra 2^{m-1} is harmless in an exponential-in-order constant. Thus the local classical coefficients with two smooth-source differences are summable without an extra $x^{-\epsilon}$. This is a statement about local coefficient norms. Summing vacuum densities over all blocks can still produce contact divergences. For separated sources, locality excludes those local terms and the remaining connected contributions have an exponential separation factor; for OS flat tests their coincidence behavior must be handled by the appropriate distribution bound.

With log radii, assume also $x_{n-r} \leq x_n + Cr$. Then for fixed $s \geq 0$ and $m \geq 2$,

$$\sum_{r \geq 1} L^{-2r} x_{n-r}^{1-m} (\log x_{n-r})^{sm} \leq C_s^m (m!)^s x_n^{1-m} (\log x_n)^{sm}. \tag{4.13}$$

For $x_n \geq e$, $\log(x_n+Cr) \leq C' \log(x_n)(1+r)$. Reduce to $\sum_r e^{-2r \log L}(1+r)^{sm}$, bounded by $C_s^m \Gamma(sm+1)$. Stirling's formula bounds $\Gamma(sm+1)$ by another $C_s^m (m!)^s$ (with polynomial factors absorbed). The resulting factorial growth still has a fixed finite exponent. This estimate is deliberately generous; fixed-order logarithmic gains can be bounded more sharply.

25.6 6. A separate regularity warning: a deep fine-coupling box is too rough

On the four-torus take the scalar energy

$$\mathcal{E}_{\epsilon h}(v) = \tfrac{1}{2} \int_{\mathbb{T}^4} (1 + \epsilon h(x)) |e_1 + \nabla v(x)|^2 \, dx, \qquad \int v = 0.$$

For real $|\epsilon| < 1$ and $\|h\|_\infty \leq 1$ it is uniformly elliptic. The derivative $w = \partial_\epsilon v|_0$ satisfies

$$-\Delta w = \partial_1 h, \qquad \partial_1 w(0) = \int_{\mathbb{T}^4} \partial_1^2 G(-y) h(y) \, dy. \tag{4.14}$$

Near zero the torus Green function has the singular part $\Gamma(y) = 1/(4\pi^2|y|^2)$. Hence

$$\partial_1^2 \Gamma(y) = \frac{4\theta_1^2 - 1}{2\pi^2 |y|^4}.$$

Choose a smooth annular cutoff supported in $\delta < |y| < r_0$, equal to one on $2\delta < |y| < r_0/2$, and multiply it by $(4\theta_1^2 - 1)/3$. Then $\|h_\delta\|_\infty \leq 1$ and

$$\partial_1 w_\delta(0) = c \log(1/\delta) + O(1), \qquad c > 0. \tag{4.15}$$

The angular integral is strictly positive; the smooth Green remainder and cutoff collars have bounded contributions. Each h_delta is smooth, but its smoothness constants grow as δ tends to zero.

Therefore a uniform holomorphic map from a fixed relative L-infinity coefficient ball into a bounded physical W^{1,infinity} minimizer ball cannot exist: Cauchy's derivative bound would contradict (4.15). The energy-norm analyticity in Section 2 is fully consistent with this failure.

This does not refute a fixed-L, current-step product domain. There the number of fine cells per block is fixed. It refutes silently promoting that statement to arbitrary independent coefficients at a deep fine resolution while keeping the same physical C^1 bound.

25.7 7. The concrete Dini repair for the linear response

The singular kernel in (4.14) has zero angular mean. If h has modulus of continuity $\omega_h(r)$, subtract h(0) before integrating its singular part. Then

$$|\partial_1 w(0)| \leq C\|h\|_\infty + C \int_0^{r_0} \frac{\omega_h(r)}{r} \, dr. \tag{4.16}$$

In particular C^α coefficients give a uniform bound $C(\|h\|_\infty + [h]_\alpha/\alpha)$, with no fine-cutoff logarithm. The same argument works for each component of the flat gradient response. It supplies a precise regularity norm for a common analytic source domain, rather than relying on arbitrary independent fine coefficients.

For the actual nonlinear Wilson construction, the local fixed-step chart and frozen-source density route may avoid demanding this deep domain altogether. That is the route being developed by the other pass-4 branches. Equations (4.1)–(4.16) delimit what such a theorem must preserve and what it need not claim.

25.8 8. Uniform Gaussian source domains survive arbitrary nested coarse maps

Let $H_0 > 0$ on a finite-dimensional real Hilbert space, extended to its complexification. Suppose

$$H = H_0^{1/2}(I+T)H_0^{1/2}, \qquad \|T\| \le \rho < 1.$$

Let Q be any fixed real surjective coarse map, complexified without changing its entries. Define the pushed-forward Gaussian precision

$$\mathsf{S}_Q(H) = (QH^{-1}Q^*)^{-1}, \qquad C_0 = QH_0^{-1}Q^*. \tag{4.17}$$

The expression exists and is holomorphic throughout this ball. In coarse coordinates normalized by C_0, it obeys

$$\left\| C_0^{1/2}\mathsf{S}_Q(H)C_0^{1/2} - I \right\| \le \frac{\rho}{1-\rho}, \qquad \Re\left(C_0^{1/2}\mathsf{S}_Q(H)C_0^{1/2}\right) \ge (1-\rho)I. \tag{4.18}$$

Proof. The map U=H_0^{-1/2}Q*C_0^{-1/2} is an isometry. Decompose the whitened fine space into ran U and its orthogonal complement, and write T in c/v blocks. The normalized coarse precision is exactly

$$I + T_{cc} - T_{cv}(I+T_{vv})^{-1}T_{vc}. \tag{4.19}$$

The norm bound follows from $\rho + \rho^2/(1-\rho) = \rho/(1-\rho)$. For its accretivity, let $v = -(I+T_{vv})^{-1}T_{vc}c$. The hidden row of $(I+T)(c,v)$ vanishes, so

$$\Re\langle c, \mathsf{S}'c\rangle = \Re\langle (c,v), (I+T)(c,v)\rangle \ge (1-\rho)(\|c\|^2 + \|v\|^2) \ge (1-\rho)\|c\|^2.$$

This also proves invertibility without restricting ρ to $1/2$. Everything is finite-dimensional and uniform in the choice or rank of Q.

For fixed successive maps Q_1, Q_2, exact covariance pushforward gives

$$\mathsf{S}_{Q_2}(\mathsf{S}_{Q_1}(H)) = \mathsf{S}_{Q_2Q_1}(H). \tag{4.20}$$

Consequently (4.18) holds after arbitrarily many Gaussian coarse steps by applying it once to the composite Q, rather than multiplying one-step bounds. The analogous statement holds for any finite number of nested maps.

For $H_\lambda = D^*\Lambda_\lambda D$ on a gauge-fixed positive subspace, $\lambda = x + h$ and $\|h\|_\infty \le \rho x$ give this hypothesis after normalizing by xD^*D. Therefore the full Gaussian precision family has a cutoff/depth-independent relative energy domain. Cauchy differentiation gives, with $S_0 = S_Q(D^*D)$,

$$\left\| S_0^{-1/2} D^m \mathsf{S}_Q(H_\lambda)|_{\lambda=x}[f_1,\dots,f_m] S_0^{-1/2} \right\| \le C_\rho^m m!\, x^{1-m} \prod_i \|f_i\|_\infty. \tag{4.21}$$

As in Section 1, no additional $x^{-\epsilon}$ holds in general. Retaining this Gaussian family exactly is a natural corrected state space for spatial source transport. The cubic and higher Wilson vertices, the non-Gaussian inherited interactions, and live histories still require their actual local estimates.

The distinction from Section 6 is essential: (4.18) is a form/energy statement and is true even when the map from rough coefficient perturbations to the pointwise gradient of the minimizing field has logarithmically large norm. It must not be reinterpreted as a uniform physical C^1 chart theorem.

25.10 10. The running Gaussian model has one exact nested-projection precision

There is an explicit representation behind the preceding induction. It makes the role of every scale and the constant-source cancellation transparent, and does not require the running x_j to be monotone.

Let $Q^{(j)} = Q_j \cdots Q_1$, put $Q^{(0)} = I$, and retain $H_j = S_{Q^{(j)}}(H_0)$. Define on the fine H_0-whitened space

$$P_j = H_0^{-1/2}(Q^{(j)})^* H_j Q^{(j)} H_0^{-1/2}, \qquad P_0 = I. \tag{4.29}$$

Each P_j is an orthogonal projection. Indeed its square reduces to itself using $H_j Q^{(j)} H_0^{-1}(Q^{(j)})^* H_j = H_j$. The ranges decrease with j because $(Q^{(j+1)})^* = (Q^{(j)})^* Q_{j+1}^*$. Therefore $P_i P_j = P_j P_i = P_{\max(i,j)}$.

Pulling every scalar counterterm in (4.26) back to the fine field gives

$$K_n = x_0 H_0 - \sum_{j=1}^{n} b_j (Q^{(j)})^* H_j Q^{(j)}. \tag{4.30}$$

This is an exact Gaussian identity, since a quadratic term depending only on an already retained coarse variable can be inserted before or after integrating the hidden fine variables. In whitened form,

$$\widetilde{K}_n = x_0(I - P_1) + x_1(P_1 - P_2) + \cdots + x_{n-1}(P_{n-1} - P_n) + x_n P_n. \tag{4.31}$$

All these identities are on the positive quotient, with both pure gauge and harmonic zero-energy modes removed or retained separately. The projection differences are mutually orthogonal. If every $x_j \geq h > 0$, then $K_n \geq hH_0$. A negative individual b_j causes no difficulty. The unconstrained fine covariance and the covariance with the final coarse variable fixed are, respectively,

$$\widetilde{K}_n^{-1} = \sum_{l=0}^{n-1} x_l^{-1}(P_l - P_{l+1}) + x_n^{-1} P_n,$$

$$G_{n,\mathrm{hid}} = H_0^{-1/2} \left[\sum_{l=0}^{n-1} x_l^{-1}(P_l - P_{l+1}) \right] H_0^{-1/2}. \tag{4.32}$$

In particular the hidden inverse has norm at most h^{-1} in the H_0 energy coordinates, uniformly in n. This weighted orthogonal sum, rather than a sum of n identical norm bounds, is the appropriate object to retain.

25.10.1 10.1 Exact all-order spatial-source response with running coefficients

Let $T_f = H_0^{-1/2} D^* \Lambda_f D H_0^{-1/2}$ on the positive gauge quotient, so $\|T_f\| \leq \|f\|_\infty$ and $T_{c\mathbf{1}} = cI$. Identify the final coarse energy space with $\operatorname{ran} P_n$ by the isometry J from Section 9, and put $V = \ker P_n$. Let $A_n = \widetilde{K}_n|_V$, so $A_n \geq hI$. For $T = T_f$ split it according to $\operatorname{ran} P_n$ plus V. The full source-dependent coarse precision, normalized by H_n, is exactly

$$x_n I + J^* T_f J - J^* T_f P_V \big(A_n + P_V T_f P_V\big)^{-1} P_V T_f J. \tag{4.33}$$

This formula follows either by one Schur complement of $K_n + E_0$ or by the repeated Gaussian integrations. It is valid on $\|f\|_\infty < h$, and includes all running counterterms in A_n.

Every homogeneous source f=c is therefore transported exactly as $x_n \mapsto x_n + c$, with no nonlinear correction. The reason is $P_V T_c J = 0$, not an estimate of an individual denominator. For a spatial source, the nonlinear terms have two endpoint factors $P_V T_f J$. These factors are unchanged if a constant is subtracted from f. Expanding the inverse in (4.33) gives the full mixed derivative formula. For $m \geq 2$ and arbitrary f1,...,fm,

$$D^m \widehat{S}_n(0)[f_1, \ldots, f_m] = (-1)^{m-1} \sum_{\sigma \in \mathfrak{S}_m} J^* T_{f_{\sigma(1)}} P_V A_n^{-1} \prod_{q=2}^{m-1} \left(P_V T_{f_{\sigma(q)}} P_V A_n^{-1} \right) P_V T_{f_{\sigma(m)}} J. \tag{4.34}$$

For m=2 the middle product is empty. The displayed product is ordered from left to right in increasing q. There are exactly m-1 hidden inverses. No factor one-half appears because this is the precision, rather than the classical energy evaluated at a fixed b.

Writing $o(f) = \inf_c \|f - c\|_\infty$, the operator norm obeys

$$\|D^m \widehat{S}_n(0)[f_1, \ldots, f_m]\| \leq 2(m-2)! h^{1-m} \sum_{i<j} o(f_i) o(f_j) \prod_{q \neq i,j} \|f_q\|_\infty. \tag{4.35}$$

To see the factor, choose the two unordered endpoint labels, order them in two ways, and permute the remaining m-2 labels. Multiplication by one-half and by a fixed coarse vector on the two sides recovers the energy version. The estimate is uniform in the number of scales and in their individual increments.

If all source variations are supported within one finite physical carrier of diameter ℓ and are compared with their values at one anchor, the endpoint oscillations are at most $C\ell\,\mathrm{Lip}(f_i)$, provided that the corresponding operators have actually been restricted/localized to that carrier. That last qualification is essential: (4.35) alone is a global oscillation estimate, and does not assert exponential localization of the fine-to-coarse projection J. A spatial kernel proof must perform that localization before replacing global o(f) by a local oscillation.

25.10.2 10.2 Why the scalar-counterterm result is stronger than a naive iteration

Suppose an estimate of each separate Schur step produces an apparent quadratic error bounded by Cr^2/x_j. Summing it without using the block/projection structure gives a logarithmically divergent upper bound along x_j of linear ultraviolet growth. Equations (4.31)–(4.35) show that this divergence is an artifact for the retained quadratic model. The hidden directions at the different steps are orthogonal channels of one fine precision. The true multi-step derivative has a single bounded hidden inverse $A_n\hat{}\{\text{-}1\}$, not n independently saturable denominators.

This observation does not erase vacuum source counterterms. The hidden determinant is a separate trace and may have dimension- dependent local divergences. Nor does it apply verbatim to source-dependent marginal subtractions, which add the W_j terms of Section 9.3. Those terms must be supplied by the interacting source induction.

26 The actual clean determinant and a real ultraviolet trajectory

This section proves a bounded cumulative one-loop coefficient for the genuine uniformly blocked, regular Wilson saddle, and then combines it with the published unmarked small-field induction to obtain a real weak-coupling trajectory. The finite determinant identities retain the nonlinear constraint, its multiplier, the gauge current, and the Haar coarea factors. The uniform estimates concern the clean, uniformly blocked class. They do not establish the observable transport through large-field histories or a continuum Yang–Mills theory.

We use the normalized Wilson action

$$A(U) = \sum_p \left(1 - \frac{1}{N} \Re \operatorname{Tr} U_p\right), \qquad x = g^{-2}, \qquad \exp[-xA(U)]. \tag{66}$$

Here g is the inverse square root of the coefficient of this normalized action. If that coefficient is instead written $2N/g_{\mathrm{phys}}^2$, then $x = 2N/g_{\mathrm{phys}}^2$. This distinction fixes the numerical factors below. Bałaban's definition of the Gaussian normalizer is in [4, (1.4)–(1.5)]; its derivation from the regular saddle is in [4, (2.10)–(2.12)]. We use those definitions, and give an independent finite-dimensional matching argument before estimating them.

26.1 The regular pushforward and exact associativity

Fix an odd block factor L, a depth k, a periodic coarse lattice of spacing one, and fine spacing $\eta = L^{-k}$. Fine forms and fine site fields have the physical pairing $\eta^4 \sum$, while coarse fields have unit volume weight. Transposes below refer to these pairings. A fixed invariant color inner product is understood. For operator estimates we divide the Wilson Hessian by its fixed representation factor so that its flat Maxwell precision is d^*d. Changing that fixed factor changes background-independent determinant constants and the fixed reference mass, but not the logarithmic coefficient expressed in (66).

Let M be the actual centered path-log/Federbush average, with its internal mean transports, and set $F_k = M^k$. On the regular branch consider the Haar pushforward

$$\rho_{k,\beta}(V) = \int \exp[-\beta A(U)]\, \delta_H(F_k(U), V)\, \mathrm{d}U. \tag{67}$$

The delta is defined relative to the displayed target Haar measure. A gauge-invariant cutoff equal to one near the regular critical orbit selects its local saddle contribution. At fixed depth and volume, finite-dimensional Laplace expansion gives

$$\rho_{k,\beta}^{\mathrm{reg}}(V) = c(\beta, k, \Lambda) e^{-\beta S_k(V)} a_k(V)(1 + O(\beta^{-1})), \qquad \Psi_k(V) = \log a_k(V) - \log a_k(1),$$

where S_k is the constrained minimum. No depth-uniform nonlinear saddle expansion is assumed in defining the finite Gaussian jet.

Lemma 26.1 (Associativity of the actual Gaussian prefactors)**.** *On compatible regular critical branches, if $p_k(V)$ is the intermediate critical field and Φ_k is the normalized logarithm of the next Gaussian prefactor, then*

$$\Psi_k = \Phi_k + \Psi_{k-1} \circ p_k. \tag{68}$$

The Hessian used at either stage is the constrained Lagrangian Hessian.

Proof. For composable smooth maps F, G, the exact distributional identity is

$$\delta_H(G(F(U)), V) = \int \delta_H(F(U), W) \delta_H(G(W), V)\, \mathrm{d}W.$$

Thus Haar pushforward is associative, with no extra Jacobian to insert: the target measures already specify each delta. Minimization is associative as well:

$$\min_{G(F(U))=V} A(U) = \min_{G(W)=V} \min_{F(U)=W} A(U).$$

Work first in a gauge-fixed chart where the regular minimum is unique. The finite implicit-function theorem identifies the nested and direct critical points. Applying Laplace expansion to the exact integral in stages evaluates the inner smooth amplitude at the outer critical point. Consequently $a_k(V) = c_k a_{\mathrm{step},k}(V) a_{k-1}(p_k(V))$ with c_k independent of V. Normalization at 1 proves (68).

For a nonlinear constraint $m(a) = 0$ and critical multiplier μ, the tangent quadratic form is the restriction of $D^2[A - \langle \mu, m \rangle]$. Its constraint-curvature term therefore appears in both orders of

expansion. Equivalently, taking the Schur complement of the complete saddle Hessian proves the same formula. The ordinary Wilson Hessian alone is insufficient. The regular charts exist near the identity at each fixed lattice: the flat average is onto, and the physical constrained Hessian has a positive gap after residual gauge fixing, as proved below. □

26.2 Straightening by an auxiliary coarse Haar gauge

The density (67) is invariant under coarse gauge transformations. Therefore it also equals

$$\int \exp[-\beta A(U)]\,\delta_H(F_k(U), V^h)\,\mathrm{d}U\,\mathrm{d}h, \tag{69}$$

where h is an independent coarse site gauge with normalized product Haar measure. At a reference field U_* with $F_k(U_*) = V$, the exact straightening identity is

$$F_k(e^{i\eta a}U_*) = (e^{iq_k(a)}V)^{v_k(a)}, \qquad q_k(0) = 0, \quad v_k(0) = 1. \tag{70}$$

To specify the actual map, write its one-step version in raw link angles:

$$u_W(a)(y) = \exp\left\{ L^{-4} \sum_{x \in B_L(y)} \log\left[T_{yx}(e^{ia}W) T_{yx}(W)^{-1} \right] \right\},$$
$$F_W(a)(c) = \frac{1}{i} \log\left[u_W(a)(c_-)^{-1} M(e^{ia}W)(c) u_W(a)(c_+) M(W)(c)^{-1} \right].$$

Here T is the internal symmetric mean transport of the specified path family. Set $W_r = M^r U_*$, $a_0 = \eta a$, and $a_{r+1} = F_{W_r}(a_r)$, and multiply the node factors in their path order. Equivariance proves (70) by induction. Both endpoint factors matter.

For fixed a, multiplication of h by $v_k(a)^{-1}$ on the appropriate side is a Haar translation. In the joint variables its Jacobian is triangular, with Haar-preserving h block, and is exactly one. This change in (69) absorbs the node factor without discarding a derivative or determinant of it.

Let $Q = Dq_k(0)$, and let D, D_c be the fine and coarse covariant gradients. Differentiating the actual node formula gives

$$QD = D_c R, \qquad (a, \sigma) \longmapsto Qa - D_c\sigma, \qquad \lambda \longmapsto (D\lambda, R\lambda), \tag{71}$$

where R is the transported site block mean. To check that this is the actual R, vary a fine gauge. The relative internal transport has derivative $\mathrm{Ad}(T_{yx})\lambda(x) - \lambda(y)$. Averaging and iterating the node product gives

$$Dv_k(0)[D\lambda] = \mathrm{sampling}(\lambda) - R\lambda.$$

Subtracting that displacement from the plain average's sampled gauge variation proves (71).

On a uniform k-block,

$$(R\lambda)(y) = L^{-4k} \sum_{x \in B^k(y)} \mathrm{Ad}(T_{yx})\lambda(x).$$

The blocks are disjoint. Its physical transpose extends a coarse field by inverse transport, so invariance of the color pairing gives the exact identity

$$RR^T = I, \qquad P = I - R^T R. \tag{72}$$

The algebraic identity continues to complex transports with the bilinear transpose; their Hermitian operator bounds are a separate estimate.

26.3 The ghost determinant and the complete constrained Hessian

Impose $\sigma = 0$ and $PD^T a = 0$. The linear Faddeev–Popov map is

$$\lambda \longmapsto (R\lambda, P\Delta\lambda), \qquad \Delta = D^T D.$$

Decompose the site space as $\operatorname{ran} R^T \oplus \ker R$. By (72), the diagonal blocks are I and $P\Delta P|_{\ker R}$. If

$$\mathbb{S}(T, Q) = \begin{pmatrix} T & Q^T \\ Q & 0 \end{pmatrix},$$

then, including the metric factors of the delta functions,

$$|\det \mathrm{FP}| = \det(P\Delta P|_{\ker R}) = |\det \mathbb{S}(\Delta, R)|. \tag{73}$$

There is no additional coarse orbit determinant: the coarse gauge variable was included in (69) and then fixed. Its R component also removes the constant stabilizer at the identity. Formula (73) is the Jacobian at the saddle. The ratio of the full nonlinear Faddeev–Popov Jacobian to this value remains in the non-Gaussian interaction.

Define

$$T_0 = (P\Delta P|_{\ker R})^{-1}, \qquad \lambda_a = T_0 P D^T a, \qquad \Pi a = a - D\lambda_a. \tag{74}$$

Then $R\lambda_a = 0$, $Q\Pi = Q$, and $PD^T\Pi = 0$. The joint critical equations give

$$J = Q^T \mu, \qquad D_c^T \mu = 0. \tag{75}$$

An invariant extension from the gauge slice to $\ker Q$ of the actual Lagrangian Hessian is

$$\mathcal{L}[a, a] = \langle \Pi a, H_W \Pi a\rangle - \langle \mu, D^2 q_k(0)[\Pi a, \Pi a]\rangle. \tag{76}$$

Changing an extension outside $\ker Q$ by terms involving Q does not change the saddle determinant: a triangular change of multiplier absorbs them.

Gauge invariance yields

$$\langle \Pi a, H_W \Pi a\rangle - \langle a, H_W a\rangle = -\langle J, E_U(a)\rangle. \tag{77}$$

Indeed take the fine gauge curve changing the velocity from a to Πa and differentiate the equality of the action values twice. The second coordinate acceleration is the covariant quadratic expression $E_U(a)$; a BCH expansion gives its usual commutator formula. Thus $j_U[J, \mu] = \mathcal{L} - H_W$ is linear in the two independent currents and contains both the gauge-coordinate and constraint-curvature terms.

The form $\mathcal{L}$ is constant along $D \ker R$, and Q is unchanged there. Replacing the hard residual gauge delta by its normalized Gaussian therefore only multiplies the integral by a background-independent Gaussian constant. In gauge coordinates both integrals have the same factor $\det(P\Delta P|_{\ker R})^{-1}$, which verifies this normalization directly. The added precision is DPD^T. Finite Gaussian integration now gives

$$\begin{aligned} \Psi_k(V) = & -\frac{1}{2} \log \left| \det \mathbb{S}(H_W + DPD^T + j_U[J, \mu], Q) \right| \\ & + \log |\det \mathbb{S}(\Delta, R)| - [\text{the same expression at } V = 1]\,. \end{aligned} \tag{78}$$

All operators are evaluated at the actual critical background $U_k(V)$. On the complex neighborhood use the coherent logarithms from the positive real branch. The site inverse in (74) exists at the identity by the block Poincaré inequality; fixed-lattice regular invertibility persists near it. Uniform current bounds on a whole background ball are not needed for the following coefficient cancellation.

26.4 Why the current terms do not change the magnetic coefficient

Let $b[\cdot]$ denote the quadratic F^2 coefficient, in the normalized action convention. Keep $z = (J, \mu)$ independent and put

$$\mathcal{F}_k(U, z) = -\frac{1}{2} \log \frac{\det \mathbb{S}(H_W + DPD^T + j_U[z], Q)}{\det \mathbb{S}(H_W + DPD^T, Q)}.$$

It vanishes for $z = 0$. Write its first current derivative as

$$D_z \mathcal{F}_k(U, 0)[j, \nu] = \langle w_J(U), j \rangle + \langle w_\mu(U), \nu \rangle.$$

Background gauge covariance makes each w an adjoint-valued covariant field at its fine or coarse endpoint. Global color invariance at $U = 1$, and the absence of an invariant vector in the simple adjoint representation, imply

$$w_J(1) = w_\mu(1) = 0, \qquad Dw(1)[d\lambda] = 0. \tag{79}$$

At fixed k the flat constrained inverses have a gap in coarse units. Their symbols and local variation vertices are analytic near real momentum, so the kernels decay exponentially. Cutoff approximation therefore extends the Ward identity to linear λ; the mixed current/background kernel annihilates constant fine forms.

The centered straight average reproduces affine potentials in midpoint coordinates: its positive weights have first moment at the coarse bond midpoint. The flat Wilson Euler operator annihilates affine potentials, as does Pd^T after the constant divergence is removed. The linearized constrained background consequently reproduces affine potentials up to gauge. For its Bloch response $H_k(p)$,

$$DJ(1)H_k(p) = O_k(|p|^2), \qquad D\mu(1)H_k(p) = O_k(|p|^2). \tag{80}$$

This follows by differentiating the finite Bloch saddle equations at $p = 0$, and holds in every alias. No uniform estimate on the second nonlinear minimizing response is asserted.

In the second derivative of $\mathcal{F}_k(U(V), z(V))$, terms with z'' vanish by (79). Two current derivatives have order $O_k(p^4)$; mixed terms have order $O_k(p)O_k(p^2)$. The pure background term is zero since $\mathcal{F}_k(U, 0) = 0$. If Ψ_k^0 is (78) with j omitted, then

$$D^2(\widehat{\Psi_k - \Psi_k^0})(p) = O_k(|p|^3), \qquad b[\Psi_k] = b[\Psi_k^0]. \tag{81}$$

The equality is exact at each depth; no uniform bound on the discarded cubic term is required.

Equivalently, take an affine constant-curvature jet. Its first current and multiplier are zero, so the correction starts at second order. Its quadratic trace is linear in the second-order adjoint current and vanishes by global color invariance. This is a bulk Bloch-jet calculation, or a calculation with each local trace centered before taking the volume limit. A sharp affine cutoff is not used, and no seam-current claim is made.

26.5 Actual all-depth averages on one physical potential ball

Now let $U = e^{i\eta A}$ be an arbitrary fine background with $\|A\|_{1,\infty} < h$ on a fixed small complex ball. Put $\xi_r = L^r \eta$. On one dependence cone, let α_r bound the raw link angle of $M^r U$ and ω_r its plaquette defect. The finite analytic one-step formulas, path products, and plaquette sliding give

$$\alpha_{r+1} \le L\alpha_r + C_L(\alpha_r^2 + \omega_r),$$
$$\omega_{r+1} \le L^2(1 + C_L\alpha_r)\omega_r + C_L\omega_r^2.$$

The linear curvature coefficient is exactly L^2: at the flat field the straightened average induces a positive average of the L by L plaquette surfaces. A comb gauge on the finite determining patch bounds the nonlinear errors by the displayed terms. Initially $\alpha_0 \le C\eta h$ and $\omega_0 \le C\eta^2(h+h^2)$. Divide by ξ_r and ξ_r^2, respectively. Since $\sum_{r<k} \xi_r \le (L-1)^{-1}$, discrete Gronwall and a first-exit argument give

$$\alpha_r \le C_L h \xi_r, \qquad \omega_r \le C_L h \xi_r^2. \tag{82}$$

These are bounds on the plain reference iterates $M^r U$ in the exact node identity, not a substitution of straightened iterates for them.

At step r, the derivative of the straightened map is the flat straight average plus a relative error $C_L h \xi_r$. In counting coordinates the flat form map has row sum L and column sum L^{-3}. Finite-patch Cauchy and the fixed number of rows influenced by a bond give the same relative error for both sums. Multiplication over the scales gives

$$\mathrm{row}(Da_k/Da_0) \le CL^k, \qquad \mathrm{col}(Da_k/Da_0) \le CL^{-3k}.$$

Physical input/output normalization and the Schur test yield

$$\|Q_k(A)\| \le C, \qquad \|Q_k(A) - Q_k(0)\| \le Ch. \tag{83}$$

The site transports have condition-number errors $O(\alpha_r)$; their sum is bounded by (82). This gives the corresponding bounds for R_k and $R_k - R_k(0)$, while $R_k R_k^T = I$ remains exact.

All these families are holomorphic on the same physical C^1 ball, so fixed background jets obey operator Cauchy bounds. Every output row and each of its jets depends only on a physical carrier of diameter at most C_L: the carrier reaches over the levels sum geometrically. This extra locality is needed for complex external momentum; a global unweighted field estimate alone would not suffice.

26.6 The projected-gauge precision has a uniform gap

Write $W_\eta = (-\Delta_\eta + 1)^{1/2}$. At the flat field set

$$\Xi_0 = d^*d + dPd^* + Q^TQ = \Delta_F - dR^TRd^* + Q^TQ. \tag{84}$$

Lemma 26.2 (Uniform physical energy comparison)**.** *Independently of depth and volume,*

$$cW_\eta^2 \le \Xi_0 \le CW_\eta^2, \qquad cW_\eta^2 \le \Delta + R^TR \le CW_\eta^2. \tag{85}$$

Proof. Use a coarse-cell Bloch fiber with $q \in [-\pi,\pi]^4$. Fine aliases have momenta $q + 2\pi n$; all nonzero aliases have physical Laplacian bounded below. Let r be the site-average row. Coisometry gives $\|r\| = 1$, and its unaliased multiplier satisfies $|r_0| \ge (2/\pi)^4$. The form average has norm at most one and diagonal unaliased multipliers of modulus at least $(2/\pi)^5$. It satisfies $Qd = d_c r$.

For a fine form v, put $w = d^*v$, $u = Pw$, and $\gamma = rw$. Then

$$w = u + r^*\gamma, \qquad |\gamma| \le |r_0|^{-1}(|d_0^* v_0| + \|u\|).$$

The flat Maxwell identity implies

$$\|\nabla v\| \le C(\|dv\| + \|Pd^*v\| + |q|\,|v_0|).$$

The unaliased lower bound of Q gives

$$|v_0| \le C(\|Qv\| + \|v_{\mathrm{high}}\|) \le C(\|Qv\| + \|\nabla v\|).$$

For sufficiently small $|q|$, the last gradient term is absorbed after multiplication by $|q|$. The high aliases and v_0 then also control $\|v\|$. This proves the lower bound near $q = 0$.

Away from zero the full flat Laplacian is invertible uniformly. Set

$$v_{\mathrm{err}} = \Delta_F^{-1}(d^* dv + du), \qquad M = r\Delta_{\mathrm{site}}^{-1} r^*.$$

Then

$$\begin{aligned} \|W v_{\mathrm{err}}\| &\le C(\|dv\| + \|u\|), \\ v &= v_{\mathrm{err}} + d\Delta_{\mathrm{site}}^{-1} r^* \gamma, \\ Qv &= Qv_{\mathrm{err}} + d_c M \gamma. \end{aligned}$$

The unaliased summand gives $M \ge |r_0|^2/\psi_0 \ge c > 0$, while $|d_c|$ is bounded below in this region. Thus γ and then $\|Wv\|$ are bounded by $\|dv\| + \|u\| + \|Qv\|$. The upper bound follows from P being a projection and $\|Q\| \le 1$. The site comparison follows from the block Poincaré inequality, or the same low/high alias split. □

Let $K = H_W + DD^T$ and

$$F = Q^T Q - DR^T RD^T, \qquad A_t = K + I + t(F - I), \quad 0 \le t \le 1. \tag{86}$$

The ordered Wilson plaquette variation estimate proved earlier gives on the common physical C^1 ball

$$\|(K(A) - K(0))W^{-1}\| + \|W^{-1}(K(A) - K(0))\| \le C(h + h^2).$$

The average bounds and covariant-gradient product rule give

$$\|W^{-1}(F(A) - F(0))W^{-1}\| \le Ch.$$

For small h, (85) and a Neumann inverse imply uniform bounds on $WA_t^{-1}W$. The analogous site interpolation from $\Delta + I$ to $\Delta + R^T R$ has the same property.

26.7 Physical Bloch shifts and background vertices

Inside a coarse cell factor out e^{iqx}. The flat fine difference is then

$$d_\mu(q) = \frac{e^{i\eta q_\mu} T_\mu - I}{\eta},$$

where T_μ is the periodic fine-cell shift. An imaginary shift changes this operator by $O(|\Im q|)$, uniformly in η. Placing the entire phase only on a boundary bond would introduce a spurious η^{-1}.

On complex momentum strips, transposes mean the analytic continuation of the physical bilinear transpose, whose symbol uses opposite momentum. They do not mean complex conjugation of a shifted symbol. On the real torus they agree with the Hermitian adjoints used for positivity.

The bounded physical carriers give weighted Schur bounds

$$\|Q(q + i\theta) - Q(q)\| + \|R(q + i\theta) - R(q)\| \le C|\theta|.$$

Together with the difference estimate and the real gap, this proves on a fixed strip

$$\|W(\Re q) A_t(q)^{-1} W(\Re q)\| \le C. \tag{87}$$

The massive and site inverses satisfy the same estimate. Crossing a coarse-zone boundary changes the alias basis by a unitary diagonal matrix and permutation; all shifted legs must use that change consistently.

An actual fine background wave $B_p(x) = e^{ipx}b$ sends a q fiber to a $q+p$ fiber. For bounded $\Re p$ and small $\Im p$, its first two precision vertices obey

$$\begin{aligned} \|W(\Re(q+p))^{-1}V_1(q;p)W(\Re q)^{-1}\| &\le C, \\ \|W(\Re(q+p_1+p_2))^{-1}V_2(q;p_1,p_2)W(\Re q)^{-1}\| &\le C. \end{aligned} \tag{88}$$

For K, use the ordered plaquette variation and the product rule; physical first differences of B_p stay bounded. For F, differentiate its displayed factorization. Each term factors through a coarse form or site space of fixed dimension and has the same energy bound.

To justify complex external waves for Q, R without applying global Cauchy to an exponentially growing field, divide into unit physical cells. For an output cell u, replace the wave on its dependence carrier by $e^{ip(x-u)}b\chi(x-u)$, where $\chi = 1$ on that carrier and has bounded physical first derivative. Its C^1 norm is uniform. Locality identifies the row-block derivative after removing e^{ipu}. Cauchy bounds that block, and a Schur sum over its fixed number of neighboring cells proves (88). Complex internal and external phases cost only a fixed factor on these carriers.

26.8 The stopped determinant remainder has a bounded quadratic coefficient

Put $G = (K+I)^{-1}$, $E = F - I$, and $A_t = K + I + tE$. Two substitutions in the exact resolvent identity and integration in t give

$$\log\det(K+F) = \log\det(K+I) - \operatorname{Tr} G - \frac{1}{2}\operatorname{Tr} G^2 + \mathcal{R}_k, \tag{89}$$

where the noncommutative remainder is

$$\begin{aligned} \mathcal{R}_k = {} & \operatorname{Tr}(FG) - \frac{1}{2}\operatorname{Tr}(FGFG) + \frac{1}{2}\operatorname{Tr}(FG^2 + GFG) \\ & + \int_0^1 t^2 \operatorname{Tr}(EGEGEA_t^{-1})\,\mathrm{d}t. \end{aligned} \tag{90}$$

The two cross terms combine by cyclicity, but their order in the derivation is retained. We estimate this representation after differentiation, not an unspecified numerical $O(1)$.

Proposition 26.3 (Uniform coefficient localization)**.** *The normalized Hessian symbol of $\mathcal{R}_k$ has a thermodynamic limit, is holomorphic for $|\Im p| < \delta$, and satisfies*

$$\sup_{|\Im p|<\delta} \|\widehat{D^2\mathcal{R}_k}(p)\| \le C, \qquad |b[\mathcal{R}_k]| \le C, \tag{91}$$

uniformly in k.

Proof. Differentiate (90) twice in waves $p, -p$. First expand each explicit E as $F - I$. Every term containing F or one of its first two derivatives factors through a coarse space of fixed dimension. For example,

$$F'[B] = Q'[B]^T Q + Q^T Q'[B] - (DR^T RD^T)'[B].$$

There are a bounded number of such factorizations. Insert the shifted energy weights at consecutive fine interfaces. Equations (87)–(88) bound each factor, and the cyclic trace can be taken on the fixed-dimensional coarse space.

For an inverse $H = A_t^{-1}$ or G,

$$H'[B] = -HA'[B]H,$$
$$H''[B,C] = HA'[B]HA'[C]H + HA'[C]HA'[B]H - HA''[B,C]H.$$

The intermediate shifted weights therefore give, for $m = 0, 1, 2$,

$$\|W_{\mathrm{out}} H^{[m]} W_{\mathrm{in}}\| \le C_m. \tag{92}$$

After the explicit expansion, the sole term with no average factor is

$$-\int_0^1 t^2 \operatorname{Tr}(GGA_t^{-1})\,\mathrm{d}t. \tag{93}$$

Its last inverse may still contain F, which is harmless. Retain its three inverse slots while differentiating. By (92), every slot or differentiated slot is $W_{\mathrm{out}}^{-1} C W_{\mathrm{in}}^{-1}$ with $\|C\|$ bounded. In a coarse Bloch cell,

$$\sup_q \operatorname{Tr} W(q)^{-6} \le C \sum_{n\in\mathbb{Z}^4} (1 + |q + 2\pi n|^2)^{-3} \le C. \tag{94}$$

Use the fine Brillouin representatives, on which the discrete sine symbol is comparable with physical momentum. The two exterior W^{-1} factors have bounded Schatten-6 norms, so each slot has bounded Schatten-3 norm. Schatten Hölder on the three slots bounds every second derivative of (93). One or two free propagators alone would not be uniformly trace class in four dimensions.

All factors are holomorphic on a common internal/external strip, and the bounds dominate the internal Bloch integral. With the simultaneous alias changes, the cyclic integrand is periodic in internal momentum. Torus sums therefore converge to the integral, uniformly on smaller strips. This proves the first claim in (91); momentum Cauchy gives the second. □

26.9 The coarse Schur determinants

Adding $Q^T Q$ to the upper-left block leaves a saddle determinant unchanged. Thus the current-free vector determinant factors, up to its fixed sign, as

$$\det \Xi \det M_v, \qquad \Xi = K + F, \quad M_v = Q\Xi^{-1}Q^T.$$

The site determinant has the analogous factors

$$\det(\Delta + R^T R) \det M_s, \qquad M_s = R(\Delta + R^T R)^{-1} R^T.$$

For $n = L^k$, the flat unaliased form multiplier is

$$q_\mu(p) = a_n(p_\mu)^2 \prod_{\nu\neq\mu} a_n(p_\nu), \qquad a_n(t) = \frac{\sin(t/2)}{n \sin(t/(2n))}.$$

Divide the isometric unaliased fine plane-wave embedding by this nonzero multiplier. It gives a fiberwise right inverse E_0 with $Q_0 E_0 = I$ and $\|WE_0\| \le C$. For sites omit the extra line factor. No global periodic choice of this one alias is needed. Cauchy–Schwarz with the energy comparison gives

$$cI \le M_v \le CI, \qquad cI \le M_s \le CI. \tag{95}$$

Small complex background perturbations and Bloch shifts preserve these inverses. Their first two vertices are products already bounded in (88) and (92). A second log determinant derivative is

$$\operatorname{Tr}(M^{-1}M'') - \operatorname{Tr}(M^{-1}M'M^{-1}M').$$

This trace is on a fixed-dimensional coarse cell. The same shifted-vertex argument gives a bounded holomorphic external Hessian symbol. In particular both coarse logarithmic determinants have uniformly bounded F^2 coefficient.

26.10 Affine jets preserve the coefficient under the actual minimizer

Let $H_k(p)$ be the linearized critical background in the residual gauge above, and let $E_{\mathrm{pw}}(p)$ embed a coarse wave as the same physical fine wave. Affine reproduction and the flat Euler equation imply at each fixed depth

$$H_k(p) - E_{\mathrm{pw}}(p) = \text{a fine gradient} + O_k(|p|^2). \tag{96}$$

Every normalized current-free determinant in the comparison is background gauge invariant. Its first derivative at 1 vanishes by color invariance; its Hessian annihilates gradients. Exponential decay at fixed depth extends this identity to constant fine forms. If $P_k(p)$ is its fine-cell Hessian, then $P_k(0)E_{\mathrm{pw}}(0) = 0$.

Insert (96) on the two legs of $H_k(-p)^T P_k(p) H_k(p)$. Gradient terms vanish, while a remaining $O_k(p^2)$ error meets $P_k(p)E_{\mathrm{pw}}(p) = O_k(p)$. It changes the expression only at order $O_k(p^3)$. A second nonlinear background response multiplies the zero first derivative. Consequently the quadratic coefficient equals the one read directly in the fine plane-wave background. This is a fixed-depth jet identity, not a depth-uniform nonlinear minimizer estimate. Under rescaling, the two form factors, two momentum factors, and four-dimensional volume factor cancel. Thus the coefficient is unchanged by each p_j in (68).

26.12 The actual fresh non-Gaussian remainder

We next import the unmarked real small-field induction of [4, 9]. In its normalized trace convention the actual recursion is

$$x_{j+1} = x_j - \beta_{j+1}, \qquad \beta_{j+1} = b^0_{j+1} + e_{j+1}. \tag{100}$$

The dependence on the entire previous coupling history is real: it is stated explicitly in the last paragraph on printed p. 298 of [4].

Freeze an admitted old state, including its age-indexed activities and already extracted marginal counterterms. The constrained classical minimum is independent of the overall current coefficient x_j. On scaling the next fluctuation by g_j, its fresh effective contribution is

$$\log Z^{(j)}(U) - \log Z^{(j)}(1) + \mathcal{F}_{j+1}(g_j, U) - \mathcal{F}_{j+1}(g_j, 1). \tag{101}$$

Here $Z^{(j)}$ is the actual Gaussian normalizer and $\mathcal{F}$ is the non-Gaussian logarithm of [4, (2.13)]. It contains the Wilson remainder, measure and constraint terms, and differences between old activities at fluctuating and central fields. The central old activities are carried separately and are not extracted again as fresh terms.

Let $\mathcal{B}$ be the bounded linear magnetic extraction of [4, (1.20)–(1.22)]. Then

$$b^0_{j+1} = \mathcal{B}(\log Z^{(j)}), \qquad e_{j+1} = \mathcal{B}(\mathcal{F}_{j+1}).$$

The paper uses one notation for the newly born non-Gaussian term in (1.3) and another in (1.6) which includes the Gaussian logarithm; these must be distinguished at zero current coupling.

For the fixed physical cutoff of [4, (2.9)], with the old state held fixed,

$$\mathcal{F}_{j+1}(0, U) = 0, \qquad e_{j+1}(0; \text{old}) = 0. \tag{102}$$

Every interaction term vanishes in that limit, the scaled cutoff tends to one, and the remaining normalized Gaussian integral is one.

The small real bound needed for existence. The age summation and exponentiated cluster expansion in [9, (2.41), printed p. 21] give newly born local terms bounded by

$$|\mathcal{F}_{j+1}(X;U,J)| \le C_3\varepsilon_1 e^{-\kappa d_{j+1}(X)}, \tag{103}$$

up to a fixed absolute factor. The Gaussian logarithm is added only afterward in the final paragraph there to recover the larger inductive E_0 allowance. Thus (103) bounds the remainder in (101), not the full one-loop term.

The constants defining C_3 around (2.38) of that paper depend on the fixed block/localization scales, field radii and old E_0 allowance, not on a positive lower bound for ε_1. Fix those constants, decrease ε_1, and then decrease the allowed coupling γ. The restrictions on these parameters are upper bounds.

The extraction $\mathcal{B}$ is bounded on this analytic local norm: differentiate twice in fixed-radius coarse background directions, use the uniform analytic minimal-background map of the imported construction, and sum the Hessian kernels with their second spatial moment. Polynomial support and moment factors are absorbed by a fixed loss in exponential rate. This is the mechanism of the uniform bound following (1.22) in [4]. Hence

$$|e_{j+1}| \le C_{\mathrm{ext}}C_3\varepsilon_1 =: \delta, \qquad \delta < c_0/2 \tag{104}$$

after choosing ε_1 small enough. The estimate is uniform over admitted old states, depth, and volume. The age sum is already included; it is not replaced by a sum of equally large errors over all old scales.

Current-parameter refinements. Immediately after (1.22), [4] states uniform bounds on all current-g derivatives on $[0,\gamma]$; Theorem 3 includes those assumptions, and [9] completes its proof. With the old state frozen, (102) and the first derivative bound give

$$|e_{j+1}| \le Cg_j. \tag{105}$$

This is not a derivative through the old state's dependence on the original bare coupling.

For the fixed physical cutoff there is also a parity refinement. All field coordinate and measure factors depend on the scaled fluctuation gB, while the classical coefficient is g^{-2}. The centered Gaussian is independent of g. The first coefficient of the fresh exponent is odd in B: it consists of classical cubic terms and first fluctuation variations of old and measure terms. Its Gaussian integral is zero. The cutoff contributes no derivative at $g = 0$; the excluded region starts at $|B| \ge c\varepsilon_1/g$, and each fixed derivative of its Gaussian tail is a power of g^{-1} times $e^{-c\varepsilon_1^2/g^2}$. Localized exponential majorants give the same conclusion for the local coefficient and background derivatives. Therefore

$$\partial_g e_{j+1}(0;\mathrm{old}) = 0, \qquad |e_{j+1}| \le Cg_j^2 = C/x_j, \tag{106}$$

using the uniform second-current-derivative bound. An even extension without this differentiability and odd-integral argument would not suffice.

These refinements are unnecessary for the existence argument below. The alternative analytic cutoff mentioned after (2.9) has a finite scaled Gaussian tail at $g = 0$; its background-dependent tail correction must be retained. Equation (102) cannot be copied to that alternative unchanged.

At each finite horizon the next coefficient is continuous on admitted real prefixes. This follows from the finite fluctuation integrals and, for the infinite-volume extraction, their uniform localized representation. The next integral exists whenever the previous couplings satisfy $0 < g_s \le \gamma$; the next output need not be known to be weak before its defining integral is performed. No complex historical product domain or bare-coupling tangent is being imported in this step.

26.13 A continuous shooting theorem with arbitrary coupling history

Theorem 26.5 (Real finite-horizon tuning)**.** *Let $b_1, b_2, \dots$ be fixed real numbers such that*

$$s_j = \sum_{\ell=1}^{j} (b_\ell - c_0), \qquad s_0 = 0, \qquad |s_j| \le C_s, \qquad c_0 > 0. \tag{107}$$

Suppose the actual forward recursion is defined on every admitted prefix $x_0, \dots, x_j \ge h$ and has the form

$$x_{j+1} = x_j - b_{j+1} - e_{j+1}(x_0, \dots, x_j), \qquad |e_{j+1}| \le \delta < c_0. \tag{108}$$

Assume the error is continuous along admitted prefixes, with any history state being the uniquely constructed actual state. For every horizon $n \ge 0$ and target $x_* \ge h + 2C_s$, there exists an actual admitted trajectory with $x_n = x_*$ and

$$\begin{aligned} x_* + (c_0 - \delta)(n - j) - 2C_s \le x_j \\ \le x_* + (c_0 + \delta)(n - j) + 2C_s, \qquad 0 \le j \le n. \end{aligned} \tag{109}$$

Proof. Set $y_j = x_j + s_j$. The recursion becomes

$$y_{j+1} = y_j - c_0 - e_{j+1}. \tag{110}$$

Use the interval

$$I = [x_* + s_n + n(c_0 - \delta),\ x_* + s_n + n(c_0 + \delta)]$$

to parameterize $x_0(t) = t$. We construct continuous auxiliary forward values on all of I. Suppose $x_0(t), \dots, x_j(t)$ have been constructed and agree with the actual recursion whenever that prefix is admitted. Its admitted set

$$D_j = \{t \in I : x_i(t) \ge h \text{ for every } 0 \le i \le j\}$$

is closed. On D_j the next actual error is continuous and lies in $[-\delta, \delta]$. Extend it continuously to I in the same interval: linearly interpolate on complementary components and continue constantly at an endpoint, using zero if D_j is empty. Define x_{j+1} by (108) with this extension. This produces continuous functions through step n. The extension is solely a shooting device, not a new physical RG definition.

For every t, the centered decrement is between $c_0 - \delta$ and $c_0 + \delta$. At the lower endpoint of I, $x_n(t) \le x_*$; at the upper endpoint, $x_n(t) \ge x_*$. The intermediate value theorem provides t_* with $x_n(t_*) = x_*$. Summing (110) backward gives

$$y_n + (c_0 - \delta)(n - j) \le y_j \le y_n + (c_0 + \delta)(n - j).$$

Since $y_n = x_* + s_n$ and $|s_n - s_j| \le 2C_s$, these are exactly (109). In particular every $x_j \ge h$. Thus t_* belongs to every D_j, and every error on the selected trajectory is the actual error. No auxiliary extension is used there. □

The proof assumes neither monotonicity of $t \mapsto x_n(t)$ nor monotonicity of the individual x_j. It proves existence, not uniqueness or differentiability of the tuning. A bounded cumulative Gaussian error is sufficient; individual increment convergence is unnecessary.

26.14 Application to the source-free Wilson construction

Take $b_\ell = b_\ell^0$ and c_0 from (99). For the most conservative direct application of the original published construction, take $G = \mathrm{SU}(2)$; the clean determinant calculation was written for $\mathrm{SU}(N)$. Fix $h \geq \gamma^{-2}$, choose the fresh fluctuation allowance so that (104) holds, and choose $x_* \geq h + 2C_s$. The published real induction is preserved whenever $x_j \geq h$. Theorem 26.5 therefore produces for each cutoff $\eta = L^{-n}$ a bare value $g_0(n) = x_0(n)^{-1/2}$ and an actual weak trajectory ending at $g_* = x_*^{-1/2}$. Every stage has $0 < g_j \leq \gamma$, and $x_0(n)$ grows linearly with n, so $g_0(n) \to 0$.

The refinements can be applied after the trajectory is known to stay in its admitted domain. Put $a = x_* - 2C_s > 0$ and $c = c_0 - \delta > 0$. Equation (109) gives $x_\ell \geq a + c(n - \ell)$. From (105),

$$\left| \sum_{\ell=j}^{n-1} e_{\ell+1} \right| \leq C \sum_{r=1}^{n-j} (a + cr)^{-1/2} \leq \frac{2C}{c} \left(\sqrt{a + c(n-j)} - \sqrt{a} \right). \tag{111}$$

Using (106) instead yields

$$|x_j - x_* - c_0(n-j) - (s_n - s_j)| \leq \frac{C}{c} \log \left(1 + \frac{c(n-j)}{a} \right). \tag{112}$$

Indeed the decreasing sum of $(a + cr)^{-1}$ is bounded by its integral over $[0, n - j]$. In particular, for a fixed terminal value,

$$x_0(n) = c_0 n + O(\log(n+1)). \tag{113}$$

Theorem 1 of [8] assumes that its actual effective couplings remain sufficiently small, and its introduction identifies the perturbative trajectory theorem as the numerical condition needed to remove that assumption. The argument here supplies a finite-horizon real tuning satisfying that condition, using the newly derived clean determinant bound and the published unmarked non-Gaussian induction. Consequently the unmarked effective-density representations and ultraviolet stability estimates of that theorem apply along these tuned cutoffs, subject to the new determinant argument above. Those published estimates have been imported, not re-proved here.

A bare-coupling tangent requires additional historical parameter control. Continuum existence further requires a correctly transported gauge-invariant observable family, noncollapse for that physical family, the required uniform distribution bounds, Euclidean covariance, and the relevant vacuum properties. Neither real tuning nor the Gaussian determinant identities establish those claims.

27.2 1. Two structural corrections that the construction must respect

27.2.1 1.1 The classical coefficient map is nonlinear

For two positive spring coefficients, eliminating their common interior variable gives the normalized coefficient

$$C(X_1, X_2) = \frac{2X_1X_2}{X_1 + X_2}.$$

At $(X_1, X_2) = (x, x)$, the second derivative in the direction $(1, -1)$ is $-2/x$. Thus a whole spatial-product theorem cannot in general have a classical remainder whose derivative is $O(x^{-1-\epsilon})$ uniformly on a fixed relative box. The actual Wilson constrained minimum has the same variational mechanism. We keep this classical response exactly.

27.2.2 1.2 A current-step product box is not a fine-cutoff product box

The local coordinate counts and constants below depend on a fixed block factor L. They do not prove that arbitrary fine-plaquette coefficient variations preserve a physical $W^{1,\infty}$ background norm as $\eta \downarrow 0$. A variable-coefficient elliptic response contains a Riesz transform, so that latter conclusion requires additional regularity of the coefficient directions. Smooth patch amplitudes or a Dini/Hölder source norm are possible domains; a bare fine-scale ℓ^∞ ball must not be silently substituted for them.

27.4 3. A volume-independent curl gap derived from local homotopy

Let $Q = Dq(0)$, and let

$$T = D_u\Phi(0,0)$$

be the free-coordinate injection into the tree-gauge subspace $\ker Q$. The key coercive estimate is

$$\|A\|_2^2 \le C_{\mathrm{hom}}(L)\big(\|dA\|_2^2 + \|QA\|_2^2\big) \tag{3.1}$$

for every fine one-form satisfying the internal tree gauge.

Here is a direct proof. For an internal non-tree bond, close it with the two tree paths to the block root. Its value equals the circulation of this closed loop, and a fixed plaquette filling inside the block expresses that circulation as a linear combination of dA in the block. For a face bond, compare it with

any other bond of the same face using the two internal tree paths. Their difference is again a plaquette circulation in the determining double block. Every path entering the coarse average is a face crossing plus internal path segments; those internal segments have already been controlled by the internal-loop formula. Thus the coarse average fixes the remaining common face value, modulo a bounded linear combination of local curls. Each fine bond is consequently a bounded linear combination of curls and coarse averages in a fixed determining neighborhood. Cauchy–Schwarz, followed by the finite overlap count, proves (3.1).

This argument also shows exactly why merely removing internal gauge modes would be insufficient: the coarse averages control the remaining common face values.

At the flat connection the Wilson quadratic form is the curl form. Hence its free-coordinate Hessian can be written, up to a fixed convention factor, as

$$H_0 = T^* d^* dT = \sum_p B_p, \qquad B_p = T^* d_p^* d_p T \geq 0. \tag{3.2}$$

The extraction of the free coordinates has norm at most one, so $\|u\| \leq \|Tu\|$. Combining this with (3.1) gives

$$m_L^2 I \leq H_0 \leq M_L I, \qquad m_L^2 = C_{\mathrm{hom}}(L)^{-1} > 0, \tag{3.3}$$

after the fixed action normalization is absorbed into the constants. The upper bound follows from the finite-range chart and finite incidence of plaquettes. Every B_p is supported on a bounded neighborhood, and H_0 has a fixed finite range R_L. These facts hold uniformly in volume.

27.5 4. Independent complex plaquette coefficients preserve ellipticity at this step

Put

$$\lambda_p = \beta(1 + h_p), \qquad \|h\|_\infty \leq \rho < 1,$$

and first consider the flat quadratic Hessian

$$H(h) = \sum_p (1 + h_p) B_p. \tag{4.1}$$

Let $\mathcal{D}u = (d_p T u)_p$, so $H_0 = \mathcal{D}^* \mathcal{D}$. The operator $\mathcal{D} H_0^{-1/2}$ is an isometry. Therefore

$$H(h) = H_0^{1/2} (I + E_h) H_0^{1/2}, \qquad \|E_h\| \leq \rho, \tag{4.2}$$

including complex independent h_p. In particular

$$\mathrm{Re}\langle u, H(h) u \rangle \geq (1 - \rho) m_L^2 \|u\|_2^2. \tag{4.3}$$

No derivative or smoothness of h is used. This statement concerns the fixed current step; it is not a fine-cutoff C^1 regularity assertion.

27.5.1 Exponential inverse kernels, with the locality proved

Let S_L bound both absolute row and column sums of $H(h)$ for $\rho \le 1/2$. Use the ambient current-lattice distance between coordinate anchors (the corresponding periodic distance on a torus), rather than an unspecified interaction-graph distance. For $W_q(u) = e^{\sigma d(u,q)}$, finite range gives

$$\|W_q H(h) W_q^{-1} - H(h)\|_{2\to 2} \le S_L(e^{\sigma R_L} - 1).$$

Choose, for example,

$$\sigma_L = R_L^{-1} \log\left(1 + \frac{m_L^2}{4S_L}\right).$$

Equations (4.3) and the resolvent identity imply

$$\|H(h)^{-1}(u,w)\| \le \frac{4}{m_L^2} e^{-\sigma_L d(u,w)}. \tag{4.4}$$

The same argument works on a torus using its ambient periodic distance; the conjugation need not be a linear coordinate weight. There are at most a fixed number of coordinate components at each ambient lattice anchor, so the number of coordinates in a shell of radius n is at most $C_L(n+1)^3$. Consequently

$$\sup_u \sum_w e^{\sigma_L d(u,w)/2} \|H(h)^{-1}(u,w)\| \le \frac{4C_L}{m_L^2} \sum_{n\ge 0} (n+1)^3 e^{-\sigma_L n/2} =: \mathcal{S}_L < \infty. \tag{4.5}$$

This derives the weighted row norm from actual Wilson coercivity and local coordinates. It is not an additional response hypothesis.

27.6 5. The actual small constrained minimizer is analytic and local

Define the current-step dimensionless weighted action

$$S_h(u,v) = \sum_p (1+h_p) a_p(e^{\Phi(v,u)}). \tag{5.1}$$

At $(u,v) = (0,0)$, its hidden gradient is zero and its hidden Hessian is (4.1). The action is a sum of bounded-range analytic terms, uniformly for $\|h\|_\infty \le \rho_0$, with any fixed $\rho_0 < 1/2$. Hence, on a sufficiently small sup box,

$$\|D_u S_h(0,v)\|_\infty \le C_L \|v\|_\infty,$$

and the nonlinear Hessian remainder has a bounded-range weighted row norm at most

$$C_L(\|u\|_\infty + \|v\|_\infty). \tag{5.2}$$

Use the fixed-point equation

$$u = -H(h)^{-1} D_u S_h(0,v) - H(h)^{-1}\{D_u S_h(u,v) - D_u S_h(0,v) - H(h)u\}. \tag{5.3}$$

By (4.5), there is an $\alpha_L > 0$, independent of volume, such that for $\|v\|_\infty \le \alpha \le \alpha_L$ this map contracts a ball $\|u\|_\infty \le C'_L \alpha$ with ratio at most $1/2$. It defines a unique analytic constrained critical point

$$u_*(h,v), \qquad \|u_*(h,v)\|_\infty \le C'_L \alpha, \tag{5.4}$$

on the full current-step coefficient product box. For real sufficiently small coefficients and data it is a strict local minimum. No claim of global uniqueness on the complete compact coarea fiber is needed here.

Let $\mathcal{H}(h,v) = D_u^2 S_h(u_*(h,v),v)$. Its difference from $H(h)$ has weighted row norm $O_L(\alpha)$. Decrease α_L so that this times $\mathcal{S}_L$ is at most $1/2$. A weighted Neumann series gives

$$\|\mathcal{H}(h,v)^{-1}\|_{\sigma_L/2} \le 2\mathcal{S}_L. \tag{5.5}$$

27.6.1 Derived coupling-derivative row bounds

Differentiating the actual critical-point equation gives

$$\partial_{h_p} u_* = -\mathcal{H}^{-1} D_u \big[a_p(e^{\Phi(v,u)})\big]_{u=u_*}. \tag{5.6}$$

The vector on the right is supported in a fixed neighborhood of p. It has magnitude $O_L(\alpha)$, because every plaquette is close to the identity and the Wilson action has a quadratic zero there. Equations (5.5)–(5.6), together with finite plaquette incidence, give

$$\sup_u \sum_p e^{c_L d(u,p)} \|\partial_{h_p} u_*(u)\| \le C_L \alpha. \tag{5.7}$$

Similarly the coarse-data derivative has a weighted row bound C_L. For additive coefficients λ_p, (5.7) has the extra factor β^{-1}. Thus the basic weighted source response is derived for this actual map.

Higher derivatives can be obtained by differentiating (5.6). A stronger absolute rooted-coefficient statement follows from the following explicit algebra, on a smaller fixed $\rho_*(L) > 0$.

27.6.2 A weighted coefficient algebra proves the rooted expansion

For finite-volume Taylor series $F_i(h,v) = \sum_\nu F_{i,\nu}(h,v)^\nu$, define

$$\|F\|_{\kappa;\rho,\alpha} = \sup_i \sum_\nu |F_{i,\nu}|\, \rho^{|\nu_h|} \alpha^{|\nu_v|} e^{\kappa \mathcal{T}(i,\operatorname{supp}\nu)}. \tag{5.8}$$

Here $\mathcal{T}$ is the length of a shortest connected graph joining the output anchor to all input anchors; finite internal Lie-algebra component sums are included in the absolute value. The constant term has tree length zero. The physical graph is the current graph, so its growth and incidence constants are fixed.

Products at one root are submultiplicative because a union of two rooted trees has length at most their sum. A product of child functions rooted within distance R_L of a new root costs at most $e^{\kappa R_L}$ per child. A linear kernel C with weighted row norm S_κ obeys

$$\|CF\|_{\kappa;\rho,\alpha} \le S_\kappa \|F\|_{\kappa;\rho,\alpha},$$

by adjoining the edge from the output to the old root. These are direct absolute coefficient estimates, not consequences of a scalar Cauchy norm.

Expand the local analytic critical equation at $(u, v, h) = (0, 0, 0)$, and invert only the fixed H_0. Every local Taylor coefficient is bounded by C_L^{m+1} at degree m, after its fixed finite number of coordinates and analytic radius are included. There is a term linear in v, a term bilinear in h, u, a term bilinear in h, v, and terms of degree at least two in (u, v), with or without one factor h. Consequently the absolute coefficient majorant satisfies, for $U = \|u\|_{\kappa;\rho,\alpha}$,

$$\|\mathcal{F}(u)\|_{\kappa;\rho,\alpha} \le S_\kappa \left[C_L\alpha + C_L\rho U + C_L\rho\alpha + \frac{C_L(U+\alpha)^2}{1 - C_L(U+\alpha)} \right]. \tag{5.9}$$

The coefficient of ρ in the last fraction can be absorbed into C_L for $\rho < 1$. The same power-series majorant bounds the difference of two inputs. Choose $\kappa < \sigma_L/2$, then $\rho_*(L)$ so $S_\kappa C_L \rho_* < 1/8$, and finally α small. The map contracts a ball $U \le C'_L\alpha$ with ratio at most $1/2$. Completeness of the absolute coefficient space gives a convergent series satisfying the critical equation; uniqueness in (5.3) identifies it with u_*.

Thus every coefficient has an actual connected rooted-tree representation and a summable absolute weighted majorant. At a vertex its children are the local derivatives of the Wilson chart/action, and inverse-Hessian entries supply its edges. Fixed-order derivative row bounds follow by using a smaller coefficient radius and differentiating the absolutely convergent series. This construction derives the spatial kernel statement rather than assuming it.

For later polymer counting, one may group current cells into fixed boxes of side M. Choosing M large makes the fixed exponential decay rate in grouped graph units exceed a prescribed entropy constant. The constants and small radii then depend on L, M, still not on volume or ultraviolet depth counted as a sequence of fixed steps.

27.9 8. A smaller relative coupling domain that fits frozen regular tests

Let $g = \beta^{-1/2}$, and write the regular background amplitude as

$$\alpha = gP_{\mathrm{bg}},$$

where P_{bg} may be a fixed power of $\log \beta$. Fix the chart, the source-zero Wilson saddle $u_0(v) = u_*(0, v)$, the reference Gaussian, and every fluctuation cutoff. Put $u = u_0 + gb$, with $\|b\|_\infty \le P$, and retain a fixed larger complex fluctuation box.

The perturbed Wilson exponent has the exact form

$$\beta\sum_p(1+h_p)a_p(u_0+gb,v)=C(h,v)+\ell_h(v)\cdot b+\tfrac12 b^{\mathsf T}H_h(v)b+\mathcal R_h(b,v), \tag{8.1}$$

where $\ell_0=0$ and the **constrained** Hessian is used. Local finite-patch Taylor estimates give

$$\begin{aligned}\|\ell_h\|_{\mathrm{local}}&\le C_L\rho P_{\mathrm{bg}},\\ \|H_h-H_0(v)\|_{\mathrm{local}}&\le C_L\rho,\\ |\mathcal R_h(Y;b,v)|&\le C_LgP\sum_{b'\subset Y^+}|b_{b'}|^2.\end{aligned} \tag{8.2}$$

The last bound is uniform for $\rho\le\rho_0$ and gP below the fixed chart radius. The coarea-density fluctuation contributes an additional local analytic potential of growth norm $O_L(g)$. No coefficient derivative of a cutoff is taken.

In the ordinary quadratic-growth norm

$$|V(Y;b)|\le\|V\|\,e^{-\kappa d(Y)}\left(1+M^{-4}\sum_{b\subset Y}|b_b|^2\right),$$

the linear term costs $C_L\rho P_{\mathrm{bg}}$, not its cutoff supremum $C_L\rho P_{\mathrm{bg}}P$: use $|b|\le(1+|b|^2)/2$. The quadratic term costs $C_L\rho$. Therefore choosing

$$\boxed{\rho_\beta=\frac{c_L}{1+P_{\mathrm{bg}}},\qquad r_\beta=\beta\rho_\beta=\frac{c_L\beta}{1+P_{\mathrm{bg}}}} \tag{8.3}$$

makes their total growth load smaller than any prescribed fixed generic allowance by taking c_L small. The remaining Wilson interaction has a genuine power gain, $O_L(gP)$.

The same radius has a direct saddle interpretation. From (5.7), the change of the critical point is $O_L(\rho\alpha)$; in canonical fluctuation units it is $O_L(\rho P_{\mathrm{bg}})$. Under (8.3) it is bounded by a fixed small constant, leaving room in the frozen fluctuation box. A fixed relative radius need not have this property when P_{bg} grows.

If $P_{\mathrm{bg}}=O((\log\beta)^q)$, the additive radius is still β divided by a logarithmic factor. Cauchy in that radius converts a power-small nonlinear remainder $O(\beta^{-p}\log^a\beta)$ into a source derivative

$$O(\beta^{-1-p}\log^{a+q}\beta),$$

which is summable along the logarithmic ultraviolet coupling trajectory. All fixed source orders retain a coupling power β^{-m} with additional logarithms.

27.10 9. Exact Gaussian completion and cutoff tails on the smaller box

Keep $C(h,v)$, ℓ_h, and H_h from (8.1) exactly. The full Gaussian integral is

$$(\det H_h)^{-1/2}\exp\left(\tfrac12\ell_h^{\mathsf T}H_h^{-1}\ell_h\right)$$

times its fixed numerical normalization. Its covariance and mean are analytic on (8.3). Equations (4.4)–(5.5) bound the covariance kernels, and

$$\|m_h\|_\infty = \|H_h^{-1}\ell_h\|_\infty \le C_L \rho P_{\mathrm{bg}}$$

is uniformly small. This keeps both the nonlinear classical open chains and the Gaussian closed cycles explicit.

No contour shift through a hard cutoff is required. A normalized complex Gaussian with a uniformly accretive precision and a bounded complex mean is dominated in absolute value, on any local coordinate set of size n, by a fixed real Gaussian with a slightly weakened precision times $e^{C_L n}$. Reserving an additional fixed quadratic fraction on v distinct failed-cutoff coordinates gives

$$e^{C_L n} e^{-c_L P^2 v}. \tag{9.1}$$

Thus the ordinary failed-cutoff expansion still has an exponentially small activity in P^2. The extensive local prefactor is part of the usual polymer estimate; it is not a uniform bound on the total variation of an infinite-volume complex measure. The companion history calculation is spelling out this local Gaussian comparison.

The natural residual interaction size after the exact classical/Gaussian extraction is therefore

$$\varepsilon_{\mathrm{new}} = C_L\{gP + \varepsilon_{\mathrm{old}} + e^{-c_L P^2}\}, \tag{9.2}$$

once the generic localized Gaussian/Mayer map is applied to the actual old analytic activities. Establishing that map and its domains is the next section to be matched with the companion note; (9.2) must not be promoted to an irrelevant contraction merely because its right side is small.

27.12 11. A geometric age estimate with an explicit full tensor extraction

The original small-field construction does not use a Markov contraction of one abstract irrelevant coordinate. Its old local terms are indexed by their birth scales. The source-free result in CMP 109 (1987), (0.29)–(0.30), is an age factor strictly better than physical dimension four; its Section 3 obtains the analytic Taylor remainder and its Section 4 performs Ward extraction. The following local argument makes the corresponding tensor-valued extraction transparent on clean contractible domains. It also explains exactly which steps are linear in a marked activity.

27.12.1 11.1 Birth-domain assumptions and actual regular readouts

Let $E_X(U,J)$ be an analytic gauge-invariant local activity at a birth scale, with its required collar included in the connected birth domain X. Subtract its deterministic flat vacuum value. Suppose it has a common analytic extension in the birth-scale coarse link coordinates and the independent current coordinates, with comparable radii R_j, and

$$\sup \|E_X\|_{\mathcal{A}_N} \le B_j e^{-\kappa|X|}. \tag{11.1}$$

This is the actual incoming coefficient-algebra norm, not the norm of the scalar coefficient alone. The analytic extension can be supplied by the original coarse-coordinate domain lemma and frozen nonlinear wall localization. No repeated use of the crude fixed-L pivot constants is involved in the age comparison.

Let $r = L^{j-k} \le 1$ be the birth spacing in the present scale's units. On the physical regular background being read out, assume the ordinary source-free minimizer estimates give, after transport to a common local frame,

$$\begin{aligned} \log U_p &= r^2 F(x_p) + O(r^{2+\zeta} A_k), \\ \|F(x) - F(y)\| &\le C A_k |x-y|^{\zeta}, \qquad \|F\| \le C A_k, \\ \|J_b^{\mathrm{own}}\| &\le C r^{2+\zeta} A_k, \end{aligned} \tag{11.2}$$

for a fixed $0 < \zeta \le 1$. The notation for the first error includes the ordinary local plaquette-to-curvature approximation in these units. One may equivalently formulate (11.2) directly using transported plaquette logarithms and their Hölder differences. The last line concerns the normalized OWN constrained current and uses the actual current response, not just a difference of arbitrary fine plaquettes. Its linear response annihilates affine constant-curvature data. Apply the localized analytic current derivatives to the old coarse datum $r^2 \mathcal{A} F + O(r^{2+\zeta} A_k)$: the first term has zero linear current, the error gives $O(r^{2+\zeta} A_k)$, and the nonlinear terms give $O(r^4 A_k^2)$, with the ordinary remote-window tail. Equivalently use the actual multiplier identity $J = Q_U^{\top} \mu$, whose coarse-data response has the same locality and zero affine jet. A mere $C^{1,\zeta}$ bound on an arbitrary fine potential would not justify this normalized-current estimate; it can lose a fine-mesh power. These are source-free actual inner-window field/current readouts, not conditions imposed on an arbitrary independent outer J. The latter uses the original reduced-current extension or the separate outer/removed-window decay. The marked observable does not cause the background to be re-minimized.

27.12.2 11.2 Tree gauge converts regularity into the engineering factor

On a contractible birth patch of diameter D, choose a tree gauge rooted at an anchor y. A non-tree link becomes the holonomy of a loop closed along tree paths. A fixed plaquette filling of that loop gives

$$a_b := \log U_b^{\mathrm{tree}} = r^2 \sum_{P=1}^{6} \mathcal{A}_b^P(X) F_P(y) + \mathcal{E}_b, \tag{11.3}$$

where $|\mathcal{A}_b^P| \le CD^2$ and

$$\|\mathcal{E}\|_\infty \le C \left[r^{2+\zeta} D^{2+\zeta} A_k + r^4 D^4 A_k^2 \right]. \tag{11.4}$$

To verify this, peel the loop across its chosen plaquette filling. The loop is an ordered product of transported plaquette holonomies. The linear sum of their logarithms yields the oriented areas in (11.3). Replacing their transported curvatures by $F(y)$ costs their number times the Hölder variation in (11.2). The difference between the logarithm of the ordered product and the sum costs the square of the sum of the plaquette sizes, giving the second term in (11.4). Every surface and path stays within the fixed collar. Complex gauge transforms are permitted only while their condition numbers and the analytic-domain margins remain bounded; on the small patches used below this follows from the same small regular fields and the common coarse-coordinate chart. Large patches are handled by their exponential locality tail.

Because the gauge group is semisimple, simultaneous constant color conjugation has no invariant linear functional on its Lie algebra. Gauge invariance therefore implies

$$DE_X(\mathbf{1}, 0) = 0. \tag{11.5}$$

This applies jointly to link and current coordinates and coefficientwise in $\mathcal{A}_N$. Consequently the leading Taylor term is quadratic. Inserting (11.3) shows that its engineering order is r^4, without using Euclidean symmetry to make its tensor scalar.

27.12.3 11.3 The complete marginal tensor

Let $\mathcal{A}_X F$ denote the link vector in the leading term of (11.3). Define

$$Q_X(F) = \tfrac{1}{2} D_a^2 E_X(0, 0)[\mathcal{A}_X F, \mathcal{A}_X F]. \tag{11.6}$$

For $SU(2)$, constant color invariance gives

$$Q_X(F) = \sum_{P,Q=1}^{6} c_X^{PQ} \langle F_P, F_Q \rangle, \qquad c_X^{PQ} = c_X^{QP}. \tag{11.7}$$

Thus the full type space has at most 21 curvature-bilinear components. A scalar-only coefficient is justified only after the appropriate Euclidean symmetries have been used. The coefficient map $E_X \mapsto c_X$ is linear and bounded by

$$\|c_X\|_{\mathcal{A}_N} \le CB_j R_j^{-2} D^4 e^{-\kappa|X|}. \tag{11.8}$$

A local gauge-invariant lattice completion is explicit. Transport the six oriented plaquettes at the anchor to that anchor, put

$$C_P(U) = \tfrac{1}{2}(U_P - U_P^{-1}),$$

and use the invariant bilinears $-\frac{1}{2}\operatorname{Tr}(C_P C_Q)$, with the fixed Lie-algebra normalization absorbed into c_X^{PQ}. They have leading value $r^4\langle F_P, F_Q\rangle$. This defines a linear local counterterm $\mathcal{M}_X[E_X]$, including all tensor types. In the Euclidean scalar sector it agrees with the usual scalar Wilson counterterm up to a higher-order local term. Symmetrizing the anchor/tree convention over the finite lattice symmetry group makes this extraction covariant without changing any estimate except its fixed constant.

27.12.4 11.4 Quantitative remainder with the true radius ratio

Set $\rho_{j,k} = A_k/R_j$. Taylor's formula, (11.3)–(11.5), and the joint current bound in (11.2) give

$$\boxed{\|E_X - \mathcal{M}_X[E_X]\|_{\mathcal{A}_N} \le CB_j e^{-\kappa'|X|}\left[\rho_{j,k}^2 r^{4+\zeta} + \rho_{j,k}^3 r^6\right], \qquad \kappa' < \kappa.} \tag{11.9}$$

Here are the terms in that estimate. The quadratic link term differs from (11.6) by a factor r^ζ through (11.4). A link-own-current quadratic term has order $r^{4+\zeta}$, and a current-current term has order $r^{4+2\zeta}$, both covered by $r^{4+\zeta}$. The Taylor terms of degree at least three have order $(\rho_{j,k}r^2)^3$. Polynomial powers of D are paid from the spare exponential weight. No claim that a J^2 polynomial is irrelevant on an unscaled arbitrary pair ball is used.

For contractible patches too large to remain in the local small chart, use the original activity bound and its locality tail. The threshold is of order $(r\sqrt{\rho_{j,k}})^{-1}$; an exponential beyond that threshold is bounded by an arbitrary power of $r\sqrt{\rho_{j,k}}$, in particular its sixth power. This pays precisely the second term in (11.9), again after spending spare locality. If $\rho_{j,k}r^2$ is already of order one or larger, that second term dominates the unexpanded bound. Nonwrapping clean patches therefore have a uniform estimate. A wrapping domain has the separate bound $CB_j e^{-c/r}$, with no factor proportional to the actual curvature. It can be absorbed into (11.9) on an ordinary running-radius system for which $\rho_{j,k} \ge c_0 > 0$, but not uniformly as $\rho_{j,k} \downarrow 0$: a flat torus field can have nontrivial holonomy. A small domain encircling a wild hole has the same obstruction and belongs to the marked boundary class, not to (11.9).

The ratio $\rho_{j,k}$ is not replaced by $\max(1, \rho_{j,k})$. Preserving its square is what retains the small field/source prefactor.

27.12.5 11.5 The source coefficient algebra does not change the proof

The tree-gauge substitution and geometric readout are source independent. Taylor expansion, the color Ward identity (11.5), and the extraction (11.6) are linear in E_X. Every estimate above uses only triangle inequalities and Banach-valued Cauchy bounds. Thus (11.9) holds in the finite source coefficient norm with the **actual incoming** B_j, including mixed source coefficients. This is a proved coefficientwise age operation; it is not inferred from a scalar unmarked value.

27.17 16. Replacing the vacuum sum by strictly local contact counterterms

The quasi-local vacuum functional $\mathcal{E}_K$ is not itself a strictly local counterterm: it has finite correlations between separated supports. A source Taylor subtraction makes the distinction precise.

Fix $0 < \alpha < 1$, and work on a sufficiently small fixed C^5 source ball. Choose a smooth real reflection-invariant cutoff χ_0, equal to one on a fixed physical ball of radius a_0 and zero outside twice that ball. The radius is smaller than a fixed fraction of every relevant torus period. At a birth root y define the bounded extension of the fourth spatial Taylor jet

$$(\mathcal{P}_y J)(x) = \chi_0(x-y) \sum_{|\gamma|\le 4} \frac{\partial^\gamma J(y)}{\gamma!}(x-y)^\gamma. \tag{16.1}$$

This is linear in the jet. Its global C^α norm is at most $C_{a_0}\|J\|_{C^4}$. Choose the C^5 ball so that every $\mathcal{P}_y J$, and every profile used below, lies well inside the genuine source domain of Section 14.

Assign each scalar carrier a fixed root $y = y(j, X)$, using a covariant pointing convention, and define

$$\mathcal{L}_K[J] = \sum_{j\le K} \sum_X \nu_j(X; \mathcal{P}_{y(j,X)} J). \tag{16.2}$$

Every summand depends only on the derivatives of J through order four AT ONE POINT. It is a holomorphic function of that finite jet and vanishes at $J = 0$. Thus $\mathcal{L}_K$ is a strictly local, finite-cutoff contact functional. Its coefficients may diverge with the cutoff; that is not a claim about the remainder.

27.17.1 16.1 The local profile comparison uses actual admissible sources

Let a carrier and its collar lie in a physical ball of radius $\rho = C\ell_j(1 + \operatorname{diam}_j X)$ around its root. If $\rho \le a_0/4$, choose a smooth cutoff χ_ρ, equal to one on that carrier and supported in its 2ρ ball, and put

$$\delta J_X(x) = \chi_\rho(x - y)\big[J(x) - T_y^4 J(x)\big].$$

Taylor's integral remainder and one derivative give

$$\|\delta J_X\|_\infty \le C\rho^5 \|J\|_{C^5}, \qquad [\delta J_X]_{C^\alpha} \le C\rho^{5-\alpha}\|J\|_{C^5}. \tag{16.3}$$

For the second inequality use the derivative bound $C\rho^4\|J\|_{C^5}$ at distances at most ρ, and the supremum bound at larger distances. The derivative of the cutoff has exactly the same size.

On the carrier, $\mathcal{P}_y J + \delta J_X = J$. Exact source locality therefore gives

$$\nu_j(X; J) = \nu_j(X; \mathcal{P}_y J + \delta J_X).$$

Both arguments lie in the ORIGINAL global C^α source ball. A one-variable Cauchy estimate in the direction δJ_X, with the fixed spare source radius and the rooted bound (15.1), proves

$$|\nu_j(X; J) - \nu_j(X; \mathcal{P}_y J)| \le Ce^{-\kappa' d_j(X)}\rho^{5-\alpha}\|J\|_{C^5}. \tag{16.4}$$

No unsupported Lipschitz estimate on an arbitrary restricted-profile space is being assumed. The displayed cutoff gives an actual small admissible global direction, and the equality follows from actual carrier locality.

For $\rho > a_0/4$, the tree length is at least c/ℓ_j. Bound both terms by (15.1) and retain the resulting e^{-c/ℓ_j} tail. This includes wrapping carriers.

27.17.2 16.2 Summing the remainder and preserving reflection positivity

Sum the polynomial diameter factor in (16.4) using spare exponential decay, and then count the physical roots. A birth contributes at most

$$CN_K\|J\|_{C^5}\left[\ell_j^{1-\alpha} + \ell_j^{-4}e^{-c/\ell_j}\right].$$

Both scale sums converge because $\alpha < 1$. Hence

$$\boxed{|\mathcal{E}_K[J] - \mathcal{L}_K[J]| \le CN_K\|J\|_{C^5},} \tag{16.5}$$

uniformly on a fixed smaller C^5 source ball, in the cutoff and in the number of scales. The same estimates give normal convergence of any compatible remainder series and uniform derivative bounds by Cauchy; they do not identify varying cutoff trajectories without a separate compactness argument.

Set

$$\mathcal{Z}_K^{\mathrm{ren}}[J] = e^{-\mathcal{L}_K[J]}Z_K[J]/Z_K[0].$$

Combining (15.10) and (16.5) gives

$$|\mathcal{Z}_K^{\mathrm{ren}}[J]| \le e^{CN_K} \tag{16.6}$$

on that genuine C^5 source ball. This is a finite-volume generating-functional bound after LOCAL contact counterterms, not after subtracting the nonlocal part of the vacuum.

For separated source supports, every mixed derivative of $\mathcal{L}_K$ that uses those supports vanishes: at a given root their jets cannot both be nonzero. Consequently (15.6) remains the exact separated derivative of $\mathcal{E}_K - \mathcal{L}_K$.

Choose the original reflection-covariant localizations and a reflection-covariant pointing (or average their allowed finite choices before the scalar filing). With real cutoff χ_0, reality and reflection covariance give $\mathcal{L}_K[\theta\bar{J}] = \overline{\mathcal{L}_K[J]}$. For sources supported strictly in the positive half-space, the two reflected supports are disjoint, so strict locality gives

$$\mathcal{L}_K[\theta\bar{J}_a + J_b] = \overline{\mathcal{L}_K[J_a]} + \mathcal{L}_K[J_b]. \tag{16.7}$$

Thus the matrix of renormalized reflection-positive exponential moments is obtained from the physical source matrix by the diagonal congruence

$$M_{ab}^{\mathrm{ren}} = e^{-\overline{\mathcal{L}_K[J_a]}}\, M_{ab}\, e^{-\mathcal{L}_K[J_b]}. \tag{16.8}$$

Its positivity is preserved. This does not identify the retained reference field with a physical pushforward: the positivity comes from the original physical functional in (15.9), whose insertions were transported exactly. Nor does it claim ordinary stochastic positivity for an arbitrary nonlinear source counterterm; the conclusion here is reflection positivity and its differentiated positive-time test identities.

27.19 18. Rooted clean-window transport and the retained two-source terms

This section controls the terms left after a FINITE clean window, rather than dropping them. It uses the fixed-core pure-clean coefficient class of Section 17 and a pair of source supports of diameter at most Cd, separated by at least d. Let the window run from scale $a' \ll d$ to scale $H \gg d$. Only the actual current-scale cell kernels and their connected carrier bounds are used.

27.19.1 18.1 A single root retains its engineering factor

Fix one source atom at one scale-a' cell. Its vacuum-subtracted analytic coefficient has zero first pair jet. If a member born at scale ℓ_t is read at scale ℓ_s, the tree-gauge/Taylor estimate of Section 14 gives

$$CR_0^2 a_t (1+s-t)^q L^{-4(s-t)}. \tag{18.1}$$

There is no translation count $L^{4(s-t)}$: the original source root is fixed. The original fluctuation-difference Cauchy estimate adds a small factor $\theta \le C\varepsilon_1/R_0$ for each newly formed descendant. Connected localization sums the unmarked branches and preserves the original root. The same estimate holds after multiplication by the final-scale weight $\exp[\kappa\,\mathrm{dist}(\cdot,y)/\ell_s]$, because every carrier contains a path from the retained root to its output. The constant is independent of the window length.

Choose any fixed $\delta \in (0,1)$ and set $\lambda = 4-\delta$. With $w_m = C(1+m)^q L^{-4m}$, the weighted convolution norm

$$W_\lambda = \sum_{m\ge 0} L^{\lambda m} w_m$$

is finite. Make the original fixed cutoff small enough that $\theta W_\lambda < 1/2$. The complete one-root readout kernel is bounded by $w + \theta w * w + \theta^2 w * w * w + \cdots$. Its weighted norm is at most $W_\lambda/(1-\theta W_\lambda)$. Hence

$$\|\text{one-root readout at } \ell_s\|_{\mathrm{loc,u}} \le CR_0^2 a_r (a'/\ell_s)^\lambda e^{-c\,\mathrm{dist}(u,y)/\ell_s}. \tag{18.2}$$

An old second-source coefficient has the additional factor g_t^2; its readout and new births satisfy the same estimate after division by g_s^2, because the running tube gives $g_t^2/g_s^2 \le C$. This constant is included in the once-chosen small θ. No assumption of a scale-by-scale monotone coupling is required.

The δ loss in (18.2) is used only over this finite window. Starting that estimate at the bare mesh and then allowing its length to go to infinity would give a false cutoff-uniform bound. The state at a' is supplied by the all-depth spatial induction of Sections 14 and 17, in units of a', before this one-root window estimate is applied.

27.19.2 18.2 First-source terminal bound without the terminal volume

At scale a', localize the already transported first source into cells y near its support, using the normally convergent current-cell filing of its complete pointed members. For a source varying on scale d, its first-degree local norm is $CR_0^2 t_i$, since $(a')^\alpha [f_i]_{C^\alpha} \le C t_i (a'/d)^\alpha$. Exponential tails can be included as extra rooted cells with summable coefficients. The number of roots is $N_i \le C(d/a')^4$, with total coefficient mass at most Ct_iN_i.

Summing (18.2), and restricting the actual physical readout to its admitted core α_H, gives the TOTAL terminal field bound

$$\boxed{|\Phi_{i,H}| \le C\alpha_H^2 t_i (d/H)^{4-\delta}(d/a')^{\delta}.} \tag{18.3}$$

The sum over output H-cells of the exponential in (18.2) is bounded uniformly. There is no factor counting all terminal cells of the physical torus. All fields and currents in this assertion belong to the actual clean readout domains; terms crossing a genuine wild region retain their separate tagged operator estimate.

27.19.3 18.3 A mixed source must connect the two roots

At a window scale ρ, a newly formed mixed coefficient has two distinguished first-source vertices. Section 17 gives its field weight $g_\rho^2 R_0^2$ and a connected carrier joining both roots and the output. Using (18.2) on each leg, its normalized coefficient at an output cell u is bounded by

$$Ct_1t_2(a'/\rho)^{2\lambda} \sum_{y\in A_1, z\in A_2} e^{-c\,\mathrm{tree}_\rho(u,y,z)}. \tag{18.4}$$

Here A_i are the scale-a' root sets, with their summable tail weights understood. This estimate follows from a connected two-vertex expansion, rather than multiplying two globally summed norms. Summing the output cell and reserving half the decay yields $CN_1N_2e^{-c'd/\rho}$. A new mixed member born there is transported to H with $(\rho/H)^\lambda$, by (18.2). The running factor g_ρ^2/g_H^2 is bounded on the weak trajectory. Finally,

$$\sum_{a'\le\rho\le H} \rho^{-\lambda} e^{-cd/\rho} \le Cd^{-\lambda} \tag{18.5}$$

is an elementary geometric-scale sum. It follows that

$$\boxed{|\Phi_{12,H}| \le Cg_H^2\alpha_H^2 t_1t_2(d/H)^{4-\delta}(d/a')^{2\delta} + g_H^2\alpha_H^2 t_1t_2\,\eta_{\mathrm{UV}}.} \tag{18.6}$$

The last term is the mixed family already present at a'. Its connected supports bridge d; Section 17 supplies $\eta_{\mathrm{UV}} \le C(d/a')^q e^{-cd/a'}$, with a fixed finite power q, after the true finer-scale root count. The polynomial is harmless for $a' \ll d$.

The estimate applies to the complete mixed source, including old mixed vertices. It is not merely the covariance born in the last step. Scalar vacuum terms created inside the window have already been retained in the separate universal vacuum; (18.6) concerns precisely the field-dependent coefficient which remains to be integrated.

27.19.4 18.4 What remains after the final field integration

For a positive source-zero retained law supported in the admitted clean core, the exact logarithmic derivative of its integral is

$$D_{12}\log\mathcal{I}(0) = -\mathbb{E}\Phi_{12,H} + \mathrm{Cov}(\Phi_{1,H}, \Phi_{2,H}). \tag{18.7}$$

The sign of the first term changes if the opposite action convention is used; both terms remain. Combining (18.3),(18.6) gives the absolute bound

$$\begin{aligned} |D_{12}\log\mathcal{I}(0)| \le Ct_1t_2[&g_H^2\alpha_H^2(d/H)^{4-\delta}(d/a')^{2\delta} \\ &+ \alpha_H^4(d/H)^{8-2\delta}(d/a')^{2\delta} + g_H^2\alpha_H^2\eta_{\mathrm{UV}}]. \end{aligned} \tag{18.8}$$

This bound holds uniformly for ANY such positive law; no independence or Gaussianity of its remaining coarse field is required. For a physical good-window decomposition it bounds the good contributions without division by the good event's probability; the complementary history/insertion terms still require their own positive numerator estimates. In particular (18.8) does not by itself assert that a sharp history event is a local function of the bare field.

Take $\alpha_H = g_H P_{\mathrm{bg}}(x_H)$, $a' = dt^{-\varepsilon}$, $H = dt^{\varepsilon}$, and $x_H \asymp x_d \asymp t$. For fixed $\delta < 1$, the first power in (18.8), divided by $g_d^4 t_1 t_2$, is bounded by a polylogarithm in t times $t^{-\varepsilon(4-3\delta)}$; the covariance term is bounded by a polylogarithm times $t^{-\varepsilon(8-4\delta)}$. Both vanish. The retained clean field cumulants are therefore negligible at the Gaussian two-point scale on this finite window, WITHOUT a whole-torus density comparison or a terminal-volume factor.

To obtain an actual physical noncollapse statement, one must still identify the universal separated vacuum coefficient inside the window with its Gaussian chains, control the accumulated Gaussian-subtracted births and the normalization of the incoming first source, and estimate all complementary tagged histories. The exact Gaussian covariance comparison in the parent note handles the Gaussian chains; (17.15) supplies a fresh clean error, but the accumulated source/insertion comparison must preserve its weights. The conclusion here is the retained-field part of that argument, with its terms explicitly kept.

27.21 20. Accumulating the clean Gaussian-subtracted vacuum comparison

The first-source matching and the two-label transfer estimates give a finite-window vacuum comparison with an explicit polynomial cost. No claim about the full R-completed physical law is made in this section. The comparison is between the universal PURE CLEAN source coefficients and the Gaussian hierarchy having the same actual flat block maps and the same reference running inverse couplings.

27.21.1 20.1 The Gaussian mixed vertex is transported exactly

At a current scale s, let H_s be the flat harmonic coarse lift and T_s the gauge-fixed hidden injection. Let C_s be the covariance of the canonical hidden coordinate B, so the physical fluctuation is $g_s T_s B$. For a Gaussian source state write

$$O_i^G(A) = \nu_i^G + \tfrac{1}{2}\langle A, K_i^G A\rangle, \qquad O_{12}^G(A) = \nu_{12}^G + \tfrac{1}{2}\langle A, K_{12}^G A\rangle. \tag{20.1}$$

Put $D_s = T_s C_s T_s^*$. Exact Gaussian integration gives

$$\begin{aligned} K_i^{G\prime} &= H_s^* K_i^G H_s, \\ K_{12}^{G\prime} &= H_s^* K_{12}^G H_s - g_s^2 H_s^* (K_1^G D_s K_2^G + K_2^G D_s K_1^G) H_s, \\ \nu_{12}^{G\prime} - \nu_{12}^G &= \tfrac{g_s^2}{2} \operatorname{Tr}(D_s K_{12}^G) - \tfrac{g_s^4}{2} \operatorname{Tr}(D_s K_1^G D_s K_2^G). \end{aligned} \tag{20.2}$$

The first-degree scalar update is $g_s^2 \operatorname{Tr}(D_s K_i^G)/2$. All traces in a local birth are current-step traces with a fixed number of hidden coordinates per cell; connected source filing precedes their sum. In particular the OLD Gaussian mixed vertex K_{12}^G contributes the first trace in the third line. Omitting that trace would break the Gaussian telescoping identity.

Start at a' with the exact all-depth Gaussian source state, not just a local scalar coupling. Its first Hessians are $F_{a'}^* M_{f_i} F_{a'}$; its mixed Hessian and scalar are the exact previously integrated Schur/determinant coefficients. Equations (20.2) then give the actual Gaussian hierarchy through H. Schur associativity identifies its scalar two-source coefficient with the Gaussian curvature covariance, with the remaining coarse-field terms retained as in Section 18. Running scalar counterterms are source independent in this frozen transport and enter only through the actual sequence g_s.

27.21.2 20.2 A local jet comparison for one clean step

Let $\mathsf{P}O$ denote the COMPLETE flat quadratic Hessian of the actual physical coarse-field function, with its own current formed before taking the jet. It is not the 21-dimensional zero-momentum marginal projection. On a localized fixed clean core its norm is bounded by a fixed field Cauchy constant, and its source anchors stay in its carrier.

For a vacuum-subtracted scalar insertion O with zero first pair jet and norm $R_0^2 a$, the rooted current-g estimate and the flat Taylor integral imply

$$\begin{aligned} \|\mathsf{P}\mathcal{T}_g O - H^*(\mathsf{P}O)H\|_{\mathrm{loc}} &\le C g^2 a, \\ |(\mathcal{T}_g O)(1,0) - \tfrac{g^2}{2} \operatorname{Tr}(D\,\mathsf{P}O)|_{\mathrm{loc}} &\le C g^4 a. \end{aligned} \tag{20.3}$$

Fixed powers of R_0 are included in the constants. For the first line, the classical composition has Hessian $H^*(\mathsf{P}O)H$, because $DO(1,0) = 0$; the difference from that composition is $O(g^2)$ on the fixed core. For the second, divide the flat insertion by g^2, use its Taylor-integral extension and bounded first two rooted current-g derivatives, and use evenness to remove the first derivative. This is the same argument as (17.15), with one insertion.

For the connected covariance map of two insertions with norms $R_0^2 a_i$, the corresponding estimates are

$$\begin{aligned} \|\mathsf{P}\operatorname{Cov}_{\mu_g}(O_1, O_2) - g^2 H^*(K_1 D K_2 + K_2 D K_1) H\|_{\mathrm{loc}} &\le C g^4 a_1 a_2, \\ |\operatorname{Cov}_{\mu_g}(O_1, O_2)(1,0) - \tfrac{g^4}{2} \operatorname{Tr}(D K_1 D K_2)|_{\mathrm{loc}} &\le C g^6 a_1 a_2, \end{aligned} \tag{20.4}$$

where $K_i = \mathsf{P}O_i$. In the first line the coefficient of g^2 is the covariance of the linear fluctuation terms. Its quadratic external-field jet uses only K_1, K_2, so it is exactly the displayed Gaussian expression. After division by g^2, evenness and the bounded second current-g derivative give the g^4 remainder. The second line is (17.15). Every error has the same connected source-carrier requirement as its left side.

These are comparisons of jets of the actual full insertion functions. A higher-field insertion is not discarded: its quadratic feedback is included in the error in (20.3), and its covariance contribution is included in (20.4). Current-dependent pair jets enter through the genuine physical window/current map. No zero off-shell AJ or JJ assertion is needed.

For a frozen logarithmic canonical cutoff instead of the fixed physical I–II cutoff, insert and remove the full Gaussian tails exactly. Let $\tau_s \le C\,\mathrm{polylog}(x_s)e^{-cP_s^2}$ be the rooted tail bound, with P_s its canonical cutoff. The same statements hold with the relative error g_s^2 replaced by $g_s^2+\tau_s$. Polynomial source insertions are paid by the same fixed Gaussian reserve. This is a cutoff correction, not an equality of two different cutoff integrals.

27.21.3 20.3 A finite-window Duhamel estimate with only polynomial counts

Let n be the number of steps from a' to H, and let $N_i \le C(d/a')^4$ be the initial rooted source-cell counts of Section 18. On this finite window the running-tube bound gives $g_s \asymp g_d$. The complete rooted resolvent in (18.2) bounds the total current first-source norms by CN_it_i, and the total normalized mixed field norm by $C(1+n)N_1N_2t_1t_2$, after division by $g_d^2R_0^2$. The latter follows by summing the connected mixed forcing over its possible birth scales; retaining its age decay would improve this crude bound but is unnecessary here. There is no factor counting all cells of the surrounding torus.

The exact Gaussian harmonic transfer has the corresponding uniformly bounded rooted propagation between any two window scales. This follows either from the actual all-depth flat lift estimates or from (18.1) with no fluctuation birth. Therefore the variation-of-constants expansion of (20.3) is a sum over the position of ONE local error insertion; it does not multiply an arbitrary norm constant once per scale.

At a', (19.5) bounds the error in the incoming first Hessian by $C\omega_{a'}N_it_i$. The incoming actual and Gaussian mixed field coefficients are both connected across the separation d. Their difference is bounded by $Cg_{a'}^2N_1N_2t_1t_2\eta_{\mathrm{UV}}$, after adjusting the polynomial in $\eta_{\mathrm{UV}} \le C(d/a')^q e^{-cd/a'}$. This explicitly retains the initial Gaussian mixed state; it is not reset to zero.

Let $e_{i,s}$ be the total rooted first-Hessian error, and let $e_{12,s}$ be the mixed-Hessian error divided by g_d^2. Put $\epsilon_{\mathrm{w}} = g_d^2 + \max_s \tau_s$. Equations (20.3)–(20.4), with their exact Gaussian transport on both sides, give the finite Duhamel bounds

$$\begin{aligned} e_{i,s} &\le CN_it_i\{\omega_{a'} + n\epsilon_{\mathrm{w}}\},\\ e_{12,s} &\le CN_1N_2t_1t_2\{\eta_{\mathrm{UV}} + n\omega_{a'} + n^2\epsilon_{\mathrm{w}}\}. \end{aligned} \tag{20.5}$$

For clarity, the second line has three sources. The old mixed insertion uses (20.3) and its already weighted g_d^2 norm. A fresh covariance uses (20.4). Replacing its two actual first Hessians by Gaussian ones produces one first-Hessian error on one of the two legs; insert the first line and sum its possible scales. The connected two-root filing and the original rooted cluster norm bound all unmarked branches. This proves the second line without assuming that the higher-field mixed function itself is a quadratic polynomial.

Finally compare the scalar updates. The first trace in (20.2) multiplies the mixed-Hessian error by g_s^2. The second trace uses the first-Hessian errors. The genuine non-Gaussian scalar errors are g_s^4 times the old mixed field norm, and g_s^6 times the two first norms, by (20.3)–(20.4). Summing these bounds yields

$$\boxed{|\Delta\mathcal{E}_{12}^{\mathrm{clean,window}} - \Delta\mathcal{E}_{12}^{G,window}| \le Cg_d^4t_1t_2N_1N_2(1+n)^3\{\omega_{a'} + \epsilon_{\mathrm{w}} + \eta_{\mathrm{UV}}\}.} \tag{20.6}$$

The already integrated scalar vacuum below a' has its own connected bridge bound

$$Cg_d^4t_1t_2\,\mathrm{poly}(d/a')e^{-cd/a'},$$

obtained with the degree-two vacuum weight of Section 17 BEFORE the fine root count. It can be added separately to (20.6). Local contact counterterms have zero mixed derivative for these separated supports.

Since $N_1N_2 \le C(H/a')^8$ and $n \le C(1+\log(H/a'))$, (20.6) has the requested explicit form

$$Cg_d^4 t_1 t_2 (H/a')^8 (1+\log(H/a'))^3 \{x_d^{-1/2}\,\mathrm{polylog}\, x_d + \max_s \tau_s\} \tag{20.7}$$

plus the ultraviolet bridge. The exponent eight is deliberately crude; the sharper connected engineering sums reduce it.

For $a' = dt^{-\varepsilon}$, $H = dt^{\varepsilon}$, $x_d \asymp t$, the relative error in (20.7) is at most $Ct^{-1/2+16\varepsilon}\,\mathrm{polylog}\, t$, apart from the superpolynomial cutoff tails. Any fixed $0 < \varepsilon < 1/32$ makes it vanish. The retained-field error (18.8) also vanishes for this choice. Combined with the exact Gaussian off-diagonal comparison, this closes the PURE CLEAN separated scalar two-point comparison on such a finite window.

The remaining step for the physical model is the literal matching of this fixed-core clean reference to the original R representation, its cutoff corrections and promoted tagged counterterms, and the positive numerator control of every complementary history contribution. Those terms are not subsumed under (20.7). The companion history note develops that matching separately; until its actual-domain bounds are established, (20.7) is a clean theorem rather than a complete Yang–Mills nontriviality proof.

28.5 4. Primitive Wilson bridges and their actual amplitude

28.5.1 4.1 The complex plaquette identity

For $W \in \mathrm{SL}(2,\mathbb{C})$,

$$a(W) := 1 - \tfrac{1}{2}\mathrm{Tr}\, W = \tfrac{1}{2}\det(1-W). \tag{4.1}$$

This agrees with the real SU(2) Wilson action. Therefore, on the actual complex regular plaquette domain,

$$\|1 - U_p(s)\| \le C\alpha\eta^2 \quad \Longrightarrow \quad |a(U_p(s))| \le C\alpha^2\eta^4. \tag{4.2}$$

If two admitted substituted plaquettes differ by at most $C\delta\eta^2$, the bilinearity of the determinant gives

$$|a(U_p^{(1)}) - a(U_p^{(0)})| \le C(\alpha\delta + \delta^2)\eta^4. \tag{4.3}$$

The bounds concern plaquettes, not merely link amplitudes. The first-difference conditions of the frozen regular core and the ordinary minimizer estimates are what supply (4.2)–(4.3).

28.5.2 4.2 Localization of the source atoms

Set $w(p) = \sum_i z_i f_i(p)$ and $\rho_p = \sum_i r_i|f_i(p)|$, $\rho = \sup_p \rho_p$. For the local atom $F_p(s,z) = w(p)a(U_p(s))$, Theorem 3.1 gives

$$\|\mathcal{L}_{Y,X_0(p)}F_p\|_{\boldsymbol{r}} \le C\rho_p\alpha^2\eta^4(e^{\kappa}-1)^{-|Y\setminus X_0(p)|}. \tag{4.4}$$

The source label remains attached to the original plaquette anchor p. At fixed final domain Y, the number of possible plaquettes is at most $CM^4\eta^{-4}|Y|$. The seed contains at most a fixed number of cubes. Taking κ larger than the desired locality exponent plus the fixed entropy margin yields

$$\|W_Y(z)\|_{\boldsymbol{r}} \le C_M \rho \alpha^2 e^{-\lambda|Y|}. \tag{4.5}$$

The fine-spacing factors cancel. The same result for a difference of two frozen reconstructions has $\alpha\delta + \delta^2$ in place of α^2. Finite successive background relocations only use a fixed number of such estimates and their declared margins.

On a real wild core, the retained Wilson source remains inside the component kernel; no regular estimate is asserted for an unrestricted bare plaquette. The source-free completed-R construction leaves the wild Wilson field real and stored. Dependence on complex exterior pairs is carried by the regular boundary/reconstruction factors, where the preceding localization applies.

28.5.3 4.3 Classical, quadratic and nonlinear fluctuation terms

At the frozen regular fiber background, write the exact source fluctuation as

$$A(w, U(gB)) - A(w, U(0)) = \ell_w(B) + \tfrac{1}{2} Q_w(B, B) + R_w(B). \tag{4.6}$$

The first two terms are retained exactly or admitted as small generic potentials. They are not declared irrelevant. For a spatially varying source the linear term need not vanish at a source-free critical background. It contains the weighted current, including its source gradient term. Gaussian integration of that term creates genuine classical multi-source paths.

On the actual fixed fluctuation box $|B| \le P$, the local growth coefficients have the respective orders

$$\rho g \alpha, \qquad \rho g^2, \qquad \rho g^3 P. \tag{4.7}$$

This notation suppresses fixed block factors and the corresponding local quadratic-growth weight. The cubic estimate is the Taylor remainder after the actual constrained chart has been inserted. It uses its unscaled analytic radius, not an unconstrained ambient Hessian in place of the constrained one.

For $\rho = x/(\log x)^M$, $g = x^{-1/2}$, $\alpha = g(\log x)^{p_{\mathrm{bg}}}$, the first two growth coefficients are log-small when M is sufficiently large; the third retains a factor g. The precise radius can be improved when the linear and quadratic parts are completed into the Gaussian. In either case, localization preserves these actual orders. It does not promote a log-small quadratic or classical term to $O(x^{-p})$.

28.8 7. Live islands remain unnormalized states

Let the state of a live island X be the coefficient-algebra kernel

$$\mathcal{T}_X(z)F(B) = \sum_{h\in\mathcal{H}_X} \int K_{X,h,0}(\xi;B)\, M_{X,h}(\xi;B,z)\, F(\xi,B)\, d\mu_{X,h}(\xi). \tag{7.1}$$

The scalar component of M is one. At an unfolded primitive history it is the exponential of the stored marks. After some internal integrations it may be a general finite-jet function. No logarithm or inverse of $\mathcal{T}_X(z)1$ is part of the definition.

The reference measures are fixed and positive. Any analytic coarea or density dependence on an exterior parameter is put in K_0 using the actual fixed chart. A large field outside an analytic chart is retained as a real stored variable, not continued to an invented product complex domain.

28.8.1 The four state transitions

Survival or renewal. Append the new stored variables and their source-free positive kernel. Multiply by the newly acquired local source factors. The old factor $M_{X,h}$ is carried once. Its primitive exponent is not charged again at every age.

Merging or swallowing. Before integrating, apply the frozen background localization to every dependence linking the children. Each resulting term contains separate child kernels and local bridges. The new state integrates their product on the connected union. Every swallowed history occurs once. The source coefficient norm is bounded by the product of the child coefficient norms and the bridge norms, using Fubini only after this localization.

Healing. Integrate the stored variables of the entire connected term. The result is a scalar coefficient-algebra activity on its final domain. It enters the ordinary convergent polymer logarithm only together with its small absolute core mass.

Drop-out. Keep the carrying Wilson action in the kernel used for its mass bound. The bare operation made only of tests, delta functions and reference weights need not be small. The source stays in the retained real action and the stored interaction factors.

These rules describe the same finite integral with a different organization of its local factors. A multiplication version of an island is allowed after its internal variables have actually been integrated, but the multiplier is then the function $m_X(B, z) = \mathcal{T}_X(z)1$ of the retained island field. It is not a scalar independent of B, and it is not removed from the history sum.

28.8.2 7.1 Why the sourced normalization can be dangerous

The issue is mathematical, even for a finite positive reference. Let a hidden variable $H = \pm 1$ have equal probabilities and let a component carry n source marks $z_1, \dots, z_n$, each equal to H. Its normalized generating jet is

$$Q(z) = \mathbb{E} e^{H \sum_i z_i} = \cosh\left(\sum_i z_i\right). \tag{7.2}$$

The unnormalized coefficient norm of $e^{H \sum_i z_i}$ is $\prod_i (1 + r_i)$, hence only exponential in the number of marks for bounded radii. The coefficients of Q^{-1} are Euler-number coefficients and exhibit factorial growth. In one ordinary source variable the corresponding function $\cosh(nt)$ has zeros at distance $\pi/(2n)$ from zero. Thus positivity at the origin and a per-site exponential reserve do not establish a uniform sourced inverse for an arbitrarily large island.

Keeping (7.1) avoids this division. When expanding small boundary bridges, use the exact identity

$$\mathcal{T}_X(z) e^V = \mathcal{T}_X(z)1 + \mathcal{T}_X(z)(e^V - 1). \tag{7.3}$$

A term with no attached bridge is the isolated island state; terms attached to several bridges are filed on their joint connected support. There is no assertion that the generating function conditioned on the presence of a fixed island has a uniform logarithmic source bound.

28.8.3 7.2 Localization commutes with the state in the needed sense

For a primitive retained core action, the real stored field in the Wilson factor is independent of the current exterior localization parameters. Its source factor can therefore stay inside (7.1) while the parameter derivatives act on the regular kernel and attached local functions.

For inherited analytic marks that do depend on those parameters, apply Theorem 3.1 to their full local coefficient-algebra factor, including its dependence on the fixed maps. Cauchy differentiates that dependence as well. The fixed measure and the integrable reserve in section 8 justify Fubini. Parameters in a disconnected wall component still do not affect the state: the primitive stored marks have no such dependence, and every inherited mark has its actual local support included in the seed.

This supplies marked bridge localization for an incoming unnormalized island state with the stated local norm and reserve. It does not replace an unproved domain inclusion of a modified background map with the word “local.”

28.13 12. A uniform complex bare disc and all constant-source jets

The product-domain result can be bootstrapped through the actual triangular recursion. This is stronger than a derivative estimate at one real parameter, and it uses the same fixed physical cutoff.

28.13.1 12.1 Half-product bounds in additive inverse couplings

Let $x_0, \dots, x_K$ be the actual real run above, ending at $x_K = x_*$, and put

$$a = x_* - 2C_s, \qquad S_u = \frac{R_u}{4g_u^3} = \frac{c_*}{4}\frac{x_u}{(\log x_u)^M}. \tag{12.1}$$

Take a larger than the fixed floors used in section 11. For $|w_u - x_u| < S_u$, use the branch $g(w_u) = w_u^{-1/2}$ continuing the positive root. Since $S_u < x_u/16$, its derivative satisfies

$$|g'(w_u)| \le g_u^3, \qquad |g(w_u) - g_u| \le R_u/4. \tag{12.2}$$

At any point of this additive product box, a coupling polydisc of radii $R_u/2$ stays inside the original coupling product. The scalar row-Cauchy argument therefore gives

$$\sum_{u \le s} R_u |\partial_{g_u} e_{s+1}| \le 2A_s. \tag{12.3}$$

Composing with (12.2), and using the real reference comparison $x_u \ge x_s/2$ from (11.11b), yields the uniform row bound

$$\sum_{u \le s} |\partial_{w_u} \mathfrak{e}_{s+1}(w_0, \dots, w_s)| \le \bar{a}_s, \tag{12.4}$$

where

$$\mathfrak{e}_{s+1}(w) = e_{s+1}(g(w_0), \dots, g(w_s)), \qquad \bar{a}_s = C\frac{(\log x_s)^M}{x_s}\{g_s P_s + p_s\}. \tag{12.5}$$

The constant in $\bar{a}_s$ includes the half-radius loss. Increase the weak stopping floor, if necessary, until

$$\mathcal{A} := \sum_{s<K} \bar{a}_s \le 1/8. \tag{12.6}$$

Equation (11.14), with its increased constant, proves this uniformly in K from the actual bounded cumulative drift.

28.13.2 12.2 Finite triangular induction on one bare disc

The function $x/(\log x)^M$ is increasing for $x > e^M$. The actual nonmonotone trajectory satisfies $x_u \ge a$, so

$$S_u \ge \frac{c_*}{4}\frac{a}{(\log a)^M} \quad \text{for every } u. \tag{12.7}$$

Define the uniform additive bare radius

$$r_* = \frac{c_*}{16}\frac{a}{(\log a)^M}. \tag{12.8}$$

It depends on the fixed weak terminal scale, not on the number of ultraviolet steps. It is comparable to $cx_*/(\log x_*)^M$ once $x_* \ge 4C_s$.

For $|\zeta| < r_*$, construct

$$\begin{aligned} X_0(\zeta) &= x_0 + \zeta, \\ X_{s+1}(\zeta) &= X_s(\zeta) - b^0_{s+1} - \mathfrak{e}_{s+1}(X_0(\zeta), \dots, X_s(\zeta)). \end{aligned} \tag{12.9}$$

This is the actual analytic continuation of the triangular recursion, not a new independent fixed-point equation.

Suppose the preceding coordinates have been constructed and lie in the additive product box. The straight segment between their reference and perturbed values remains in that box. Integrating (12.4) along the segment gives

$$|\mathfrak{e}_{s+1}(X) - \mathfrak{e}_{s+1}(x)| \le \bar{a}_s \max_{u \le s} |X_u - x_u|. \tag{12.10}$$

Consequently, with $D_s = \max_{u \le s} |X_u - x_u|$,

$$D_{s+1} \le (1 + \bar{a}_s) D_s, \qquad D_s \le e^{\mathcal{A}} |\zeta| < 2|\zeta|. \tag{12.11}$$

Equations (12.7),(12.8) put the last quantity strictly below $S_u/2$ for every coordinate. Thus the hypothesis needed to perform the next step is preserved, with a fixed margin. Finite induction proves joint validity on the whole disc:

$$X_s \text{ is holomorphic on } |\zeta| < r_*, \qquad |X_s(\zeta) - x_s| \le e^{\mathcal{A}} |\zeta|. \tag{12.12}$$

This proof uses a lower floor for all reference x_u; it does not assume pointwise monotonicity of that run.

28.13.3 12.3 Higher bare jets and analytic tuning

Differentiate (12.9). The same row bound gives

$$\max_s |X_s'(\zeta)| \le e^{\mathcal{A}}, \qquad \max_s |X_s'(\zeta) - 1| \le \prod_{v<K} (1 + \bar{a}_v) - 1 \le \eta := e^{\mathcal{A}} - 1 < 1/4. \tag{12.13}$$

The telescoping product identity proves the second inequality; there is no extra factor counting past scales. Since $X_s(0) = x_s$,

$$|X_s(\zeta) - x_s - \zeta| \le \eta |\zeta|. \tag{12.14}$$

Cauchy therefore gives, at the reference bare value,

$$|X_s'(0) - 1| \le \eta, \qquad |X_s^{(m)}(0)| \le m!\, \eta\, r_*^{1-m} \quad (m \ge 2). \tag{12.15}$$

In particular the real bare tangent is positive and bounded away from zero. The associated coupling has

$$\left| \frac{d^m}{d\zeta^m} g(X_s(\zeta)) \Big|_{\zeta=0} \right| \le C m!\, g_s^3 r_*^{1-m} \quad (m \ge 1), \tag{12.16}$$

by (12.2), (12.12), and Cauchy on the difference from its reference value.

Let $F(\zeta) = X_K(\zeta) - x_*$. On the convex disc, (12.13) implies

$$|F(\zeta_1) - F(\zeta_2) - (\zeta_1 - \zeta_2)| \le \eta |\zeta_1 - \zeta_2|. \tag{12.17}$$

Hence F is one-to-one there. Its image contains $|w| < (1 - \eta) r_*$: on a circle $|\zeta| = r < r_*$ with $|w| < (1 - \eta) r$, the inequality $|F(\zeta) - w - \zeta| < r$ permits Rouché's theorem. It has exactly one preimage inside that circle. The inverse is holomorphic and satisfies

$$|\zeta(w)| \le |w|/(1 - \eta), \qquad |\zeta'(w)| \le (1 - \eta)^{-1}. \tag{12.18}$$

Thus the bare value tuning the actual terminal coefficient has a locally unique analytic dependence on that coefficient, on a disc whose radius is uniform in ultraviolet depth. No uniqueness claim outside this neighborhood is needed.

28.13.4 12.4 Actual local birth activities have the same bare disc

Every coupling prefix produced by (12.9) lies strictly inside its product domain. Compose each birth activity, its own marginal extraction, and its vacuum subtraction with that prefix. They are holomorphic on the same bare disc and remain in the same fixed analytic/locality spaces. In particular, for the genuinely new non-Gaussian birth part,

$$\left\| \frac{d^m}{d\zeta^m} E_{s+1}^{\mathrm{new}}(\zeta)\Big|_{\zeta=0} \right\|_{\mathrm{loc}} \le C m! \left\{ g_s P_s + p_s \right\} r_*^{-m} \quad (m \ge 1). \tag{12.19}$$

The same bound holds for its paired marginal and pointed remainder, with the fixed extraction constants. A weaker $Cm!E_0 r_*^{-m}$ bound holds for the full birth object. The Gaussian field-dependent birth term has zero derivative. Field-independent vacuum factors are explicit holomorphic local contributions and are retained whenever a full density or source partition identity is asserted; they are absent only from the F^2 extraction itself.

28.13.5 12.5 Compatibility with the frozen large-field allowance

On this bare disc, the current inverse-coupling change is at most $2r_*$. For every stored real layer,

$$\operatorname{Re} X_s(\zeta) \ge x_s - 2r_*, \qquad \frac{2r_*}{x_s} \le \frac{c_*}{8(\log a)^M}. \tag{12.20}$$

This is precisely a uniformly small fractional-action loss. If its regular field radius obeys $\alpha_s^2 \le C x_s^{-1} (\log x_s)^{2p_{\mathrm{bg}}}$ and $M > 2p_{\mathrm{bg}}$, then

$$2r_* \alpha_s^2 \le C c_* (\log a)^{2p_{\mathrm{bg}} - M}. \tag{12.21}$$

The eventual decrease of $x^{-1}(\log x)^{2p_{\mathrm{bg}}}$ and $x_s \ge a$ justify the bound without monotonicity. Equations (12.20),(12.21) supply the actual scalar parameter loads used in sections 8.1 and 8.4. The unnormalized live-state convention keeps the reference denominators fixed, and the ordinary action/coordinate credit recurrence pays the resulting costs once per stored layer.

Accordingly the constant-coupling family is compatible with the marked completed-R step on its original admitted domains. This supplies the constant-source parameter allowance; it does not convert an unverified modified seam chart into an admitted one, and it does not substitute for a spatial nonconstant-source or continuum-observable construction.

28.18 17. Exact local draws, approximate kernels, and sharp-test witnesses

The proposed chessboard route requires a local nonnegative function of the ORIGINAL bare field (and independently localized auxiliary seeds). A history event is not such a function merely because its background is exponentially insensitive to remote data.

28.18.1 17.1 What equality of local reference laws actually gives

The original completed R denominator in R-II (1.2) has fixed axial coordinates B' on Λ, with the cutoff $\chi\{|B'| < M_0 g_k^{-1}\varepsilon_k\}$. Its local background is defined on the finite region Z; the minimization constraint is (1.3). The completion later factors the relevant operations into components, as stated explicitly after (1.71).

Consequently, if two runs have the SAME

- component and buffer geometry Λ, Z;
- local determining fields on that whole geometry;
- selected axial graph and local gauge/coordinate choice;
- source-free coupling and cutoff constants,

then their reference integrands, denominators, and maps from integration coordinates to the new local field agree exactly. A fixed measurable sampling map for that common probability law, applied to the same seed, then gives exactly the same draw.

This is a finite-data equality. It does not say that almost equal backgrounds produce equal draws. For example, the usual common quantile realization of $N(m, 1)$ gives $m + \Phi^{-1}(u)$; two different means give unequal draws for every u. A proof of tile agreement must establish the displayed finite-data equality BEFORE invoking common seeds. In particular, approximate equality of the centers of a moving chart is insufficient.

28.18.2 17.2 A rigorous common-minorant repair when only proximity is known

Let $\{K_p : p \in P\}$ be probability densities relative to one fixed positive measure on one common coordinate/support space. Assume a locally chosen reference K_0 satisfies

$$|\log(K_p/K_0)| \le \epsilon \quad \text{on that common support, for all } p. \tag{17.1}$$

Then $K_p \ge e^{-\epsilon}K_0$. The exact decomposition

$$K_p = e^{-\epsilon}K_0 + (1 - e^{-\epsilon})\widetilde{K}_p \tag{17.2}$$

has $\widetilde{K}_p$ a probability density. It gives a simultaneous coupling of EVERY admitted parameter p: with probability $e^{-\epsilon}$ draw from the local K_0, using the common seed; only on the remaining seed event is remote data read. The failure probability is at most ϵ. Pairwise total variation bounds alone would not give this one local common part for an arbitrary family of parameters.

For normalized kernels

$$K_p(B) = D(p)^{-1}\,\chi(B)J_p(B)e^{-S_p(B)}, \tag{17.3}$$

with a fixed χ, suppose

$$|S_p - S_0| + |\log J_p - \log J_0| \le \delta.$$

The unnormalized densities have ratios in $[e^{-\delta}, e^{\delta}]$, and so do their normalizers. Thus (17.1) holds with $\epsilon = 2\delta$. This elementary normalization comparison has no small-denominator loss.

On a fixed current-scale block, bounded Wilson action derivatives give $\delta \le Cx_kN\,\Delta$, with the appropriate fixed gauge/chart derivative constant included, when the admitted background difference is Δ and there are N current coordinates. A decaying background response $\Delta \le Ce^{-cD}$ can therefore be made to give any prescribed polynomially small coupling failure by choosing

$$D \ge c^{-1}\log(Cx_kN/\epsilon_{\text{goal}}). \tag{17.4}$$

The exact original component-local law of 17.1 is stronger when its hypotheses hold, but (17.2) is available if the algorithm genuinely uses a nearby family of densities on a common support.

28.18.3 17.3 Why moving sharp supports and post-test conditioning matter

If the characteristic function in (17.3) depends on p, (17.1) can fail even for an arbitrarily small parameter difference. For example, uniform laws on $[0, 1]$ and $[\delta, 1 + \delta]$ have nearby centers but their log density ratio is infinite on their noncommon parts. Their common mass is $1 - \delta$; proving a useful analogue requires an explicit boundary-layer estimate.

A narrow-window estimate must be proved for the PRETEST conditional law. Conditioning on the completed branch can force the tested value into that window. Even a standard Gaussian X, conditioned on $t < X < t + \delta$, gives that window probability one. Neither Gaussian tails nor the unconditioned density bound permits such a conditional step.

A usable finite-block shell lemma is elementary. If a vector $Y \in \mathbb{R}^m$ has density

$$p(y) \le C_0e^{-c_0|y-\mu|^2}, \qquad |\mu| \le M,$$

then for $t \ge 2M + 2$, $0 < \delta \le 1$,

$$\mathbb{P}\{t - \delta \le |Y| \le t + \delta\} \le C_{m,c_0}C_0\,\delta(t + 1)^{m-1}e^{-c_0t^2/16}. \tag{17.5}$$

This follows by integrating the density over the annulus and bounding its radial surface measure. For a linear image of a nondegenerate Gaussian, the required density bound follows from its actual marginal covariance; for a nonlinear tested coordinate it requires the corresponding Jacobian or coarea estimate. A covariance on a constrained fiber can be singular in a tested direction, so that nondegeneracy is not automatic.

The primary local denominator estimates establish finite-dimensional Gaussian domination for eligible regular replacement blocks. They do not, by themselves, establish (17.5) for EVERY historical test under its actual pretest law, particularly when the determining set or available free coordinates change at a seam.

28.18.4 17.4 Present status of the volume-uniform route

The component-local sampling equality above is available for the original reference operation, and the common-minorant lemma repairs the generic shared-seed fallacy. A complete reflection witness still needs a simultaneous induction showing:

1. the exact determining data needed by each draw are common before that draw is made;
2. every nonlocal sharp comparison is covered by a quantitatively bounded pretest shell;
3. the exceptional shells and long dependency components form local tile witnesses with replication estimates in the ORIGINAL Wilson law.

These are not consequences of the final scalar large-field mass bound. The finite-torus estimate in section 16 does not use them.

28.25 24. Fixed-order cumulants transfer from one physical reference torus

The common holomorphic normalizer in (18.10) controls differences of every FIXED cumulant order. An appropriate sign packing then retains the volume gain used for the two-point function. This is useful at order four because it transfers the fixed-reference result without proving volume-uniform rarity of an adaptive history.

28.25.1 24.1 A local logarithm with a common counterterm

Fix one admitted exponent $0 < \alpha \le 1$, one reference torus, and the compatible periodic volumes of section 18. For profiles supported in a fixed interior cell ball, that section gives a common holomorphic functional $\mathcal{E}^\circ[J]$, with $\mathcal{E}^\circ[0] = 0$, such that

$$G_\rho(z) = e^{-\mathcal{E}^\circ[\sum_{i=1}^m z_i F_i]} \left\langle e^{\sum_{i=1}^m z_i O(F_i)} \right\rangle_\rho, \qquad |G_\rho(z)| \le Q, \quad G_\rho(0) = 1, \tag{24.1}$$

whenever $\sum_i |z_i| B_i < r_0$, where $B_i = \|F_i\|_{C^\alpha}$. The same $\mathcal{E}^\circ$ and Q serve the reference law τ, the larger compatible periodic law ρ, and their fixed-cutoff periodic weak limits. They depend on the fixed reference data and source budget, but not on the larger volume or the fine cutoff.

For completeness, Schwarz on the complex ray through z, applied to $G_\rho - 1$, gives

$$|G_\rho(z) - 1| \le (Q+1) r_0^{-1} \sum_i |z_i| B_i. \tag{24.2}$$

Thus on the polydisc

$$|z_i| < \frac{r_0}{4m(Q+1)B_i} \tag{24.3}$$

the function is within $1/4$ of one. Its logarithm with value zero at the origin is bounded by a fixed constant there. Cauchy yields, for the physical connected cumulant difference,

$$\begin{aligned} D_m(F_1, \dots, F_m) &:= \kappa_m^\rho(F_1, \dots, F_m) - \kappa_m^\tau(F_1, \dots, F_m), \\ |D_m(F_1, \dots, F_m)| &\le C_m(Q, r_0) \prod_{i=1}^m \|F_i\|_{C^\alpha}. \end{aligned} \tag{24.4}$$

The common $\mathcal{E}^\circ$ cancels in the difference of the logarithmic derivatives. It can contain divergent local source coefficients; none is estimated separately in (24.4). Only the small zero-free disc (24.3) has been used, not a global source zero-free theorem.

28.25.2 24.2 Signs which select one common packed location

Let $f_1, \dots, f_m$, with $m \geq 2$, be fixed-shape profiles at scale d, all contained in a common ball of radius Cd, and suppose

$$\|f_i\|_{C^\alpha} \leq Cd^{-\alpha} t_i, \qquad \|f_i\|_\infty \leq Ct_i. \tag{24.5}$$

The supports inside a tuple may be separated as required by the physical cumulant being compared. Pack $N_d \geq cd^{-4}$ translates of the ENTIRE tuple into the fixed cell ball, with the different tuple carriers separated by at least cd. Choose translations on the common bare lattice; the packing is available for all sufficiently fine cutoffs $a \leq c'd$, as needed for the window limit.

For each packed location v and each $i < m$, choose independent symmetric signs $\epsilon_{v,i}$. Set

$$\sigma_{v,i} = \epsilon_{v,i} \quad (i < m), \qquad \sigma_{v,m} = \prod_{i<m} \epsilon_{v,i}, \qquad F_i^\epsilon = \sum_v \sigma_{v,i} \tau_v f_i. \tag{24.6}$$

The separation and the vanishing collars give

$$\|F_i^\epsilon\|_{C^\alpha} \leq Cd^{-\alpha} t_i \tag{24.7}$$

with no factor of N_d. This also bounds differences between points in two distinct packed carriers: their distance is at least cd, while the two profile values are bounded by Ct_i.

The choice of signs eliminates every mixed-location term exactly:

$$\mathbb{E}_\epsilon \prod_{i=1}^m \sigma_{v_i,i} = \prod_{i<m} \mathbb{E}_\epsilon(\epsilon_{v_i,i}\epsilon_{v_m,i}) = \prod_{i<m} \mathbf{1}_{\{v_i = v_m\}}. \tag{24.8}$$

In particular using one common sign for all four labels would be insufficient at order four, because it would retain two-pair assignments. Formula (24.8) retains only a single common packed location.

Connected cumulants are multilinear in their m arguments. Both laws are invariant under common bare translations. Therefore

$$\mathbb{E}_\epsilon D_m(F_1^\epsilon, \dots, F_m^\epsilon) = N_d D_m(f_1, \dots, f_m). \tag{24.9}$$

Apply (24.4), (24.7), and $N_d^{-1} \leq Cd^4$ to get

$$\boxed{|\kappa_m^\rho(f_1, \dots, f_m) - \kappa_m^\tau(f_1, \dots, f_m)| \leq C_m d^{4-m\alpha} \prod_{i=1}^m t_i.} \tag{24.10}$$

This is a physical cumulant comparison. No reflection or monotonicity has been applied to a history event.

28.25.3 24.3 The fourth-order application and its scope

For the four-label application fix $\alpha = 1/2$ once in the source construction. Then (24.10) is

$$|\kappa_4^\rho - \kappa_4^\tau| \leq C_4 d^2 \prod_{i=1}^4 t_i. \tag{24.11}$$

On the same weak trajectory $x_d = O(\log(1/d))$; hence its error relative to $g_d^8 = x_d^{-4}$ is $O(d^2 x_d^4)$, which tends to zero. Together with (23.19), a separately verified clean four-label asymptotic on the ONE reference torus therefore transfers to all compatible larger periodic volumes and their admitted limits.

At fixed separated profiles, the already proved fixed-order moment bounds pass the cumulants along the chosen ultraviolet and periodic subsequences: each cumulant is a finite polynomial in moments. The estimate is uniform in the larger volume and in every sufficiently fine cutoff. Taking these limits preserves (24.11), after which d may decrease. There is no exchange of an unbounded near-diagonal source family with the cutoff limit.

For general m, (24.10) has a positive physical power only when $m\alpha < 4$. The formula itself holds for every fixed m for which the admitted source bound is available, but one cannot deduce positive-power transfer at arbitrary orders from one fixed α. Choosing a new exponent for another order may require different source and analysis constants. That further choice is not needed for (24.11).

29.1 1. A pointwise bound from the actual harmonic equation

Let the fine spacing be a and a retained spacing be ℓ. Normalize the quadratic Wilson action as $x\|F\|_2^2/2$, so that the running Gaussian strength in these conventions is x^{-1}. The constant change from this convention to a specified trace normalization of the physical coupling is kept fixed throughout. A raw fine curvature coordinate is a^2 times its physical curvature coordinate.

Here is that change explicitly for $SU(N)$. Let $\operatorname{tr}(T_aT_b) = \frac{1}{2}\delta_{ab}$ and use the orthonormal Lie algebra basis $e_a = \sqrt{2N}\,T_a$ for the inner product $N^{-1}\operatorname{tr}(XY)$. If $F = F^a e_a$, then

$$1 - \frac{1}{N}\operatorname{Re}\operatorname{tr} e^{ia^2F} = \frac{a^4}{2}\sum_a (F^a)^2 + O(a^8|F|^4).$$

Thus the Wilson coefficient and the Gaussian strength in this chapter are

$$x = \frac{2N}{g_{\mathrm{YM}}^2}, \qquad g^2 := x^{-1} = \frac{g_{\mathrm{YM}}^2}{2N}.$$

Writing $F^a = gF^a_{\mathrm{can}}$ makes the quadratic part of the plaquette density a_p/a^4 equal to g^2Q, where $Q = \frac{1}{2}\sum_a (F^a_{\mathrm{can}})^2$. Equivalently, a smeared plaquette insertion has quadratic part $g^2\sum_p a^4 f(p)Q(p)$, with the orientation sum understood. The canonically normalized components have the Maxwell kernel used below. In particular the coefficient $3\dim G/\pi^4$ multiplies $g^4 = x^{-2}$, not g_{YM}^4 without its corresponding factor $(2N)^{-2}$. The determinant calculation uses this same x-convention.

Let $F_\ell B$ be the curvature of the actual flat constrained harmonic lift of coarse raw bond data B. The weighted energy estimate already proved for this lift is

$$\|W_v F_\ell e_v\|_{L^2_a} \le C, \qquad W_v(p) = e^{\kappa\,\mathrm{dist}(p,v)/\ell}, \tag{G1}$$

for each unit coarse bond vector e_v, with the corresponding periodic distance on a torus. The physical curvature is used in the weighted L^2_a norm.

The constrained Euler equation gives

$$d_a^* F = Q^*_{\mathrm{plain}} \mu, \qquad \mu = S_\ell B, \qquad d_c^* \mu = 0. \tag{G2}$$

Here S_ℓ is the actual coarse classical precision, rather than the inverse of an augmented covariance with a coarse identity term still present. The final identity is the coarse Ward identity. It cancels the node-gradient correction between the raw and straightened averages. Consequently the right side is also $Q^*_{\mathrm{st}}\mu$. Gauge-fixing multipliers do not contribute: variation along a fine gauge gradient annihilates the constrained precision by the same Ward identity, while the chosen gauge condition is nonsingular on the removed gauge directions.

The actual averaging weights, the exponentially local rows of S_ℓ, and their physical normalization give

$$\|W_v Q^*_{\mathrm{st}} S_\ell e_v\|_\infty \le C\ell^{-3}. \tag{G3}$$

This is the same adjoint averaging estimate used for the smooth lift, rescaled from retained spacing one. It is uniform in a/ℓ. The Bianchi identity $d_a F = 0$ and (G2) imply

$$\Delta_a F = d_a J, \qquad J = Q^*_{\mathrm{st}} \mu. \tag{G4}$$

Apply the elementary heat-semigroup identity at time ℓ^2:

$$F = e^{-\ell^2 \Delta_a} F + \int_0^{\ell^2} d_a e^{-s\Delta_a} J \, ds. \tag{G5}$$

For small fixed κ, the weighted lattice heat estimates are

$$\begin{aligned} \|W e^{-\ell^2 \Delta_a} W^{-1}\|_{2\to\infty} &\le C\ell^{-2}, \\ \|W d_a e^{-s\Delta_a} W^{-1}\|_{\infty\to\infty} &\le C(s^{-1/2} + \kappa/\ell) e^{C\kappa^2 s/\ell^2}. \end{aligned} \tag{G6}$$

These follow from the product heat kernel and its first difference, using the exponential moment of the underlying nearest-neighbor walk. The extra κ/ℓ term is needed after conjugating a derivative by the weight. For $s < a^2$, the elementary difference norm Ca^{-1} is bounded by $Cs^{-1/2}$, so no small-time cutoff loss occurs. Periodization preserves the estimates at scales at most the torus diameter.

The integral in (G6) is $O(\ell)$. Combining (G1), (G3), and (G5) therefore proves

$$|(F_\ell e_v)(p)|_{\mathrm{physical}} \le C\ell^{-2} e^{-\kappa \operatorname{dist}(p,v)/\ell}. \tag{G7}$$

In raw fine curvature coordinates this is $C(a/\ell)^2 e^{-\kappa \operatorname{dist}(p,v)/\ell}$. This argument supplies the pointwise factor which an energy norm alone would not supply.

29.2 2. Point kernels of the orthogonal bands

Let $\mathcal{E}_j$ be the fine-curvature image of the harmonic lift from spacing ℓ_j. These spaces decrease with j. Let Γ_j project onto $\mathcal{E}_j \ominus \mathcal{E}_{j+1}$. The actual current-step Gaussian identity gives

$$\Gamma_j = F_j C_j F_j^*, \tag{G8}$$

where the injections of the hidden coarse coordinates are included in F_j, and C_j is their current-scale covariance in raw coarse angle coordinates. Its exponential row bound and fixed number of coordinates per coarse cell were proved from the actual constraint gap. Set $\rho_j = \ell_{j+1}$; the fixed factor L between this scale and ℓ_j only changes constants.

Using (G7) on both sides of (G8) and summing the coarse coordinates gives the raw fine kernel estimate

$$|\Gamma_j(p,q)| \le C(a/\rho_j)^4 e^{-c\,\mathrm{dist}(p,q)/\rho_j}. \tag{G9}$$

Thus its PHYSICAL curvature covariance kernel obeys

$$|\Gamma_j^{\mathrm{phys}}(p,q)| \le C\rho_j^{-4} e^{-c\,\mathrm{dist}(p,q)/\rho_j}. \tag{G10}$$

There is no count of all fine modes in this estimate: the factorization uses only the fixed finite number of current-scale hidden coordinates per cell. For finitely many colors and curvature orientations their matrix norms merely change C.

Writing P_a for the full exact-curvature projection, the telescoping orthogonal identity gives

$$P_a = \sum_j \Gamma_j \tag{G11}$$

when the zero modes are removed and the flat Gaussian hierarchy is continued to the torus scale. The remaining one-cell abelian coarse curvature is zero. Equivalently one can retain the last coarse projection; its kernel has the corresponding torus-scale bound. This concerns exact curvatures on the trivial linearized bundle and does not discard topological sectors of the interacting group model.

29.3 3. Uniform off-diagonal sums

For two points separated by $r > 0$, the geometric scales satisfy

$$\begin{aligned} \sum_j \rho_j^{-4} e^{-cr/\rho_j} &\le Cr^{-4}, \\ \sum_{\rho_j \le a'} \rho_j^{-4} e^{-cr/\rho_j} &\le C(a')^{-4} e^{-c'r/a'}, \qquad a' \le r, \\ \sum_{\rho_j \ge H} \rho_j^{-4} &\le CH^{-4}. \end{aligned} \tag{G12}$$

For the first inequality, split at $\rho_j \approx r$. Above that scale the powers form a geometric sum; below it the exponential dominates the increasing inverse powers. The same change of variable $u = r/\rho_j$ proves the second inequality. These bounds are uniform in the fine cutoff and the total number of bands.

Let x_j be the real trajectory. Pick x_d at a band whose scale is comparable to a separation d. On a finite logarithmic window $a' \le \rho_j \le H$, its proved increment/partial-sum estimates imply

$$|x_j - x_d| \le C\{1 + \log_L(H/a')\}. \tag{G13}$$

Assume the right side is at most $x_d/2$, and put

$$\varepsilon_{\mathrm{run}} = C\frac{1+\log_L(H/a')}{x_d}. \tag{G14}$$

All still finer coefficients satisfy $x_j \geq x_d/2$, after increasing the common weak floor to absorb the bounded nonmonotone partial-sum error. The hidden curvature covariance through scale H, normalized by the strength x_d^{-1}, is

$$C^{\mathrm{run}}_{a,H} = \sum_{\rho_j \leq H} \frac{x_d}{x_j}\Gamma^{\mathrm{phys}}_j. \tag{G15}$$

On a separation range $d \leq |x-y| \leq Rd$, (G10)–(G14) imply

$$|C^{\mathrm{run}}_{a,H}(x,y) - P^{\mathrm{phys}}_a(x,y)| \leq C_R d^{-4}\Big[\varepsilon_{\mathrm{run}} + (d/H)^4 + (d/a')^4 e^{-c'd/a'}\Big]. \tag{G16}$$

The important point is that (G16) estimates a separated kernel. It does not lose the self-variance factor $(d/a')^4$ except in front of an exponentially small ultraviolet tail. No uniform approximation of the running weights on the infinitely many finer bands was required.

29.4 4. The free lattice limit in this normalization

At noncoincident points, the flat lattice exact-curvature kernel converges to the Maxwell curvature kernel. With fields placed at their link and plaquette centers, its Fourier symbol is the standard curvature projection with k_μ replaced by $2a^{-1}\sin(ak_\mu/2)$. On $|k| \leq c/a$, the difference of the centered symbols has derivative bounds

$$|\partial^\beta(P_a(k) - P_0(k))| \leq C_\beta a^2 |k|^{2-|\beta|}. \tag{G17}$$

A smooth dyadic momentum partition and integration by parts $m > 6$ times bound an annulus of radius λ by $Ca^2\lambda^6(1+\lambda|x-y|)^{-m}$. Summing gives $Ca^2|x-y|^{-6}$. The high annuli at the boundary of the Brillouin zone are handled before changing to centered coordinates: the uncentered cochain symbols are smooth and periodic there, and their placement phases are retained. Integration by parts has no boundary term in that convention. Restoring the exact link and plaquette placement phases gives the centered kernel at its actual displacement. Equivalently the quasiperiodic centered symbols have matching boundary phases, which cancel rather than being discarded. The low-frequency second-order estimate (G17) is unchanged. Thus on the infinite lattice

$$|P^{\mathrm{phys}}_a(x,y) - K(x-y)| \leq Ca^2|x-y|^{-6}, \qquad |x-y| \geq Ca. \tag{G18}$$

Keeping actual plaquette centers in this statement avoids an artificial first-order displacement error. If a different fixed sampling rule is used, its explicit $O(a/|x-y|)$ relative error is still harmless.

On a fixed torus of size T, the continuum torus Green function is the Euclidean fundamental solution plus a smooth correction near the diagonal. To verify this, cut off the fundamental solution inside a unit torus, solve for its smooth mean-zero error by the nonzero Fourier modes, and then rescale. Two

derivatives of that correction are bounded by CT^{-4}. The lattice version converges to it uniformly on separated compact sets. Hence the additional short-distance relative error is $C(d/T)^4$.

For $q = \dim G$ independent colors, the canonical free potential has covariance $\delta^{ab}\delta_{\mu\nu}/(4\pi^2|x-y|^2)$. Its curvature covariance K therefore gives, for $Q = \frac{1}{2}\sum_{a,\mu<\nu} : F^a_{\mu\nu}F^a_{\mu\nu} :$,

$$\operatorname{Cov}(Q(x), Q(y)) = \frac{3q}{\pi^4|x-y|^8}. \tag{G19}$$

This is the coefficient explicitly computed in the earlier section; the present argument matches the actual constrained Gaussian bands to that coefficient.

29.5 5. Coarse backgrounds with no Gaussian assumption

Let X have covariance (G15), and let a coarse mean m, independent of X, have an arbitrary law satisfying

$$\sup_x \mathbb{E}|m(x)|^4 \le B^4 H^{-8}. \tag{G20}$$

For $F = X + m$, Wick contraction conditionally on m, followed by the law of total covariance, gives

$$\begin{aligned}\operatorname{Cov}(Q(x), Q(y)) = {} & \tfrac{1}{2}\|C^{\mathrm{run}}_{a,H}(x,y)\|^2_{\mathrm{HS}} \\ & + \mathbb{E}[m(x)^T C^{\mathrm{run}}_{a,H}(x,y)m(y)] \\ & + \tfrac{1}{4}\operatorname{Cov}(|m(x)|^2, |m(y)|^2).\end{aligned} \tag{G21}$$

If $f, h \ge 0$ have positive masses and all pairs of support points have separation in $[d, Rd]$, define

$$V_0(f,h) = \frac{3q}{\pi^4}\iint f(x)h(y)|x-y|^{-8}\,dx\,dy.$$

Equations (G16), (G18), and (G20)–(G21) give

$$\begin{aligned}\frac{|\operatorname{Cov}(Q_f, Q_h) - V_0(f,h)|}{V_0(f,h)} \le C_{q,R}\Big[& \varepsilon_{\mathrm{run}} + (d/a')^4 e^{-c'd/a'} + (a/d)^2 + (d/T)^4 \\ & + (1+B^2)(d/H)^4 + (1+B^4)(d/H)^8\Big].\end{aligned} \tag{G22}$$

The result is uniform in the law of the coarse mean subject to (G20). For an actual regular harmonic background at scale H, its deterministic regularity bound gives $B \le C(x_d/x_H)^{1/2}P_{\mathrm{bg}}(x_H)$, a fixed power of $\log x_d$ on the window. This only identifies the size of a regular background; it is not a bound on the probability of the exceptional histories.

29.6 6. A compatible short-distance choice

Let $t = \log(1/d)$, with $x_d \asymp t$, and choose

$$a' = dt^{-\epsilon}, \qquad H = dt^{\epsilon}, \qquad \epsilon > 0. \tag{G23}$$

Rounding to adjacent actual lattice scales changes only constants. Then $\varepsilon_{\mathrm{run}} = O(\log t/t)$, the ultraviolet band error is $t^{4\epsilon}e^{-ct^{\epsilon}}$, and every coarse-mean error in (G22) tends to zero when B grows only as a fixed power of $\log t$. The original fine cutoff is removed first at fixed d, a', H.

Suppose, additionally, the actual interacting conditional law on this regular window has a normalized density oscillation and insertion error bounded by

$$Cx_d^{-\eta/2}(\log x_d)^u (H/a')^v t^{w\epsilon}, \qquad \eta > 0, \tag{G24}$$

with fixed finite exponents. The already proved normalized-density comparison loses at most $(d/a')^4 = t^{4\epsilon}$. Choosing $\epsilon < \eta/[2(2v+w+4)]$ makes that remaining error vanish. Thus polynomial finite-window costs are compatible with the exact Gaussian signal.

Equation (G24), its exact observable normalization, and a separately controlled exceptional-history numerator are the interacting inputs. Neither the heat equation nor the Gaussian bands prove them. In particular the independent mean in (G21) is part of the comparison measure; it is not asserted for the full interacting measure. The result here closes the actual Gaussian covariance comparison and its scale choices, while keeping that distinction explicit.

30.1 1. A precise finite-order one-step statement

Let γ be the centered constrained Gaussian of one clean step. Freeze the background, the old regular activities, and every old coupling. The current fluctuation law is

$$d\mu_g(B) = Z_g^{-1}\chi(gB)e^{-V_g(B)}\,d\gamma(B), \tag{HS1}$$

with the original physical cutoff χ. The statements below use the following concrete current-step hypotheses. After the saddle linear term and the source-free quadratic Gaussian have been removed, the Taylor vertices of the primitive interaction with $l \geq 3$ fluctuation legs have power at least g^{l-2}. Jacobian and old regular activity vertices have the additional powers coming from their ordinary substitution in gB. A residual old quadratic vertex, for example, has at least g^2, and a residual linear vertex has at least g. At the flat background the invariant linear jet vanishes.

Let $O_1, \dots, O_r$ be regular invariant insertions with zero constant and first pair jets. Subtract each zero-fluctuation value and write

$$X_i(g, B) = O_i \circ \Psi_g - O_i \circ \Psi_0. \tag{HS2}$$

A term with l fluctuation legs in this substitution has power at least g^l. At a flat background the first possible term is $g^2 Q_i(B)$, with Q_i quadratic. The transformation $(g, B) \mapsto (-g, -B)$ leaves the complete integral unchanged. For the fixed integer r, require the rooted local logarithm and these coefficients to have uniformly bounded real g-derivatives through order $2r + 2$. The bounds use fixed polynomial Gaussian reserves and a fixed smaller analytic core, as in the two-label proof.

These are finite-order requirements. At the original fixed physical cutoff, the extra boundary terms are powers of g^{-1} times e^{-c/g^2}, hence have zero Taylor jet at $g = 0$. In the clean source construction, taking a fixed higher number of current derivatives inserts only a fixed higher-degree Gaussian polynomial; the same connected rooted reserve bounds its integral. This is the relevant uniformity, rather than a volume-dependent bound for a normalized global expectation.

Under these hypotheses, for each fixed $r \geq 2$,

$$\left|\kappa_{\mu_g}(X_1, \ldots, X_r)\right|_{\mathrm{loc}} \leq C_r g^{2r-2} \prod_i a_i, \tag{HS3}$$

where a_i is the fixed-core insertion norm. At the flat pair,

$$\boxed{\kappa_{\mu_g}(X_1, \ldots, X_r) = g^{2r} \kappa_\gamma(Q_1, \ldots, Q_r) + O_r\left(g^{2r+2} \prod_i a_i\right).} \tag{HS4}$$

The field-dependent coefficient obtained after removing the flat value has zero first pair jet. On a smaller physical core of radius α, radial Cauchy therefore adds its usual α^2 factor to the appropriate fixed-core bound. Its complete quadratic physical jet, not merely its zero-momentum marginal, must still be retained in the Gaussian recursion.

30.2 2. Why finite jets give the powers in these formulas

At a finite cutoff introduce one nilpotent label for each source and differentiate

$$\log \int \exp\left(-V_g + \sum_{i=1}^{r} z_i X_i\right) d\gamma. \tag{HS5}$$

To compute a specified finite g-jet, only finitely many interaction Taylor vertices and their Gaussian moments are needed. Expansion of the logarithm keeps exactly the connected Wick graphs: the exponential formula for set partitions cancels every graph whose vertex set decomposes into two unconnected parts. The denominator in (HS1) removes graphs without a source vertex. This is an identity of finite Taylor coefficients, not an assertion that the full power series converges.

Consider a connected graph with r source vertices, p primitive interaction vertices, and E contraction edges. Ignoring extra favorable powers from Jacobians or old activities, its coupling degree is

$$\sum_{\text{source}} l_i + \sum_{\text{interaction}} (l_v - 2) = 2E - 2p. \tag{HS6}$$

Connectivity implies $E \geq r + p - 1$, giving degree at least $2r - 2$. Old regular vertices can be counted instead among the non-source vertices with their actual extra powers; the same lower bound remains valid. Equivalently, an old linear or quadratic vertex has two more powers than the deliberately pessimistic assignment g^{l-2}. Thus it cannot lower (HS6).

At the flat background each source has at least two legs and each primitive interaction has at least three. Consequently

$$2E \geq 2r + 3p, \qquad 2E - 2p \geq 2r + p. \tag{HS7}$$

At degree exactly $2r$, there can be no primitive interaction vertex. An old regular vertex with at least two legs also increases the degree by at least two; the flat first-jet condition excludes old linear vertices. Every source must therefore contribute precisely its quadratic Q_i. The surviving connected graphs are exactly the Gaussian cumulant in (HS4). All smaller coefficients vanish.

The complete integral is an even function of g, by the stated simultaneous sign change, so its coefficient of degree $2r + 1$ also vanishes. Taylor's theorem with the bounded $(2r + 2)$nd derivative proves the remainder in (HS4). The same argument through order $2r - 2$ proves (HS3). Polynomial moments at the finitely many distinguished vertices are paid before integration; the original connected-tree estimate sums the remaining unmarked vertices. Thus the locality and source-root information survive the power-count argument.

The fixed-order constant C_r has not been estimated here with the uniform factorial dependence needed for an independent all-order reconstruction theorem. The earlier scalar ordinary-source bound supplies that separate reconstruction estimate.

30.3 3. Old mixed vertices cannot be removed from the leading term

Suppose an incoming vertex carrying a label set I has field weight $g^{2(|I|-1)}$. In a term with a partition of the m labels into r incoming vertices, their combined prefactor is $g^{2(m-r)}$. Equation (HS3) then gives the field weight $g^{2(m-1)}$; at the flat pair, (HS4) gives vacuum weight g^{2m}. The running-tube comparison between old and current couplings changes only fixed-order constants over the window.

This explains the correct grading for a higher-label induction. It also explains why the Gaussian leading term is the determinant of the complete transported quadratic source state. Already at two labels, the trace of the old mixed Hessian is part of that determinant. At higher labels there is a mixed Hessian for every nonempty subset. Replacing that collection by repeated fresh covariances would give the wrong answer.

To obtain a physical higher-point ultraviolet theorem one must still close that graded old-vertex induction, control every retained-field term in the total-cumulant formula, and extend the finite-window distinguished-history estimate to that fixed number of labels. The one-step result (HS4) is an input to that work, not its conclusion.

30.4 4. The Gaussian scalar targets in four dimensions

Let F_I^a, $I = 1, \ldots, 6$, be a free Maxwell curvature with independent color copies $a = 1, \ldots, d_G$, in the normalization used by the two-point window calculation. At separated points its covariance is

$$\langle F_I^a(x) F_J^b(y) \rangle = \delta^{ab} K_{IJ}(x - y), \qquad K(x) = \frac{1}{\pi^2 |x|^4} \bigwedge{}^2 R_x, \quad R_x = I - 2\hat{x}\hat{x}^{\mathsf{T}}. \tag{HS8}$$

Define the centered action-density quadratic

$$Q(x) = \frac{1}{2} \sum_{a,I} : F_I^a(x)^2 : . \tag{HS9}$$

For distinct $x_1, \dots, x_m$, finite-dimensional Gaussian integration, followed by differentiation of the logarithmic determinant, gives

$$\kappa_m(Q(x_1), \dots, Q(x_m)) = \frac{d_G}{2m} \sum_{\sigma \in S_m} \operatorname{Tr}[K_{\sigma_1\sigma_2} \cdots K_{\sigma_m\sigma_1}], \quad K_{ij} = K(x_i - x_j). \tag{HS10}$$

Normal ordering removes self-contractions, which do not occur in these connected cycles for $m \geq 2$. Formula (HS10) gives $3d_G/(\pi^4|x-y|^8)$ at order two, consistently with the previous computation.

There is a useful cancellation at every odd order. The Euclidean Hodge star on two-forms anticommutes with $\bigwedge^2 R_x$, because $\det R_x = -1$. A product of an odd number of such factors therefore also anticommutes with the star and has trace zero. Thus

$$\kappa_{2k+1}(Q(x_1), \dots, Q(x_{2k+1})) = 0 \quad \text{at separated points.} \tag{HS11}$$

In particular, the leading Gaussian three-point function cannot serve as a nonzero ultraviolet test. This statement concerns separated distributions; it does not set their contact extensions to zero.

At order four, cyclicity and reversal reduce (HS10) to

$$\begin{aligned} \kappa_4(Q_1, Q_2, Q_3, Q_4) = d_G \operatorname{Tr}\big(&K_{12}K_{23}K_{34}K_{41} + K_{12}K_{24}K_{43}K_{31} \\ &+ K_{13}K_{32}K_{24}K_{41}\big). \end{aligned} \tag{HS12}$$

For four distinct collinear points, all the reflection matrices are the same, their exterior square has square one, and each trace in (HS12) is strictly positive. More explicitly it is six divided by π^8 times the product of the four fourth powers of the corresponding edge distances. Continuity then gives nonnegative smooth profiles in four small separated neighborhoods for which the integrated fourth cumulant is positive.

A completed physical four-label comparison would therefore prove a non-Gaussian scalar sector, with leading fourth cumulant of order g_d^8. That statement would exclude a generalized free scalar description. It would still not distinguish interacting Yang–Mills from a non-Gaussian composite of free fields without the additional running and joint-operator identification. The full invariant family remains part of the target.

31.2 2. The physical-Hessian version of the connected power count

For r invariant inputs $O_1, \dots, O_r$, subtract their zero-fluctuation values and denote the resulting fluctuations by $X_i(A, g, B)$. At the flat pair, $DO_i = 0$, so

$$X_i(0, g, B) = g^2 Q_i(B) + O(g^3 |B|^3). \tag{FL6}$$

The full physical Hessian of the leading fluctuation term is determined by the source Hessian:

$$D_A\{g\, DO_i(A) T_A B\}_{A=0} = g K_i(HA, TB). \tag{FL7}$$

Here H is the actual linear harmonic lift, T the hidden injection and C the hidden Gaussian covariance; put $D_0 = TCT^*$. Dependence of T_A, C_A on A, and nonlinear terms in the actual current/background map, contributes at a strictly higher coupling order in the statement below.

The finite connected-jet calculation gives, for $2 \le r \le 4$,

$$\begin{aligned} \mathsf{P}\kappa_{\mu_g}(X_1, \dots, X_r) &= g^{2r-2} \mathcal{C}_r(K_1, \dots, K_r) + O_r(g^{2r} \prod_i a_i), \\ \kappa_{\mu_g}(X_1, \dots, X_r)\big|_0 &= g^{2r} \kappa_\gamma(Q_1, \dots, Q_r) + O_r(g^{2r+2} \prod_i a_i). \end{aligned} \tag{FL8}$$

The multilinear $\mathcal{C}_r$ is the physical Hessian of the corresponding Gaussian cumulant. Its connected graphs are chains with two linear endpoint legs and $r-2$ quadratic internal vertices. The non-Gaussian field-dependent coefficient on the fixed core is bounded by $C_r g^{2r-2}\prod a_i$, and its value on a physical core α has the usual factor α^2.

Here is a direct check of the first formula, beyond the flat scalar power count. At the full physical Hessian there are two external coarse-field legs. Suppose a finite connected Wick graph has r source vertices, p primitive action vertices, and E internal contractions. Its coupling power is at least $2E-2p$. Each source has total field degree at least two, and each primitive interaction has total degree at least three. Therefore

$$2E+2\geq 2r+3p, \qquad 2E-2p\geq 2r-2+p. \tag{FL9}$$

Old activity and Jacobian vertices have the additional favorable powers of their ordinary gB substitution. At degree $2r-2$ there are no primitive or old interaction vertices, all source vertices use their quadratic jets, and the two external legs must be on sources. Since X_i has at least one fluctuation leg, the result is exactly the chain just described. Two derivatives of the background covariance leave all source fluctuation degrees present and begin at degree $2r$. One covariance derivative combined with one external source leg can have formal degree $2r-1$, which is odd. Neither case supplies a missing leading chain. The full integral is even under $(g,B)\mapsto(-g,-B)$, so degree $2r-1$ vanishes. Bounded finite current-g derivatives give the remainder in (FL8). At the flat value there are no external legs; the same count gives the second formula.

For $r=1$, include the zero-fluctuation composition. Since the first source jet vanishes, its physical Hessian is H^*K_IH. The first line of (FL8) is then replaced by

$$\mathsf{P}\mathcal{T}_gO_I=H^*K_IH+O(g^2a_I), \qquad (\mathcal{T}_gO_I)|_0=\tfrac{1}{2}g^2\,\mathrm{Tr}(D_0K_I)+O(g^4a_I). \tag{FL10}$$

These are the previously proved one-insertion bounds. The cutoff tail (FL1) adds τ_s to each relative remainder in (FL8)–(FL10). All statements retain the actual source anchors and the rooted spatial decay of the connected construction.

31.3 3. The prefix grading and the original source roots

Split each f_i into own-a'-cell pieces. Each piece satisfies

$$\|f_{i,q}\|_\infty+(a')^\alpha[f_{i,q}]_{C^\alpha}\leq Ct_i. \tag{FL11}$$

There are $N_i\leq CD^4$ pieces. This operation precedes the finite-window rooted propagation; no one-root decay is started at the bare mesh.

The finite grading through four labels closes in the ultraviolet prefix. To see it, keep every birth on its own core and origin scale j, with normalized bound

$$g_j^{-2(|I|-1)}R_0^{-2}\|O_{I,j}\|_{\mathrm{root}}. \tag{FL12}$$

In a connected formation term whose r incoming label sets partition I, the incoming powers are $g_j^{2(|I|-r)}$, up to the bounded old-to-current ratios on the actual weak tube. Formula (FL8) supplies the missing g_j^{2r-2}. Thus the output has precisely the field grade in (FL12); its flat scalar output has grade $g_j^{2|I|}$. The $r=1$ old member is carried, not born again.

For the spatial induction first freeze the entire homogeneous family, including its own old counterterms, at the pointed source anchor. The constant-source coefficient of order $m\leq 4$ is controlled by its

own-endpoint bare disc:

$$|Y^{(m)}| \le C_m g_j^{2m-1} \operatorname{polylog} x_j \quad (m \ge 2), \qquad |Y' - 1| \le C g_j \operatorname{polylog} x_j. \tag{FL13}$$

The old homogeneous birth derivatives have the stronger ordinary Cauchy powers of that disc and their paired Ward age bound. The heterogeneous difference has at least one factor $\ell_j^{\alpha}[f_i]_{C^\alpha}$. At degree m, its only linear unknown is the degree-m old defect, multiplied by the small actual fluctuation/core factor. Products involve proper subsets and are already bounded by the lower-degree induction. The same geometric recurrence used for two labels therefore closes at degrees three and four: choose the physical/core margin once small enough for these finitely many recurrences. This proves (FL12) with the own-scale loads (FL11), rather than obtaining its coupling powers from an unweighted source Cauchy bound.

Exact source-carrier filing gives an additional simplification for separated tests. Any regular mixed member with $|I| \ge 2$ must have a completed carrier meeting two of their supports. Below a', its exponential therefore bridges distance d. Consequently, at the prefix endpoint,

$$\begin{aligned} \|O_{i,a'}\|_{\mathrm{root}} &\le C R_0^2 M_i, \\ \|K_{i,a'} - K^G_{i,a'}\|_{\mathrm{root}} &\le C M_i \omega, \\ g_{a'}^{-2(|I|-1)} R_0^{-2} \|O_{I,a'}\|_{\mathrm{root}} + g_{a'}^{-2(|I|-1)} \|K^G_{I,a'}\|_{\mathrm{root}} &\le C_I M_I \eta_{\mathrm{UV}} \quad (|I| \ge 2), \end{aligned} \tag{FL14}$$

where $\eta_{\mathrm{UV}} = P_4(x_{a'}, D)e^{-cD}$ is allowed a fixed polynomial loss. The first Gaussian Hessian is the exact all-depth one $F^*_{a'} M_{f_i} F_{a'}$, in the common normalization. All Gaussian mixed Hessians are retained, even though their incoming separated-carrier bound is exponentially small.

The mixed scalar prefix also has a uniform ultraviolet bridge. Choose a first-source cell as root before counting. Its total bound is

$$C g_{a'}^{2|I|} \prod_{i \in I} t_i \sum_{\ell_j \le a'} (d/\ell_j)^4 \operatorname{poly}(x_j/x_{a'}) e^{-cd/\ell_j} \le C_I g_{a'}^{2|I|} \prod_{i \in I} t_i P_I(x_{a'}, D) e^{-c'D}. \tag{FL15}$$

The fixed tube ratios and any polylogarithms can be included in the displayed harmless polynomial. Since $x_j \le x_{a'} + C(1 + \log_L(a'/\ell_j))$, the bridge pays those older-scale powers. There is one connected root count, not an independent ultraviolet volume for every label. A carrier through a genuine live hole is not treated by (FL15); that is part of the separate physical history matching.

31.4 4. Exact Gaussian transport for every subset

For a pure quadratic source state $\nu(z) + \frac{1}{2} A^* K(z) A$, let $K(z) = \sum_{I \ne \varnothing} z_I K_I$. Gaussian completion at one step gives

$$\begin{aligned} K'_I &= \sum_{r=1}^{|I|} (-1)^{r-1} g^{2r-2} \sum_{(I_1,\dots,I_r) \in \mathrm{OP}_r(I)} H^* K_{I_1} D_0 K_{I_2} \cdots D_0 K_{I_r} H, \\ \nu'_I - \nu_I &= \frac{1}{2} \sum_{r=1}^{|I|} \frac{(-1)^{r+1} g^{2r}}{r} \sum_{(I_1,\dots,I_r) \in \mathrm{OP}_r(I)} \operatorname{Tr}(D_0 K_{I_1} \cdots D_0 K_{I_r}). \end{aligned} \tag{FL16}$$

Here $\mathrm{OP}_r(I)$ is the set of ordered partitions into nonempty disjoint blocks. There are no factorial conventions hidden in (FL16): the coefficient of z_I equals the mixed derivative at zero in the squarefree algebra. The $r = 1$ term in the scalar line is the trace of the OLD mixed Hessian K_I.

Formula (FL16) follows by the finite nilpotent Neumann series for the Schur complement and the finite logarithmic determinant series. In particular it is an exact identity, not an approximation by repeated fresh covariances. The powers in (FL5) close term by term. The source-zero harmonic compositions and the nested Gaussian precision are those of the actual maps; their all-depth bounds are used as such. No locality assertion about a massless $S^{-1/2}$ is needed.

31.5 5. Accumulated clean Hessian and vacuum errors

Normalize a source coefficient of degree m by $g_d^{2(m-1)} R_0^2 M_I$. The rooted one-origin propagation has a bounded resolvent, independent of the length of the regular run. After this linear propagation has been summed, a marked nonlinear formation joins at least two proper label sets. There can be at most $m-1$ such joining vertices in a degree-m graph. Choosing their times in an n-step window and summing their connected spatial kernels gives

$$g_d^{-2(m-1)} R_0^{-2} M_I^{-1} \|O_{I,s}\|_{\mathrm{root}} \le C_m (1+n)^{m-1}, \qquad m \le 4. \tag{FL17}$$

This is also proved by induction on m: a partition into r proper blocks contributes at most $(1+n)^{m-r}$, and its birth-time sum supplies one more power. The worst case is $r=2$. Linear descendants are already in the one-root resolvent and do not introduce a constant to the power n.

Let

$$e_I(s) = g_d^{-2(|I|-1)} M_I^{-1} \|K_{I,s} - K_{I,s}^G\|_{\mathrm{root}}. \tag{FL18}$$

The physical-Hessian statement (FL8) identifies the leading map with (FL16). Telescope the exact finite compositions at the first different formation. A Gaussian leading term containing an error has a proper-subset error unless it is the single linear composition $H^* \Delta K_I H$; the latter has the uniform all-depth propagation just used. A nonlinear joining increases the label degree, so at most $m-1$ such joining times occur. Each genuine quantum discrepancy has relative size ϵ_{w}, and its position contributes at most one additional time sum. Thus

$$e_I(s) \le C_m (1+n)^{m-1} \{\omega + n\epsilon_{\mathrm{w}} + \eta_{\mathrm{UV}}\}. \tag{FL19}$$

The same estimate follows by subtracting the two multilinear partition recursions and using the elementary telescoping identity for a product. Only one error factor is needed; all other factors are bounded by (FL17). This does not assume that two independently small live masses can be multiplied: this section concerns regular analytic activities and their genuine Gaussian connected kernels.

For the scalar output, (FL8) gives the exact Gaussian cycle in (FL16), plus a relative ϵ_{w} remainder. The old mixed trace contains e_I and is retained. Emitting a scalar adds one final time choice, and a scalar already emitted is never regenerated at later steps. Therefore

$$\begin{aligned} |\Delta\nu_I^{\mathrm{cl}} - \Delta\nu_I^G| \le C_m g_d^{2m} M_I (1+n)^m \\ \times \{\omega + n\epsilon_{\mathrm{w}} + \eta_{\mathrm{UV}}\}. \end{aligned} \tag{FL20}$$

For $m=4$, using $M_I \le C(H/a')^{16} \prod_i t_i$, a convenient weaker form is

$$C g_d^8 \prod_i t_i \, (H/a')^{16} (1+\log(H/a'))^5 \{x_d^{-1/2} \operatorname{polylog} x_d + \max_s \tau_s + \eta_{\mathrm{UV}}\}. \tag{FL21}$$

This bound concerns the accumulated vacuum, not just a fresh four-point birth.

31.6 6. A connected-root bound for every retained subset

Fix $0 < \delta < 1$, and set $\lambda = 4 - \delta$. Choose the physical/core margin so that the one-root regular resolvent has rate $L^{-\lambda m}$, as in the two-label argument. For a specified leaf source cell at scale a', its readout at scale ρ has factor $(a'/\rho)^{\lambda}$, with an exponentially summable spatial kernel at scale ρ.

The following finite branching estimate keeps every original leaf root. For a label block I, $m = |I| \le 4$, after summing its output location,

$$\|\Phi_{I,H}\|_{\infty,\mathrm{regular}} \le C_m g_H^{2(m-1)} \alpha_H^2 \prod_{i\in I} t_i \, (d/H)^{\lambda} (d/a')^{m\delta} + \text{ultraviolet-bridge remainder.} \tag{FL22}$$

Here the physical core is $\alpha_H = g_H P_{\mathrm{bg}}(x_H)$. The functions are the actual retained coefficients, including their full classical higher field terms, on their admitted own core.

To prove (FL22), first keep one specified a'-cell for each label and suppress its amplitude. For $m = 1$ the bound before summing source cells is $(a'/H)^{\lambda}$. For a joining vertex at scale ρ whose $r \ge 2$ child blocks partition I, the induction bound and subsequent linear propagation give, before the leaf sums,

$$C_m (a')^{m\lambda} H^{-\lambda} d^{-(m-r)\lambda} \rho^{-(r-1)\lambda} \chi_m(\rho/d). \tag{FL23}$$

For $\rho \ge d$, take $\chi_m \le C_m$. For $\rho < d$, either a child already contains two separated labels, or all children are single labels and the new connected kernel joins two separated source supports. The retained complete source tree then supplies $\chi_m(\rho/d) \le C_m e^{-c_m d/\rho}$, after using spare decay to sum polynomial carrier costs. This statement uses the completed REGULAR carrier, not a relative tree crossing a hole.

The geometric scale sum is

$$\sum_{\rho \le H} \rho^{-(r-1)\lambda} \chi_m(\rho/d) \le C_{m,\lambda} d^{-(r-1)\lambda}. \tag{FL24}$$

Above d it is a decreasing geometric series; below d the bridge exponential dominates every increasing power. Combining (FL23)–(FL24) proves

$$C_m (a')^{m\lambda} H^{-\lambda} d^{-(m-1)\lambda} \tag{FL25}$$

for the specified leaves. The sum over their at most $\prod N_i \le C(d/a')^{4m}$ original cells turns (FL25) into $(d/H)^{\lambda}(d/a')^{m\delta}$, exactly (FL22). Input mixed packets from (FL14) already have the ultraviolet bridge and are included by the same linear propagation. The number of joining vertices is bounded by $m - 1$, so a fixed loss of spatial decay suffices throughout.

31.7 7. The terminal total-cumulant formula

After all hidden integrations through H, retain the ACTUAL positive coarse/background law μ_H. No Gaussian law for that remainder is assumed. In this section set $\Phi_{I,H} = O_{I,H}$: all emitted scalar constants ν_I have already been placed in the separate vacuum sum. Thus $\Phi(z) = \sum_{I \ne \varnothing} z_I \Phi_{I,H}$ in the next formula contains only the retained zero-flat members, unlike the combined incoming notation (FL2). For the four labels,

$$D_I \log \int e^{-\Phi(z)} d\mu_H\big|_0 = \sum_{\pi \in \mathfrak{P}(I)} (-1)^{|\pi|} \kappa_{\mu_H}(\Phi_B : B \in \pi). \tag{FL26}$$

This is the finite partition identity for the logarithm. It includes the single mixed insertion, every covariance and third cumulant of mixed insertions, and the fourth cumulant of the first insertions. None is discarded.

For r bounded random variables, $|\kappa_r| \le C_r \prod \|\cdot\|_\infty$, with a constant given by the finite partition formula. If π has r blocks, (FL22) gives its contribution bounded by

$$C_4 g_H^{2(4-r)} \alpha_H^{2r} \prod_i t_i \, (d/H)^{r\lambda} (d/a')^{4\delta}. \tag{FL27}$$

Since $\alpha_H = g_H P_{\mathrm{bg}}$, every term has the correct total strength g_H^8, times a fixed polylogarithm. This is why the mixed coefficient grading is needed; a fixed unweighted source Cauchy bound would lose inverse powers of the coupling here.

Take

$$t = \log(1/d), \qquad a' = d\, t^{-\varepsilon}, \qquad H = d\, t^{\varepsilon}, \qquad x_d \asymp t. \tag{FL28}$$

The worst terminal exponent is $r = 1$:

$$(d/H)^{\lambda} (d/a')^{4\delta} = t^{-\varepsilon(4-5\delta)}. \tag{FL29}$$

Choose $\delta < 4/5$, for example $\delta = 1/2$. Every retained-field term is then $o(g_d^8 \prod t_i)$, uniformly on the admitted regular core, independently of the coarse law.

32 Compact coarse-field limits, reference entropy, and physical observables

Three conclusions must be kept distinct. A continuous completion of the clean block map gives projective subsequential limits of the actual Wilson measures by compactness. The effective-density bounds give ordinary noncollapse for the renormalized reference laws. Relating that noncollapse to physical Wilson observables requires the correctly carried insertion through the large-field R operation. The first two conclusions do not by themselves supply the third.

32.1 A globally continuous completion of the regular block map

For an oriented coarse bond c, take a finite determining carrier containing every path in its centered average and finite plaquette fillings comparing those paths. Let U_c be the central straight-path transporter. There is a fixed $\delta_{\mathrm{out}} > 0$ such that, when every determining plaquette satisfies $\|U_p - 1\| < \delta_{\mathrm{out}}$, all internal group means and relative logarithms lie on their unique regular branches. To verify this, fix a tree on the finite patch. Each relative path is a product of at most C_L transported plaquettes; choose $C_L\delta_{\mathrm{out}}$ smaller than the logarithmic chart radius. Write this regular average as $M_{\mathrm{reg},c}$ and put

$$a_c(U) = \mathrm{Log}\big(M_{\mathrm{reg},c}(U)U_c^{-1}\big).$$

It is small, analytic, and transforms by conjugation at c_-.

Choose a smooth gauge-invariant cutoff χ_c, equal to one on a smaller plaquette core and supported strictly inside the regular domain. A product of identical smooth functions of squared Hilbert–Schmidt plaquette deviations can be made invariant under inversion and the determining carrier's lattice symmetries. Define

$$\widehat{M}_c(U) = \exp[\chi_c(U)a_c(U)]U_c, \tag{114}$$

extending $\chi_c a_c$ by zero outside its support. This is a globally smooth, local, gauge-covariant map into G. On reversing c, the logarithm becomes $-\mathrm{Ad}(U_c^{-1})a_c$, so the result is the inverse bond value. The symmetric choice of carrier and cutoff preserves the lattice isometries.

The completion equals the original mean on the smaller clean core and changes none of its Gaussian jets. The regular-domain definition of the group mean is the one in [4, (0.10)–(0.12)]. We do not assume that the entire published large-field construction automatically covers this particular global completion. The compactness argument below does not require that.

32.2 Actual Wilson projective limits and their bounded reflection form

Fix a physical torus with nested lattice spacings $a_r = L^{-r}$, and put $X_r = G^{E_r}$ for its finite compact link spaces. Let $M_r : X_{r+1} \to X_r$ be (114) at that scale. If μ_n is the actual Wilson probability law at fine cutoff n, define

$$\nu_{n,r} = (M_r \cdots M_{n-1})_*\mu_n, \qquad r \le n. \tag{115}$$

Proposition 32.1 (Compact physical projective system)**.** *There is a subsequence and probability laws ν_r such that $\nu_{n,r} \Rightarrow \nu_r$ at every fixed level and $M_{r*}\nu_{r+1} = \nu_r$. They define a joint probability law on the countable product of link spaces, supported on all deterministic blocking relations. Gauge invariance passes to that law.*

Proof. Each X_r is compact metric, so its probability laws are weakly sequentially compact. Diagonalize over r. Continuity of M_r passes the finite-cutoff identity to the limit. The joint law at levels $0, \dots, r$ is the pushforward of ν_r by

$$v \longmapsto (M_0 \cdots M_{r-1}v, \dots, M_{r-1}v, v).$$

These finite laws are consistent. Countable product extension gives the stated law, and the blocking graphs are closed. Gauge covariance and finite-cutoff gauge invariance give the last assertion. □

The argument works for any positive bare coupling sequence. In particular, the tuned trajectory does not turn compactness alone into a noncollapse theorem.

Let θ be an aligned coordinate reflection. A coarse positive-side cylinder whose complete determining carriers lie in the positive half pulls back to a fine positive-side cylinder. Reflection covariance of the map and Wilson reflection positivity give

$$\int \overline{F(\theta U)} F(U)\, \mathrm{d}\nu_{n,r}(U) \geq 0. \tag{116}$$

For continuous F, the inequality passes to the limit. Put a finite family from different resolutions on a common finer resolution to obtain its full Gram matrix. This proves reflection positivity for that bounded cylinder algebra with the stated carrier convention. It supplies neither continuity of physical translations nor a shrinking-block curvature distribution bound.

32.3 A stronger extraction lemma for bounded densities

Lemma 32.2 (Measurable projective passage and ordinary noncollapse)**.** *Let λ_r be normalized Haar measure on X_r. Suppose at each fixed level*

$$0 \leq p_{n,r} \leq M_r < \infty, \qquad p_{n,r} \geq m_r \mathbf{1}_{O_r},$$

where $m_r > 0$ and O_r is nonempty open. A diagonal subsequence converges against every bounded measurable test at each level, with the same limiting bounds. Any fixed Borel pushforward identity between these marginals passes to the limit. For a continuous real F nonconstant on O_r,

$$\mathrm{Var}_{p_r \lambda_r}(F) \geq m_r \lambda_r(O_r)\, \mathrm{Var}_{\lambda_r(\cdot|O_r)}(F) > 0. \tag{117}$$

Proof. The $L^\infty(\lambda_r)$ ball is weak-star sequentially compact because $L^1(\lambda_r)$ is separable. Diagonal extraction and testing positive L^1 functions preserve the upper and lower bounds. All bounded measurable tests belong to L^1. If M_r is merely Borel, both a bounded test f and $f \circ M_r$ are among these tests, so its pushforward identity passes. Finally use $\mathrm{Var}_\mu F = \inf_c \int |F - c|^2 \, \mathrm{d}\mu$ and the lower density on O_r. A coarse plaquette trace is a suitable nonconstant gauge-invariant example near the identity. □

This stronger topology, rather than an unjustified global choice of a group mean branch, handles Borel block maps when the density bounds are available.

32.4 A common lower-density core over all finer scales

We give a uniformly bounded smooth lift, including the correction from straightened to raw averaging. Work in retained spacing one and fine spacing $\eta = L^{-k}$. Let Q_k be the flat derivative of the cumulative straightened map. Put

$$G_\eta = (I + \Delta_\eta)^{-1}, \quad B_k = Q_k G_\eta^2 Q_k^T, \quad E_k = G_\eta^2 Q_k^T B_k^{-1}. \tag{118}$$

Then $Q_k E_k = I$. The unaliased multiplier bound used in the clean determinant comparison gives $cI \leq B_k \leq CI$ uniformly in depth and volume. The physical Bloch strip for the average and massive resolvent gives the same strip for B_k^{-1}; its coarse kernel is exponentially decaying, hence bounded on ℓ^∞.

The physical ℓ^∞ norm of Q_k^T is uniform: its averaging coefficients are divided by η^4, while each fine bond meets only a fixed number of physical carriers. Moreover

$$\|\nabla_\eta^\alpha G_\eta^2\|_{\infty \to \infty} \leq C_\alpha, \qquad |\alpha| \leq 2. \tag{119}$$

Indeed $G_\eta^2 = \int_0^\infty te^{-t}e^{-t\Delta_\eta}\,\mathrm{d}t$. The heat semigroup is a positive contraction. In one dimension its first kernel difference has ℓ^1 norm twice its maximum, by unimodality; the Fourier formula bounds that maximum by $C\min(1,\eta/\sqrt{t})$. Divide by η, use the semigroup identity to split repeated differences, and tensor over the four coordinates. The bound is $C_\alpha t^{-|\alpha|/2}$, integrable against te^{-t} for these orders. Periodic summation preserves the ℓ^1 estimate. Thus

$$\|E_k v\|_{C^2_\eta} \le C\|v\|_\infty. \tag{120}$$

The clean all-depth averaging estimate supplies a common holomorphic C^1 ball for q_k. Consequently $w \mapsto q_k(E_k w)$ has derivative I at zero and uniformly bounded second derivative on a smaller fixed ball. The contraction form of the analytic inverse theorem produces, for $\|v\|_\infty < \delta$,

$$q_k(E_k w(v)) = v, \qquad \|w(v)\|_\infty \le 2\|v\|_\infty,$$

uniformly in k and volume. With $A = E_k w(v)$ and $U_b = e^{i\eta A_b}$, each fine plaquette is $O(\eta^2\delta)$ from the identity, and hence

$$A_{mWilson}(U) \le C\delta^2|T|. \tag{121}$$

This is initially a lift for the straightened map. The exact identity gives $M^k(U) = (e^{iv})^{v_k(A)}$. The accumulated node factor is $O(\delta)$ from the identity: its one-step logarithms are $O(\delta L^s\eta)$ and sum geometrically. Extend its inverse from coarse nodes to a smooth fine gauge by fixed cardinal interpolation of the small logarithm. Its physical derivatives are uniformly bounded. Gauge-transforming U gives the raw constraint $M^k(U) = e^{iv}$ and leaves (121) unchanged. This correction cannot be replaced by simply calling a right inverse of Q_k a right inverse of the raw map.

The original regular variational theorem supplies the actual minimizing branch on a fixed small coarse neighborhood. Its action is at most that of the lift, and its own regularity estimates place it in the lower-cutoff core after decreasing δ. A small action alone would not prove that pointwise regularity; this is an explicit imported variational input.

For fixed retained physical resolution r, let t count finer scales. The tuned trajectory bounds give $x_{mfine}(t) \le A_r + Bt$, so $g_{mfine}(t) \ge (A_r + Bt)^{-1/2}$. The smooth background has raw link size $C\delta L^{-t}$ and plaquette size $C\delta L^{-2t}$. Since

$$\sup_{t\ge 0} L^{-t}\sqrt{A_r + Bt} < \infty, \tag{122}$$

one fixed $\delta_r > 0$ meets every old threshold of the form $g_s P_s$ with $P_s \ge 1$. Fixed extra powers or logarithms are also dominated by the geometric factor. Nested critical-minimum fluctuations are zero at their centers. Thus old tests do not force this fixed-resolution core to shrink with the ultraviolet cutoff; it may depend on r and the terminal coupling.

32.5 Reference-law bounds and an infinite-volume entropy argument

At a fixed retained resolution, the effective coupling remains in a compact weak interval. The upper and lower effective-density estimates of [8, Theorem 1], together with the preceding core, have the form

$$0 \le \rho_{n,r}(V) \le e^{C_{+,r}|T|}, \qquad \rho_{n,r}(V) \ge e^{-C_{-,r}|T|} \quad (V \in O_{r,T}),$$

where $O_{r,T}$ is a product of fixed small Haar balls. Its Haar volume is at least $e^{-C_{\mathrm{ball},r}|T|}$. Thus the normalizing integral is bounded above and below exponentially in volume, and the normalized reference density satisfies

$$p_{n,r,T} \le e^{C_r|T|}. \tag{123}$$

On a fixed torus, Lemma 32.2 gives nondegenerate reference-law limits. A positive history representation can also provide a joint law of successive reference fields and stored data; its marginal consistency is not the deterministic blocking relation of Proposition 32.1.

Proposition 32.3 (Local entropy and variance, uniformly in volume)**.** *Suppose* (123) *holds and the reference law is invariant under a fixed translation sublattice, independent of volume and cutoff. Then every fixed finite window A has*

$$D(\mu_{n,r,T,A}\|\lambda_A) \le C_{r,A}. \tag{124}$$

Subsequential limits have the same bound and converge against bounded measurable tests in that window. Every fixed local continuous observable with nonatomic Haar distribution has a strictly positive ordinary variance, uniformly over these laws and their limits.

Proof. The global entropy is at most $C_r|T|$. Enclose A in a box B fitting the translation sublattice and tile tori by its disjoint translates B_i. Entropy relative to a product reference is superadditive over disjoint coordinate groups:

$$D(\mu_T\|\lambda_T) \ge \sum_i D(\mu_{T,B_i}\|\lambda_{B_i}).$$

This is the chain rule plus the nonnegative entropy relative to the product of the marginals. Translation invariance makes the summands equal. Dividing by the number of boxes gives a uniform bound on B, and marginalization gives (124). Taking tori divisible by the box avoids a boundary packing issue. Translation averaging of the joint reference laws can supply this symmetry if necessary; it is an explicit reference-state choice.

For an event E with $q = \lambda_A(E) \in (0,1)$, binary coarse graining gives

$$\mu_A(E) \le \min\left(1, \frac{C_{r,A} + \log 2}{\log(1/q)}\right). \tag{125}$$

Weak compactness and entropy lower semicontinuity preserve the bound in a limit. For a bounded measurable test use Lusin approximation by a continuous function outside a Haar-small set. Equation (125) controls that exceptional set uniformly; weak convergence then proves convergence for the measurable test.

If F has nonatomic Haar distribution, compactness of its range implies $q_F(\epsilon) = \sup_c \lambda_A(|F-c| \le \epsilon) \to 0$. Choose $\epsilon > 0$ so that $\log(1/q_F(\epsilon)) \ge 4(C_{r,A} + \log 2)$. Every interval of that radius has μ_A-mass at most $1/4$, so

$$\mathrm{Var}_{\mu_A} F = \inf_c \mathbb{E}|F-c|^2 \ge \tfrac{3}{4}\epsilon^2. \tag{126}$$

A nonconstant real analytic plaquette trace has nonatomic Haar distribution: each of its level sets is a proper analytic subset and has Haar measure zero. □

Diagonalize over windows and resolutions for infinite-volume reference laws. The stronger convergence passes a fixed Borel pushforward identity, or a fixed Borel transition kernel tested on continuous products, through its controlled input marginal. It does not pass an arbitrary Borel relation involving an uncontrolled joint law. No deterministic relation absent at finite cutoff may be inserted by this argument.

32.6 The physical/reference distinction and a support criterion

The complete averaging transformation T in the introduction of [10] is followed by the additional R operation. The later large-field paper explicitly describes a gauge-fixed density as equivalent under

integration, rather than pointwise equal, and retains operators on live islands [8, printed p. 378 and (1.103)–(1.104)]. This distinction matters for observables.

For a positive reference replacement with retained boundary data b,

$$(Rq)(y,b) = \nu(\mathrm{d}y \mid b) \int q(\xi,b)\,\mathrm{d}\xi,$$

the physical numerator is

$$R(qF)(y,b) = \nu(\mathrm{d}y \mid b) \int q(\xi,b)F(\xi,b)\,\mathrm{d}\xi. \tag{127}$$

It is not generally $(Rq)(y,b)F(y,b)$. For a finite example, on $[0,1]$ take $p_n = n\mathbf{1}_{[0,1/n]}$ and replace the coordinate by independent uniform measure. Every reference density is one, while the original laws converge to a point mass. Positivity and mass preservation cannot transfer reference noncollapse to the original coordinate.

A support-preserving positive kernel would provide another route. Suppose an input law is reflection positive and an output kernel factors, for two reflected regions, into a positive-side kernel depending only on positive-side input and its reflected copy. Boundary variables fixed by reflection may be retained. For output F define $A_F(x_+) = \int F(y_+)K_+(\mathrm{d}y_+ \mid x_+)$. Then

$$\mathbb{E}_{\mu K}[\overline{F(\theta y)}F(y)] = \mathbb{E}_{\mu}[\overline{A_F(\theta x)}A_F(x)] \geq 0, \qquad \|A_F\|_\infty \leq \|F\|_\infty. \tag{128}$$

The identity gives every finite Gram matrix. If the two-region output kernel is within input-averaged total variation ϵ of such a factorized kernel, the form is at least $-2\epsilon\|F\|_\infty^2$.

For actual histories, one prospective implementation retains islands meeting a mirror collar and performs replacements separately on the two sides. It needs a rooted dependency-cluster estimate, for example $C_r e^{-c\delta/(a_r P_r)} \to 0$, linking an observable at distance δ from the mirror to that collar. Positivity and local charts do not prove this estimate; retained boundary interactions and old histories must be included. Moreover joint products must be transported together: $\mathbb{E}[FG \mid U] \neq \mathbb{E}[F \mid U]\mathbb{E}[G \mid U]$ in general. Separate conditional expectations cannot silently replace the output algebra.

Ordinary variance is not an Osterwalder–Schrader excitation witness. With independent fields on the two halves, $\mathbb{E}[\overline{F(\theta U)}F(U)] = |\mathbb{E}F|^2$, so every centered positive-side observable has zero reflection norm despite positive ordinary variance. Product Haar fields give a compact gauge example. A nonzero reflected connected form, local distribution bounds, translation continuity, full Euclidean covariance, and the relevant vacuum property remain additional requirements.